```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

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PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
#  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
#  因此系统管理组只留一个入口。
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
]
@app.context_processor
def inject_nav():
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    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count()
== 0],
    }
```

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    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
```

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    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
```

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        "gender": self.gender,                                          301
                    "role": self.role, "active": self.active}           302
class OperationLog(db.Model):                                           303
    """操作日志。"""                                                    304
    __tablename__ = "operation_logs"                                    305
    id = db.Column(db.Integer, primary_key=True)                        306
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"            307
    ), nullable=True)                                                   308
    action = db.Column(db.String(128), nullable=False)                  309
    # 操作类型                                                          310
    target_type = db.Column(db.String(64), nullable=True)               311
    # 对象类型                                                          312
    target_id = db.Column(db.Integer, nullable=True)                    313
    # 对象主键                                                          314
    detail = db.Column(db.Text, nullable=True)                          315
    # 操作详情                                                          316
    created_at = db.Column(db.DateTime,                                 317
    default=datetime.utcnow, index=True)                                318
class Project(db.Model):                                                319
    """评价项目（项目与矿山档案）。"""                                  320
    __tablename__ = "projects"                                          321
    id = db.Column(db.Integer, primary_key=True)                        322
    name = db.Column(db.String(128), nullable=False)                    323
    code = db.Column(db.String(64), nullable=True)                      324
    # 项目编号                                                          325
    description = db.Column(db.Text, nullable=True)                     326
    status = db.Column(db.String(20), default="进行中")                 327
    # 进行中 / 已完成 / 已归档                                          328
    created_by = db.Column(db.Integer,                                  329
    db.ForeignKey("users.id"), nullable=True)                           330
    created_at = db.Column(db.DateTime,                                 331
    default=datetime.utcnow)                                            332
    patches = db.relationship("MinePatch",                              333
    backref="project", lazy="dynamic",                                  334
                                    cascade="all, delete-orphan")       335
class MinePatch(db.Model):                                              336
    """矿山图斑。"""                                                    337
    __tablename__ = "mine_patches"                                      338
    id = db.Column(db.Integer, primary_key=True)                        339
    project_id = db.Column(db.Integer,                                  340
    db.ForeignKey("projects.id"), nullable=False)                       341
    name = db.Column(db.String(128), nullable=False)                    342
    admin_region = db.Column(db.String(128), nullable=True)             343
    # 所在行政区                                                        344
    center_x = db.Column(db.Float, nullable=True)                       345
    # 中心经度 / 横坐标                                                 346
    center_y = db.Column(db.Float, nullable=True)                       347
    # 中心纬度 / 纵坐标                                                 348
    area_ha = db.Column(db.Float, nullable=True)                        349
    # 图斑面积（公顷）                                                  350
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    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)  # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
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    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
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    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)        # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
```

```
    """权重明细：方案内的各项权重。"""                                    501
    __tablename__ = "weight_items"                                      502
    id = db.Column(db.Integer, primary_key=True)                        503
    scheme_id = db.Column(db.Integer,                                   504
    db.ForeignKey("weight_schemes.id"), nullable=False)                 505
    indicator_code = db.Column(db.String(32),                           506
    nullable=False)       # 指标编码                                    507
    weight = db.Column(db.Float, nullable=False)                        508
    # 权重值                                                            509
class ExpertMatrix(db.Model):                                           510
    """专家判断矩阵：多专家打分的原始矩阵。"""                            511
    __tablename__ = "expert_matrices"                                   512
    id = db.Column(db.Integer, primary_key=True)                        513
    scheme_id = db.Column(db.Integer,                                   514
    db.ForeignKey("weight_schemes.id"), nullable=True)                  515
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             516
    unit_type = db.Column(db.String(32), nullable=True)                 517
    # 微地质单元类型                                                    518
    l1_name = db.Column(db.String(64), nullable=True)                   519
    # 所属一级指标（二级矩阵用）                                        520
    expert_name = db.Column(db.String(64), nullable=True)               521
    # 专家姓名                                                          522
    level = db.Column(db.Integer, nullable=False,                       523
    default=1)          # 矩阵对应指标层级                              524
    matrix_json = db.Column(db.Text, nullable=False)                    525
    # 判断矩阵（JSON 二维数组）                                         526
    cr = db.Column(db.Float, nullable=True)                             527
    # 该专家矩阵的一致性比例                                            528
    created_at = db.Column(db.DateTime,                                 529
    default=datetime.utcnow)                                            530
class IndicatorData(db.Model):                                          531
    """指标实测数据：按微地质单元组织。"""                               532
    __tablename__ = "indicator_data"                                    533
    id = db.Column(db.Integer, primary_key=True)                        534
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni            535
    ts.id"), nullable=False, unique=True)                               536
    soil_ratio = db.Column(db.Float, nullable=True)                     537
    # 土石比                                                            538
    thickness = db.Column(db.Float, nullable=True)                      539
    # 土层厚度 cm                                                       540
    ph = db.Column(db.Float, nullable=True)                             541
    organic_matter = db.Column(db.Float, nullable=True)                 542
    # 有机质 g/kg                                                       543
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null   544
    heavy_metals = db.Column(db.Text, nullable=True)                    545
    coverage = db.Column(db.Float, nullable=True)                       546
    # 植被覆盖度 %                                                      547
    shannon = db.Column(db.Float, nullable=True)                        548
    # Shannon-Wiener 多样性指数                                         549
    slope = db.Column(db.Float, nullable=True)                          550
```

```
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
```

```
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator,
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  #  动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("账户名称", validators=[DataRequired(
    ), Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
```

```
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                       Length(min=2, max=64,
                                              message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码
    "),
                                         Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",

    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,

    message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
```

```
        EqualTo("new_password",
        message="两次输入的新密码不一致")])
        submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ---------------------------------------------------------------------
    ------------------
# 一、微地质单元类型
# ---------------------------------------------------------------------
    ------------------
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ---------------------------------------------------------------------
    ------------------
# 二、指标标准化理想极值（论文 4.2.2 节）
# ---------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 （mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
```

```
#  坡度分界，论文  4．2．2  节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２  号
SLOPE_BREAK  =  25．0
#  坡向离散赋值，论文  4．2．2  节："阳坡、阴阳坡、阴坡分别赋值  0．2、0．6、1"
ASPECT_SCORES  =  {"阳坡":  0．2，  "阴阳坡":  0．6，  "阴坡":  1．0}
ASPECT_TYPES  =  ("阳坡",  "阴阳坡",  "阴坡")
#  pH  适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW  =  4．0
PH_TOL_HIGH  =  10．0
#  ----------------------------------------------------------------
      ------------------
#  三、适宜性等级划分（论文表  4）
#  ----------------------------------------------------------------
      ------------------
GRADE_LEVELS  =  (
      {"grade":  "不适宜",  "min":  None,  "max":  0．3，
        "meaning":  "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复"，
        "measure":  "生态重建、人工辅助"}，
      {"grade":  "基本适宜",  "min":  0．3，  "max":  0．7，
        "meaning":  "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求"，
        "measure":  "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"}，
      {"grade":  "适宜",  "min":  0．7，  "max":  None，
        "meaning":  "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快"，
        "measure":  "自然恢复"}，
)
def  grade_of(score):
      """按论文表  4  判定适宜性等级。"""
      for  g  in  GRADE_LEVELS:
            lo，  hi  =  g["min"]，  g["max"]
            if  lo  is  None  and  score  <  hi:
                  return  g
            if  hi  is  None  and  score  >=  lo:
                  return  g
            if  lo  is  not  None  and  hi  is  not  None  and  lo  <=
      score  <  hi:
                  return  g
      #  score  恰好等于  0．7  时落入"适宜"
      return  GRADE_LEVELS[-1]
#  ----------------------------------------------------------------
      ------------------
#  四、平均随机一致性指标  RI（论文  4．2．1  节给出  n=2，3，4，5，其余为常用扩展值）
#  ----------------------------------------------------------------
      ------------------
RI_TABLE  =  {1:  0．0，  2:  0．0，  3:  0．58，  4:  0．90，  5:  1．12，  6:
      1．24，
                  7:  1．32，  8:  1．41，  9:  1．45，  10:  1．49}
#  一致性比例阈值（论文：当  CR<0．1  时，判断矩阵满足一致性要求）
CR_THRESHOLD  =  0．1
#  ----------------------------------------------------------------
      ------------------
#  五、默认权重方案（论文表  3）
```

```
#  ----------------------------------------------------------------------  801
    ----------------                                                        802
#  论文　４．２．２　节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  803
#  分别计算评价指标权重"（论文摘要、图　２　流程与表　５　均作"碎石边坡"，  804
#  系统统一采用"碎石边坡"）。权重由　１０　位专家打分后经　ＡＨＰ　方法确定。  805
#  下表为论文表　３　所载的归一化权重，作为系统默认方案。                  806
DEFAULT_WEIGHTS　=　{                                                      807
    "岩质边坡"：　{                                                        808
        "ｌ１"：　{"植被指标"：　０．４５，　"地形指标"：　０．５５},        809
        "ｌ２"：　{                                                        810
            "植被指标"：　{"植被覆盖度"：　０．６７，　"生物多样性"：　０．３３},  811
            "地形指标"：　{"坡度"：　０．６４，　"坡向"：　０．３６},       812
        },                                                                 813
    },                                                                     814
    "碎石边坡"：　{                                                        815
        "ｌ１"：　{"土壤指标"：　０．４４，　"植被指标"：　０．３１，　"地形指标"：　０．２５},  816
        "ｌ２"：　{                                                        817
            "土壤指标"：　{"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  818
                          "有机质"：　０．１７，　"污染指标"：　０．１７},  819
            "植被指标"：　{"植被覆盖度"：　０．６４，　"生物多样性"：　０．３６},  820
            "地形指标"：　{"坡度"：　０．６７，　"坡向"：　０．３３},       821
        },                                                                 822
    },                                                                     823
    "平台"：　{                                                            824
        "ｌ１"：　{"土壤指标"：　０．５４，　"植被指标"：　０．４６},        825
        "ｌ２"：　{                                                        826
            "土壤指标"：　{"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  827
                          "有机质"：　０．１７，　"污染指标"：　０．１７},  828
            "植被指标"：　{"植被覆盖度"：　０．６１，　"生物多样性"：　０．３９},  829
        },                                                                 830
    },                                                                     831
}                                                                          832
#  ----------------------------------------------------------------------  833
    ----------------                                                        834
#  六、限制性指标（论文表　２　+　４．１　节流程）                          835
#  ----------------------------------------------------------------------  836
    ----------------                                                        837
#  论文摘要与　４．１　节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  838
CONSTRAINT_ITEMS　=　("政策"，　"区位约束"，　"土壤污染"，　"地质灾害"，　"耕地属性")  839
#  ----------------------------------------------------------------------  840
    ----------------                                                        841
#  七、账号职务                                                            842
#  ----------------------------------------------------------------------  843
    ----------------                                                        844
#  只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。  845
ROLE_LABELS　=　{"admin"：　"管理员"，　"operator"：　"操作员"}            846
ROLE_CODES　=　("admin"，　"operator")                                     847
DEFAULT_ROLE　=　"operator"　　#　新注册账号的默认职务                    848
ROLE_ORDER　=　("admin"，　"operator")　　#　列表排序用：管理员在前        849
#  性别（注册与个人资料中选填）                                            850
```

```
GENDERS = ("男", "女", "保密")
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    # 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    # 用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
# 职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
# 旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                                #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
```

```
    return added                                                    901
def _migrate_roles(db):                                             902
    """把旧角色名统一为新的两种，返回改动的行数。"""                    903
    changed = 0                                                     904
    with db.engine.connect() as conn:                               905
        rows = conn.execute(text("PRAGMA                            906
    table_info(users)")).fetchall()                                 907
        if not rows:                                                908
            return 0                                                909
        for old, new in ROLE_MIGRATION.items():                     910
            r = conn.execute(text("UPDATE users SET role =          911
    :new "                                                          912
                                  "WHERE role = :old"),             913
                                 {"new": new, "old": old})          914
            changed += r.rowcount or 0                              915
        conn.commit()                                               916
    return changed                                                  917
def ensure_schema(db):                                              918
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""        919
    try:                                                            920
        changed = _add_columns(db)                                  921
        n = _migrate_roles(db)                                      922
        if n:                                                       923
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)  924
        return changed                                              925
    except Exception as e:                                  #       926
    迁移失败不应阻断启动                                            927
        print("[schema] 结构检查跳过：%s" % e)                      928
        return []                                                   929
# ==== app/utils/uploads.py ====                                    930
# -*- coding: utf-8 -*-                                             931
"""上传文件处理（遥感影像、批量导入）。"""                           932
import os                                                           933
import re                                                           934
import uuid                                                         935
from datetime import datetime                                       936
from werkzeug.utils import secure_filename                          937
def _safe_ext(filename):                                            938
    return os.path.splitext(filename)[1].lower().lstrip(".")        939
def ext_allowed(filename, allowed):                                 940
    return _safe_ext(filename) in allowed                           941
def save_upload(file_storage, subdir, allowed_ext,                  942
    upload_root):                                                   943
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。  944
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。      945
    """                                                             946
    if file_storage is None or not file_storage.filename:           947
        return None, "未选择文件"                                   948
    ext = _safe_ext(file_storage.filename)                          949
    if ext not in allowed_ext:                                      950
```

```
            return None, "不支持的文件类型：.%s" % ext                    951
        target_dir = os.path.join(upload_root, subdir)                    952
        os.makedirs(target_dir, exist_ok=True)                            953
        # 仅保留安全字符，并加时间戳与随机串防重名                          954
        base = secure_filename(os.path.splitext(file_storage.fil          955
        ename)[0]) or "file"                                              956
        base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or               957
        "file"                                                            958
        name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M          959
        %S"),                                                             960
                                    uuid.uuid4().hex[:6], ext)            961
        abs_path = os.path.join(target_dir, name)                         962
        file_storage.save(abs_path)                                       963
        rel = os.path.join("uploads", subdir,                             964
        name).replace("\\", "/")                                          965
        return rel, None                                                  966
def remove_upload(rel_path, upload_root):                                 967
        """删除已保存的上传文件（相对 static 的路径）。"""                 968
        if not rel_path:                                                  969
            return                                                        970
        abs_path = os.path.join(upload_root,                              971
        rel_path.replace("uploads/", "", 1))                              972
        try:                                                              973
            if os.path.isfile(abs_path):                                  974
                os.remove(abs_path)                                       975
        except OSError:                                                   976
            pass                                                          977
# ==== app/utils/charts.py ====                                           978
# -*- coding: utf-8 -*-                                                   979
"""轻量图表。                                                             980
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                               981
- 服务端渲染，页面与打印（PDF）中都能稳定显示                             982
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）               983
- 单色方案，符合界面「一个视图一个强调色」的约定                          984
"""                                                                       985
from html import escape                                                   986
ACCENT = "#1f3a5f"                                                        987
ACCENT_SOFT = "#7f93ad"                                                   988
INK = "#23262b"                                                           989
MUTED = "#6b7280"                                                         990
LINE = "#dfe3e8"                                                          991
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            992
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      993
        "#bcc9da", "#dbe3ec"]                                             994
def _fmt(v, digits=4):                                                    995
        try:                                                              996
            return ("%%.%df" % digits) % float(v)                         997
        except (TypeError, ValueError):                                   998
            return str(v)                                                 999
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,               1000
```

```
    value_w=76):                                                    1001
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                    1002
    if not items:                                                   1003
        return '<p class="muted">暂无数据</p>'                       1004
    n = len(items)                                                  1005
    height = n * (bar_h + gap) + 10                                 1006
    max_v = max((abs(float(v)) for _, v, _ in items),               1007
    default=1) or 1                                                 1008
    body_w = width - label_w - value_w                              1009
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                1010
    height="%d" '                                                   1011
             'role="img" aria-label="条形图" '                       1012
             'style="max-width:%dpx;font-family:inherit;">'         1013
    % (width, height, height, width)]                               1014
    for i, (label, value, disp) in enumerate(items):                1015
        y = i * (bar_h + gap) + 5                                   1016
        frac = (abs(float(value)) / max_v) if max_v else 0          1017
        w = max(2.0, body_w * frac)                                 1018
        fill = GRAYS[i % len(GRAYS)]                                1019
        parts.append(                                               1020
            '<text x="%d" y="%d" text-anchor="end"                  1021
    font-size="12" fill="%s">%s</text>'                             1022
            % (label_w - 10, y + bar_h * 0.72, INK,                 1023
    escape(str(label))))                                            1024
        parts.append(                                               1025
            '<rect x="%d" y="%d" width="%.1f" height="%d"           1026
    rx="3" fill="%s"/>'                                             1027
            % (label_w, y, w, bar_h, fill))                         1028
        parts.append(                                               1029
            '<text x="%d" y="%d" font-size="12" fill="%s" '         1030
            'style="font-variant-numeric:tabular-nums;">%s</        1031
    text>'                                                          1032
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,            1033
    escape(str(disp))))                                             1034
    parts.append('</svg>')                                          1035
    return "".join(parts)                                           1036
def donut(items, size=210, thickness=34):                           1037
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]              1038
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                1039
    """                                                             1040
    vals = [(l, float(v), d) for l, v, d in items if                1041
    float(v) > 0]                                                   1042
    if not vals:                                                    1043
        return '<p class="muted">暂无数据</p>'                       1044
    total = sum(v for _, v, _ in vals) or 1                         1045
    r = (size - thickness) / 2                                      1046
    cx = cy = size / 2                                              1047
    import math                                                     1048
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                   1049
    height="%d" role="img" '                                        1050
```

```
            'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
           % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
              100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
```

```
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
```

```
                exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
                "--no-first-run", "--no-default-browser-check",
                "--no-pdf-header-footer",
                "--virtual-time-budget=4000",
                "--print-to-pdf=%s" % pdf_path,
                "file:///" + html_path.replace("\\", "/"),
            ]
            p = subprocess.run(cmd, capture_output=True,
    text=True,
                               timeout=timeout,
    encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
    except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
            return None, "生成 PDF 失败：%s" % e
    finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
```

```
                A[j, i] = 1.0 / upper[idx]
                idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
```

```
                expected = w[i] / w[j]
                if expected == 0:
                    continue
                deviation = abs(A[i, j] - expected) / expected
                if best is None or deviation >
    best["deviation"]:
                    best = {"i": i, "j": j, "value": A[i, j],
                            "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
```

```
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
               tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染(=1)处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
```

```
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"),
*IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter")
, IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_meta
ls") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ---- 植被指标 ----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"),
IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
                           std_benefit(data.get("shannon"),
IDEAL_SHANNON),
                           l2w["植被指标"]["生物多样性"]),
        }
```

```
    #  ----  地形指标  ----                                                    1401
    if  "地形指标"  in  l1_enabled:                                           1402
        result["地形指标"]  =  {                                              1403
            "坡度":  (data.get("slope"),                                      1404
                      std_slope(data.get("slope")),                           1405
                      l2w["地形指标"]["坡度"]),                               1406
            "坡向":  (data.get("aspect"),                                     1407
                      aspect_score(data.get("aspect")),                       1408
                      l2w["地形指标"]["坡向"]),                               1409
        }                                                                     1410
    return  result                                                            1411
#  ====  app/services/weight_service.py  ====                                1412
#  -*-  coding:  utf-8  -*-                                                   1413
"""权重方案的存取与生效权重解析。                                             1414
系统有两套权重来源：                                                         1415
1．  论文表  3  给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），      1416
    是**权威取值来源**，案例复算与报告结论以此为准；                         1417
2．  通过层次分析法在本系统中实际算出的权重，保存在                          1418
    weight_schemes  /  weight_items  两张表中，可标记为「生效」。            1419
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                    1420
以论文表  3  为底，用生效方案中已保存的条目覆盖。                           1421
没有任何生效方案时，即完全使用论文表  3  的权重。                           1422
这样既保证默认行为与论文一致，又使  AHP  计算结果真正参与评价，             1423
而不是算完即弃。                                                             1424
"""                                                                          1425
import  copy                                                                 1426
from  app.models  import  db,  WeightScheme,  WeightItem                     1427
from  app.utils.constants  import  DEFAULT_WEIGHTS                           1428
from  app.services.indicator_service  import  TREE                           1429
#  方案名称：每个微地质单元类型固定一个可生效的方案                          1430
ACTIVE_NAME  =  "自定义权重方案（AHP  计算结果）"                            1431
def  _code_maps():                                                           1432
    """由指标树导出  编码  →  名称  的映射，避免名称在两处各写一遍。"""      1433
    l1,  l2  =  {},  {}                                                      1434
    for  row  in  TREE:                                                      1435
        code,  name,  level,  parent  =  row[0],  row[1],  row[2],           1436
    row[3]                                                                   1437
        if  level  ==  1:                                                    1438
            l1[code]  =  name                                                1439
        elif  level  ==  2:                                                  1440
            l2[code]  =  (l1.get(parent,  parent),  name)                    1441
    return  l1,  l2                                                          1442
def  effective_weights(unit_type):                                           1443
    """返回该单元类型当前生效的权重。                                        1444
    返回  (weights,  source_label,  scheme)                                  1445
        weights        嵌套权重结构，可直接交给标准化与评价引擎              1446
        source_label   权重来源说明，用于界面与报告标注                     1447
        scheme         生效的权重方案对象，无自定义方案时为  None            1448
    """                                                                      1449
    base  =  DEFAULT_WEIGHTS.get(unit_type)                                  1450
```

```
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme

def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
                              name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
```

```
                continue
            item = WeightItem.query.filter_by(scheme_id=scheme.i
        d,

        indicator_code=code).first()
            if item is None:
                item = WeightItem(scheme_id=scheme.id,
        indicator_code=code)
                db.session.add(item)
            item.weight = float(w)
            written += 1
        db.session.commit()
        return scheme, written
def reset_weights(unit_type):
        """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""
        schemes = WeightScheme.query.filter_by(unit_type=unit_ty
        pe,

        name=ACTIVE_NAME).all()
        n = 0
        for s in schemes:
            WeightItem.query.filter_by(scheme_id=s.id).delete(
                synchronize_session=False)
            db.session.delete(s)
            n += 1
        db.session.commit()
        return n
def scheme_items(unit_type):
        """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""
        scheme = WeightScheme.query.filter_by(unit_type=unit_typ
        e,

        name=ACTIVE_NAME).first()
        if scheme is None:
            return {}, None
        items = {i.indicator_code: i.weight
                 for i in WeightItem.query.filter_by(scheme_id=s
        cheme.id).all()}
        return items, scheme
def summary():
        """各单元类型的权重来源概览，供权重首页展示。"""
        rows = []
        for unit_type in DEFAULT_WEIGHTS:
            weights, label, scheme = effective_weights(unit_type
        )
            rows.append({
                "unit_type": unit_type,
                "label": label,
                "custom": scheme is not None,
            })
```

```
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式（９） 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标： 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    # 一级指标权重按实际参与计算的指标归一化。
```

```
    #  论文表  3  的权重本身已归一（和为  1），此处为恒等变换；
    #  若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子  F  与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0～1），data 为实测数据  dict。
    weights_map : dict，{单元类型：  权重结构}。未提供的单元类型
        使用论文表  3  的默认权重。权重可来自  AHP  计算并经
        weight_service 保存生效，从而真正参与评价。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]))
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
#  ====  app/services/report_service.py  ====
```

```
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
```

```
    ----                                                                    1701
    patch : MinePatch，图斑对象                                             1702
    mine_result : evaluation_service.score_mine 的返回 dict                 1703
    """                                                                     1704
    doc = Document()                                                        1705
    # 页边距                                                                1706
    for section in doc.sections:                                            1707
        section.top_margin = Cm(2.5)                                        1708
        section.bottom_margin = Cm(2.5)                                     1709
        section.left_margin = Cm(2.5)                                       1710
        section.right_margin = Cm(2.5)                                      1711
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)              1712
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",       1713
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                     1714
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d           1715
    %H：%M：%S"),                                                           1716
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                     1717
    space_after=12)                                                         1718
    # 图斑信息                                                              1719
    _heading(doc, "一、图斑基本信息", 1)                                    1720
    info = [                                                                1721
        ("图斑名称", patch.name),                                           1722
        ("所在行政区", patch.admin_region or "—"),                          1723
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is         1724
    not None else "—"),                                                     1725
        ("开采矿种", patch.mineral_type or "—"),                            1726
        ("开采方式", patch.mining_method or "—"),                           1727
    ]                                                                       1728
    _table(doc, ["项目", "内容"], info)                                     1729
    # 评价结论                                                              1730
    _heading(doc, "二、评价结论", 1)                                        1731
    gi = mine_result["grade_info"]                                          1732
    concl = [                                                               1733
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),            1734
        ("适宜性等级", mine_result["grade"]),                               1735
        ("判定依据", gi["meaning"]),                                        1736
        ("推荐措施", gi["measure"]),                                        1737
    ]                                                                       1738
    _table(doc, ["项目", "结果"], concl)                                    1739
    # 各单元得分                                                            1740
    _heading(doc, "三、各微地质单元得分", 1)                                1741
    rows = []                                                               1742
    for u in mine_result["unit_results"]:                                   1743
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]                 1744
    * 100),                                                                 1745
                     "%.4f" % u["unit_score"], "%.4f" %                     1746
    u["weighted"]))                                                         1747
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],  1748
    rows)                                                                   1749
    # 完整得分明细                                                          1750
```

```
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
    u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f"
    % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else
    "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std,
    w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
    "贡献"],
               detail_rows)
        # 一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,
    space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
```

```
        "论文表 2 评价指标之一级指标", False),
        ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
        "论文表 2 评价指标之一级指标", False),
        ("terrain", "地形指标", 1, None, None, None, None,
        "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
        # ---- 二级指标：土壤 ----
        ("soil_ratio", "土石比", 2, "soil", "—",
        str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
          "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
        ("thickness", "土层厚度", 2, "soil", "cm",
        str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
          "论文 4.2.2 节：理想值 20 cm", False),
        ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
        "mid", "碎石边坡,平台",
          "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
        IDEAL_PH_RANGE, False),
        ("organic_matter", "有机质", 2, "soil", "g/kg",
        str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
          "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
        ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
        "cost", "碎石边坡,平台",
          "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
        # ---- 二级指标：植被 ----
        ("coverage", "植被覆盖度", 2, "veg", "%",
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
          "论文 4.2.2 节：参照生态系统取值 75%", False),
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
        "benefit", ALL_UNITS,
          "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
        # ---- 二级指标：地形 ----
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
          "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
        为陡坡重要分界", False),
        ("aspect", "坡向", 2, "terrain", "—",
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
          "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
        False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
```

```
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
```

```
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
```

```
                out.append({header[i]: (r[i] if i < len(r) else
    None)
                                for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict，{单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict]，每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str]，逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
```

```
                                % (idx, unit_type,                    2001
    "、".join(unit_type_map) or "无"))                                 2002
            continue                                                   2003
        if unit_type not in UNIT_TYPES:                                2004
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"   2005
                          % (idx, unit_type))                          2006
            continue                                                   2007
        fields, heavy = {}, {}                                         2008
        ok = True                                                      2009
        for key, val in rec.items():                                   2010
            if key in ("unit_type",):                                  2011
                continue                                               2012
            if key.startswith("hm_"):                                  2013
                n, err = _num(val)                                     2014
                if err:                                                2015
                    errors.append("第 %d 行：%s %s" % (idx,            2016
    key[3:], err))                                                     2017
                    ok = False                                         2018
                elif n is not None:                                    2019
                    heavy[key[3:]] = n                                 2020
                continue                                               2021
            if key == "aspect":                                        2022
                s = str(val).strip() if val is not None                2023
    else ""                                                            2024
                if s == "":                                            2025
                    fields["aspect"] = None                            2026
                elif s in ASPECT_TYPES:                                2027
                    fields["aspect"] = s                               2028
                else:                                                  2029
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"  2030
                                  % (idx, s,                           2031
    "、".join(ASPECT_TYPES)))                                          2032
                    ok = False                                         2033
                continue                                               2034
            n, err = _num(val)                                         2035
            if err:                                                    2036
                errors.append("第 %d 行：%s %s" % (idx, key,           2037
    err))                                                              2038
                ok = False                                             2039
            else:                                                      2040
                fields[key] = n                                        2041
        if ok:                                                         2042
            parsed.append({"unit_type": unit_type,                     2043
    "fields": fields,                                                  2044
                           "heavy_metals": heavy})                     2045
    return parsed, errors                                              2046
# ==== app/services/workflow.py ====                                   2047
# -*- coding: utf-8 -*-                                                2048
"""评价流程前置条件（定量评价与报告输出的流程门禁）。                    2049
论文 4.1 节确定的评价流程存在严格的先后关系：                            2050
```

```
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
                "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units         {done, count, ratio_sum, detail}
        constraint    {done, passed, detail}
        data          {done, ready, missing, detail}
        blocking      未满足的前置步骤列表（按流程顺序）
        ready         是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    #  ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    #  ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
```

```
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    missing = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if not _has_data(ind):
            missing.append(u.unit_type)
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    ratio_ok = ratio_sum > 0.999 if units else False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按论文 ４．１ 节的流程顺序：限制性判定优先） ----
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去复核判定结果",
```

```
            })
        if not units_step["done"]:
            blocking.append({
                "step": "units", "label": "未划分微地质单元",
                "reason": "该图斑下没有任何微地质单元。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去划分",
            })
        elif not ratio_ok:
            blocking.append({
                "step": "units", "label": "面积占比合计不为 100%",
                "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去核对面积",
            })
        if units_step["done"] and not data_step["ready"]:
            blocking.append({
                "step": "data", "label": "指标数据未录入完整",
                "reason": data_step["detail"],
                "fix_url": "/indicator/", "fix_label": "去录入",
            })
        return {
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
    def latest_result(patch_id):
        """取该图斑最近一次的评价记录。"""
        return EvaluationResult.query.filter_by(patch_id=patch_i
        d) \
            .order_by(EvaluationResult.id.desc()).first()
    def report_progress(patch_id):
        """报告输出的前置条件：评价流程齐备，且已有评价记录。
        报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
        导出时读取已保存的评价结果，而不是在导出环节重新计算。
        否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
        同一份评价出现两个数字，结论无法追溯。
        返回 dict：{ready, blocking, result, flow}
            result 为最近一次评价记录（EvaluationResult），无记录时为空。
        """
        flow = patch_progress(patch_id)
        blocking = list(flow["blocking"])
        record = latest_result(patch_id)
        if not blocking and record is None:
            blocking.append({
                "step": "evaluate", "label": "尚未执行适宜性评价",
                "reason": "该图斑还没有评价记录。",
                "fix_url": "/evaluate/%d" % patch_id,
```

```
                "fix_label": "去执行评价",                                    2201
            })                                                                2202
    return {                                                                  2203
        "ready": not blocking,                                                2204
        "blocking": blocking,                                                 2205
        "result": record,                                                     2206
        "flow": flow,                                                         2207
    }                                                                         2208
# ==== app/services/dashboard.py ====                                         2209
# -*- coding: utf-8 -*-                                                       2210
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。                    2211
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」      2212
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。            2213
"""                                                                           2214
from flask import g                                                           2215
from app.models import (db, Project, MinePatch, MicroUnit,                   2216
    IndicatorData,                                                            2217
                            ConstraintCheck, EvaluationResult,                2218
    Standard)                                                                 2219
from app.services import workflow as wf                                       2220
# 评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）                           2221
#   key          步骤标识                                                     2222
#   no           步骤编号（01~08）                                            2223
#   title        名称                                                         2224
#   icon         图标名（见 templates/_macros.html 的 ICONS）                 2225
#   desc         一句话说明                                                   2226
#   href         入口地址                                                     2227
STEPS = [                                                                     2228
    dict(key="project", no="01", title="项目与矿山档案",                      2229
    icon="folder",                                                            2230
         href="/project/",                                                    2231
         desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),  2232
    dict(key="standard", no="02", title="政策与标准知识库",                   2233
    icon="book",                                                              2234
         href="/standard/",                                                   2235
         desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),    2236
    dict(key="constraint", no="03", title="限制性指标判定",                   2237
    icon="shield",                                                            2238
         href="/constraint/",                                                 2239
         desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),    2240
    dict(key="unit", no="04", title="微地质单元", icon="layers",              2241
         href="/unit/",                                                       2242
         desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),      2243
    dict(key="ahp", no="05", title="指标体系与权重",                          2244
    icon="sliders",                                                           2245
         href="/ahp/",                                                        2246
         desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),2247
    dict(key="indicator", no="06", title="指标数据采集",                      2248
    icon="database",                                                          2249
         href="/indicator/",                                                  2250
```

```
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
        dict(key="evaluate", no="07", title="适宜性定量评价",
        icon="target",
            href="/evaluate/",
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
        dict(key="report", no="08", title="报告与结果输出",
        icon="file",
            href="/report/",
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
# 系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
```

```
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
    l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
    passed_constraint),
        "unit": sum(1 for p in patches if not
    unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
    us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
```

```
            "constraint": "需先建立矿山图斑",
            "unit": "需先建立矿山图斑",
            "indicator": "需先划分微地质单元",
            "evaluate": "需前置步骤齐备的图斑",
            "report": "需先执行适宜性评价",
        }
        steps = []
        for s in STEPS:
            steps.append(dict(
                s,
                available=avail[s["key"]],
                pending=pending.get(s["key"], 0),
                note=None if avail[s["key"]] else
        notes.get(s["key"], ""),
            ))
        unit_types = len({u.unit_type for u in units})
        stats = {
            "projects": projects,
            "patches": len(patches),
            "unit_types": unit_types,
            "standards": Standard.query.count(),
        }
        return {
            "stats": stats,
            "steps": steps,
            "badges": {k: v for k, v in pending.items() if v},
            "progress": {
                "patches": len(patches),
                "units": len(units),
                "data_units": len(data_unit_ids),
                "ready": ready_patches,
                "evaluated": len(evaluated),
            },
        }
    def nav_badges():
        """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
        snap = snapshot()
        return {
            "/constraint/": snap["badges"].get("constraint", 0),
            "/unit/": snap["badges"].get("unit", 0),
            "/indicator/": snap["badges"].get("indicator", 0),
            "/evaluate/": snap["badges"].get("evaluate", 0),
        }
    # ==== app/services/audit.py ====
    # -*- coding: utf-8 -*-
    """操作审计。
    在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
    且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
    动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
    """
```

```
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
#  端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准 / 政策文件",
    "standard.edit": "编辑标准 / 政策文件",
    "standard.delete": "删除标准 / 政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
#  无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    #  个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
```

```
def extract_target_id():                                                2451
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""                2452
    args = request.view_args or {}                                      2453
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",             2454
    "result_id", "user_id"):                                            2455
        if key in args:                                                 2456
            try:                                                        2457
                return int(args[key])                                   2458
            except (TypeError, ValueError):                             2459
                return None                                             2460
    return None                                                         2461
def record(endpoint, status_code):                                      2462
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""                    2463
    if endpoint in SKIP_ENDPOINTS:                                      2464
        return                                                          2465
    try:                                                                2466
        detail = "路径 %s" % request.path                               2467
        if status_code >= 400:                                          2468
            detail += ";结果 HTTP %d" % status_code                     2469
        db.session.add(OperationLog(                                    2470
            user_id=current_user.id if                                  2471
    current_user.is_authenticated else None,                            2472
            action=action_name(endpoint),                               2473
            target_type=(endpoint or "").split(".")[0] or               2474
    None,                                                               2475
            target_id=extract_target_id(),                              2476
            detail=detail,                                              2477
        ))                                                              2478
        db.session.commit()                                             2479
    except Exception:                                                   2480
        # 审计失败不应影响业务响应                                          2481
        db.session.rollback()                                           2482
# ==== app/views/auth.py ====                                           2483
# -*- coding: utf-8 -*-                                                 2484
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。                    2485
页面组织：                                                               2486
  /login          登录页（含「点我注册」入口）                             2487
  /register       自助注册（默认「操作员」）                               2488
  /profile        个人设置，分类切换：info / security / users / logs       2489
                  后两个分类仅管理员可见，由本模块统一管控。                 2490
"""                                                                     2491
from flask import (Blueprint, render_template, redirect,                2492
    url_for, request,                                                   2493
                       flash, abort, current_app)                       2494
from flask_login import login_user, logout_user,                        2495
    login_required, current_user                                        2496
from werkzeug.security import check_password_hash,                      2497
    generate_password_hash                                              2498
from app.forms import (LoginForm, EmptyForm, ProfileForm,               2499
    PasswordForm,                                                       2500
```

```
                                RegisterForm, email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,
    GENDERS
bp = Blueprint("auth", __name__)
#  「个人设置」的四个分类；后两个是管理员专属
PROFILE_TABS = ("info", "security", "users", "logs")
ADMIN_TABS = ("users", "logs")
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
def _intro_data():
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "error")
                return render_template("auth/login.html",
    form=form,
                                       **_intro_data())
            login_user(user, remember=form.remember.data)
            #  用 toast 类别：渲染为页面正中的浮层，淡入停留后淡出。
            #  登录成功只是告知性反馈，不需要用户停下来阅读。
            flash("登录成功", "toast")
            return redirect(url_for("auth.index"))
        flash("账户名称或密码错误", "error")
    return render_template("auth/login.html", form=form,
    **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
    """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
```

```
    if current_user.is_authenticated:                                  2551
        return redirect(url_for("auth.index"))                        2552
    form = RegisterForm()                                              2553
    if form.validate_on_submit():                                      2554
        username = form.username.data.strip()                          2555
        if User.query.filter_by(username=username).first():            2556
            form.username.errors.append("该账户名称已被使用")          2557
        else:                                                          2558
            user = User(username=username,                             2559
                        role=DEFAULT_ROLE,                             2560
                        password_hash=generate_password_hash           2561
    (form.password.data),                                              2562
                        display_name=(form.display_name.data           2563
    or "").strip() or None,                                            2564
                        gender=(form.gender.data or                    2565
    "").strip() or None,                                               2566
                        email=(form.email.data or                      2567
    "").strip() or None,                                               2568
                        phone=(form.phone.data or                      2569
    "").strip() or None)                                               2570
            rel, err = _save_avatar(request.files.get("avata           2571
    r"))                                                               2572
            if err:                                                    2573
                flash(err, "error")                                    2574
            else:                                                      2575
                user.avatar = rel                                      2576
                db.session.add(user)                                   2577
                db.session.commit()                                    2578
                login_user(user)                                       2579
                flash("注册成功，欢迎使用", "toast")                   2580
                return redirect(url_for("auth.index"))                 2581
    return render_template("auth/register.html", form=form)            2582
@bp.route("/logout")                                                   2583
@login_required                                                        2584
def logout():                                                          2585
    logout_user()                                                      2586
    flash("已退出登录", "ok")                                          2587
    return redirect(url_for("auth.login"))                             2588
# ------------------------------ 个人设置 ------------------------------2589
    ----                                                               2590
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。                2591
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。                    2592
@bp.route("/profile", methods=["GET", "POST"])                         2593
@login_required                                                        2594
def profile():                                                         2595
    is_admin = current_user.role == "admin"                            2596
    tab = request.values.get("tab") or "info"                          2597
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and               2598
    not is_admin):                                                     2599
        tab = "info"                                                   2600
```

```
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit():
            current_user.display_name = (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or "").strip() or None
            current_user.email = (info_form.email.data or "").strip() or None
            current_user.phone = (info_form.phone.data or "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
            return redirect(url_for("auth.profile", tab="info"))
        if action == "password" and pwd_form.validate_on_submit():
            if not check_password_hash(current_user.password_hash,
                                       pwd_form.old_password.data):
                pwd_form.old_password.errors.append("当前密码不正确")
            else:
                current_user.password_hash = generate_password_hash(
                    pwd_form.new_password.data)
                db.session.commit()
                flash("密码已修改", "ok")
                return redirect(url_for("auth.profile", tab="security"))
    ctx = dict(info_form=info_form, pwd_form=pwd_form,
               avatar_form=EmptyForm(),
               tab=tab, role_labels=ROLE_LABELS,
               genders=GENDERS)
    if tab == "users":
        ctx.update(_users_context())
    elif tab == "logs":
        ctx.update(_logs_context())
    return render_template("auth/profile.html", **ctx)
```

```
def _users_context():                                                    2651
    """管理员分类「账号」所需的数据。"""                                  2652
    return {"users": User.query.order_by(User.id).all(),                 2653
            "form": EmptyForm()}                                         2654
def _logs_context():                                                     2655
    """管理员分类「操作日志」所需的数据。"""                              2656
    page = request.args.get("page", 1, type=int)                         2657
    per_page = 50                                                        2658
    q = OperationLog.query.order_by(OperationLog.id.desc())              2659
    total = q.count()                                                    2660
    records = q.offset((page - 1) * per_page).limit(per_page2661
    ).all()                                                              2662
    users_map = {u.id: u.username for u in User.query.all()}2663
    pages = max(1, (total + per_page - 1) // per_page)                   2664
    return {"records": records, "page": page, "pages":                   2665
    pages,                                                               2666
            "total": total, "users_map": users_map}                      2667
# ---------------------------- 头像 ----------------------------2668
    --                                                                   2669
def _avatar_length_ok(file_storage):                                     2670
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""  2671
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *  2672
    1024 * 1024)                                                         2673
    stream = file_storage.stream                                         2674
    pos = stream.tell()                                                  2675
    stream.seek(0, 2)                                                    2676
    size = stream.tell()                                                 2677
    stream.seek(pos)                                                     2678
    return size <= limit, size                                           2679
def _save_avatar(file):                                                  2680
    """保存头像文件，返回 （相对路径， 错误信息）。没有文件时返回 （None， None）。"""  2681
    if file is None or not file.filename:                                2682
        return None, None                                                2683
    ok, size = _avatar_length_ok(file)                                   2684
    if not ok:                                                           2685
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"              2686
                      % (current_app.config["MAX_AVATAR_LENG2687
    TH"] / 1024 / 1024,                                                  2688
                         size / 1024 / 1024))                            2689
    return up.save_upload(file, "avatars",                               2690
                          current_app.config["ALLOWED_AVATAR2691
    _EXT"],                                                              2692
                          current_app.config["UPLOAD_FOLDER"2693
    ])                                                                   2694
@bp.route("/profile/avatar", methods=["POST"])                           2695
@login_required                                                          2696
def upload_avatar():                                                     2697
    form = EmptyForm()                                                   2698
    back = url_for("auth.profile", tab=request.form.get("tab2699
    ") or "info")                                                        2700
```

```
    if not form.validate_on_submit():                                  2701
        flash("表单校验失败，请重试", "error")                          2702
        return redirect(back)                                          2703
    rel, err = _save_avatar(request.files.get("avatar"))               2704
    if err:                                                            2705
        flash(err, "error")                                            2706
        return redirect(back)                                          2707
    if rel is None:                                                    2708
        flash("请选择头像图片", "error")                                2709
        return redirect(back)                                          2710
    # 换新头像时删掉旧文件，避免残留                                    2711
    up.remove_upload(current_user.avatar,                              2712
    current_app.config["UPLOAD_FOLDER"])                               2713
    current_user.avatar = rel                                          2714
    db.session.commit()                                                2715
    flash("头像已更新", "ok")                                          2716
    return redirect(back)                                              2717
@bp.route("/profile/avatar/delete", methods=["POST"])                  2718
@login_required                                                        2719
def delete_avatar():                                                   2720
    form = EmptyForm()                                                 2721
    back = url_for("auth.profile", tab=request.form.get("tab2722
    ") or "info")                                                      2723
    if form.validate_on_submit():                                      2724
        up.remove_upload(current_user.avatar,                          2725
    current_app.config["UPLOAD_FOLDER"])                               2726
        current_user.avatar = None                                     2727
        db.session.commit()                                            2728
        flash("头像已移除", "ok")                                      2729
    return redirect(back)                                              2730
# ---------------------------- 账号维护（仅管理员）                    2731
    ------------------------------                                     2732
@bp.route("/users")                                                    2733
@login_required                                                        2734
def users():                                                           2735
    """旧地址保留，统一并入「个人设置 · 账号」。"""                    2736
    return redirect(url_for("auth.profile", tab="users"))              2737
@bp.route("/users/add", methods=["GET", "POST"])                       2738
@login_required                                                        2739
def add_user():                                                        2740
    require_admin()                                                    2741
    if request.method == "POST":                                       2742
        username = request.form.get("username", "").strip()            2743
        password = request.form.get("password", "")                    2744
        role = request.form.get("role", DEFAULT_ROLE)                  2745
        if role not in ROLE_LABELS:                                    2746
            role = DEFAULT_ROLE                                        2747
        if not username or not password:                               2748
            flash("账户名称和密码不能为空", "error")                   2749
            return redirect(url_for("auth.add_user"))                  2750
```

```
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
"").strip() or None,
            gender=request.form.get("gender", "").strip()
or None,
            email=request.form.get("email", "").strip() or
None,
            phone=request.form.get("phone", "").strip() or
None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        #  保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        #  个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
```

```
    user_id=user.id))                                                    2801
        user.role = role                                                 2802
        user.active = active                                             2803
        user.display_name = request.form.get("display_name",             2804
    "").strip() or None                                                  2805
        user.gender = request.form.get("gender",                         2806
    "").strip() or None                                                  2807
        user.email = email or None                                       2808
        user.phone = request.form.get("phone", "").strip()               2809
    or None                                                              2810
        new_password = request.form.get("password",                      2811
    "").strip()                                                          2812
        if new_password:                                                 2813
            user.password_hash = generate_password_hash(new_             2814
    password)                                                            2815
        db.session.commit()                                              2816
        flash("账号信息已保存", "ok")                                      2817
        return redirect(url_for("auth.profile",                          2818
    tab="users"))                                                        2819
    return render_template("auth/edit_user.html", user=user,             2820
                           role_labels=ROLE_LABELS,                      2821
    genders=GENDERS,                                                     2822
                           form=EmptyForm())                             2823
@bp.route("/users/<int:user_id>/delete", methods=["POST"])               2824
@login_required                                                          2825
def delete_user(user_id):                                                2826
    require_admin()                                                      2827
    user = db.get_or_404(User, user_id)                                  2828
    if user.id == current_user.id:                                       2829
        flash("不能删除当前登录的账号", "error")                           2830
        return redirect(url_for("auth.profile",                          2831
    tab="users"))                                                        2832
    if user.role == "admin":                                             2833
        others = User.query.filter(User.role == "admin",                 2834
                                   User.id != user.id,                   2835
                                   User.active.is_(True)).co             2836
    unt()                                                                2837
        if others == 0:                                                  2838
            flash("系统至少需保留一名启用状态的管理员", "error")           2839
            return redirect(url_for("auth.profile",                      2840
    tab="users"))                                                        2841
    db.session.delete(user)                                              2842
    db.session.commit()                                                  2843
    flash("账号「%s」已删除" % user.username, "ok")                        2844
    return redirect(url_for("auth.profile", tab="users"))                2845
# ---------------------------- 操作日志（仅管理员）                       2846
    ------------------------------                                       2847
@bp.route("/logs")                                                       2848
@login_required                                                          2849
def logs():                                                              2850
```

```
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""
    return redirect(url_for("auth.profile", tab="logs"))
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
```

```
    if request.method == "POST" and form.validate_on_submit(
    ):
        p.name = request.form.get("name", "").strip()
        p.code = request.form.get("code", "").strip() or
    None
        p.description = request.form.get("description",
    "").strip() or None
        p.status = request.form.get("status", "进行中")
        db.session.commit()
        flash("项目已保存", "ok")
        return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=p)
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
    p = db.get_or_404(Project, pid)
    db.session.delete(p)
    db.session.commit()
    flash("项目已删除", "ok")
    return redirect(url_for("project.index"))
# ---------------------------- 矿山图斑 ----------------------------
    ----
@bp.route("/<int:pid>/patches")
@login_required
def patches(pid):
    p = db.get_or_404(Project, pid)
    return render_template("project/patches.html",
    project=p,
                           form=EmptyForm())
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])
@login_required
def patch_delete(patch_id):
    """删除矿山图斑，及其微地质单元与指标数据。"""
    from app.models import MicroUnit, IndicatorData,
    EvaluationResult, \
        ScoreDetail, ConstraintCheck, ReferenceEcosystem
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_
    (unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch
    .id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).dele
```

```
    te(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet
    e(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
```

```
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = fv("area_ha", float)
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
        patch.stop_years = fv("stop_years", float)
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
        return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                           project=patch.project,
    patch=patch,
                           reference=ReferenceEcosystem.quer
    y.filter_by(
                               patch_id=patch.id).first())
# ---------------------------- 遥感影像挂接
    ----------------------------
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])
@login_required
def upload_image(patch_id):
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
```

```
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    rel, err = up.save_upload(request.files.get("image"),
    "patches",
                              current_app.config["ALLOWED_IM
    AGE_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    # 替换旧影像时清理原文件，避免残留
    if patch.image_path:
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
    patch.image_path = rel
    patch.image_uploaded_at = datetime.utcnow()
    note = request.form.get("image_note", "").strip()
    if note:
        patch.image_note = note
    db.session.commit()
    flash("遥感影像已挂接", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
@bp.route("/patch/<int:patch_id>/image/delete",
    methods=["POST"])
@login_required
def delete_image(patch_id):
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
        patch.image_path = None
        patch.image_uploaded_at = None
        db.session.commit()
        flash("遥感影像已移除", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
```

```
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
```

```
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        # 改类型时同样不允许与本图斑已有的其他单元重复
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,


    unit_type=new_type).first()
        if dup is not None and dup.id != u.id:
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,
    "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
```

```
            u.unit_type = new_type                                          3201
            area = request.form.get("area", "").strip()                     3202
            u.area = float(area) if area else None                          3203
            u.description = request.form.get("description",                 3204
        "").strip() or None                                                 3205
            recalc_ratio(u.patch_id)                                        3206
            db.session.commit()                                             3207
            flash("微地质单元已保存", "ok")                                  3208
            return redirect(url_for("unit.index",                           3209
        patch_id=u.patch_id))                                               3210
        return render_template("unit/edit.html", unit=u,                    3211
        patch=u.patch,                                                      3212
                                    unit_types=UNIT_TYPES, form=form)       3213
@bp.route("/delete/<int:unit_id>", methods=["POST"])                        3214
@login_required                                                             3215
def delete(unit_id):                                                        3216
        u = db.get_or_404(MicroUnit, unit_id)                               3217
        pid = u.patch_id                                                    3218
        name = u.unit_type                                                  3219
        # 一并删除该单元的指标数据，避免残留                                3220
        IndicatorData.query.filter_by(unit_id=u.id).delete(synch            3221
        ronize_session=False)                                               3222
        db.session.delete(u)                                                3223
        db.session.flush()                                                  3224
        recalc_ratio(pid)                                                   3225
        db.session.commit()                                                 3226
        flash("微地质单元「%s」已删除" % name, "ok")                         3227
        return redirect(url_for("unit.index", patch_id=pid))                3228
# ==== app/views/standard.py ====                                           3229
# -*- coding: utf-8 -*-                                                     3230
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""          3231
from flask import Blueprint, render_template, request,                      3232
    redirect, url_for, flash                                                3233
from flask_login import login_required                                      3234
from app.forms import EmptyForm                                             3235
from app.models import db, Standard, StandardRule                           3236
bp = Blueprint("standard", __name__, url_prefix="/standard")                3237
@bp.route("/")                                                              3238
@login_required                                                             3239
def index():                                                                3240
        standards = Standard.query.order_by(Standard.category,              3241
        Standard.province).all()                                            3242
        local = [s for s in standards if s.category == "地方标准"]          3243
        policy = [s for s in standards if s.category != "地方标准"]         3244
        return render_template("standard/index.html",                       3245
        local=local, policy=policy,                                         3246
                                    form=EmptyForm())                       3247
@bp.route("/add", methods=["GET", "POST"])                                  3248
@login_required                                                             3249
def add():                                                                  3250
```

```
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        s = Standard(
            province=request.form.get("province", "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number", "").strip() or None,
            publish_date=request.form.get("publish_date", "").strip() or None,
            implement_date=request.form.get("implement_date", "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip() or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form, standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
def edit(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        s.province = request.form.get("province", "").strip()
        s.name = request.form.get("name", "").strip()
        s.doc_number = request.form.get("doc_number", "").strip() or None
        s.publish_date = request.form.get("publish_date", "").strip() or None
        s.implement_date = request.form.get("implement_date", "").strip() or None
        s.category = request.form.get("category", "地方标准")
        s.summary = request.form.get("summary", "").strip() or None
        db.session.commit()
        flash("已保存", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    return render_template("standard/edit.html", form=form, standard=s)
```

```
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
    s = db.get_or_404(Standard, sid)
    StandardRule.query.filter_by(standard_id=sid).delete()
    db.session.delete(s)
    db.session.commit()
    flash("已删除", "ok")
    return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        r = StandardRule(
            standard_id=s.id,
            item=request.form.get("item", "").strip(),
            operator=request.form.get("operator", ">="),
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
            flash("判定项不能为空", "error")
        else:
            db.session.add(r)
            db.session.commit()
            flash("规则已添加", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    rs = StandardRule.query.filter_by(standard_id=s.id).all(
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
```

```
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch
    _id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_
    id) \
```

```
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit(
    ):
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field)
    == "1")
        record.passed = all(getattr(record, f) for f, _ in
    FIELDS)
        failed = [name for f, name in FIELDS if not
    getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check",
    patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else
    getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
```

```
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
```

```
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
    range(n)] for i in range(n)],
```

```
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
    j), "1")) 
                     for i in range(n) for j in range(i +
    1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
    "error")
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
    names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
    pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
    "ok")
                return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                        level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
    it_type)
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
        source_label=source_label, custom_scheme=custom_sche
    me,
    )
@bp.route("/weights/reset", methods=["POST"])
```

```
def weights_reset():                                                    3601
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""                    3602
    unit_type = request.form.get("unit_type")                           3603
    form = EmptyForm()                                                  3604
    if form.validate_on_submit():                                       3605
        if unit_type not in DEFAULT_WEIGHTS:                            3606
            flash("无效的微地质单元类型", "error")                          3607
        else:                                                           3608
            n = ws.reset_weights(unit_type)                             3609
            if n:                                                       3610
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,           3611
    "ok")                                                               3612
            else:                                                       3613
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,    3614
    "ok")                                                               3615
    return redirect(url_for("ahp.index"))                               3616
# ---------------------------- 指标维护 ----------------------------    3617
    ----                                                                3618
@bp.route("/indicators")                                                3619
def indicators():                                                       3620
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                          3621
    if Indicator.query.count() == 0:                                    3622
        ind_svc.seed_indicators()                                       3623
    all_ind = Indicator.query.order_by(Indicator.level,                 3624
    Indicator.id).all()                                                 3625
    by_parent = {}                                                      3626
    for i in all_ind:                                                   3627
        by_parent.setdefault(i.parent_code, []).append(i)               3628
    return render_template("ahp/indicators.html",                       3629
    by_parent=by_parent,                                                3630
                                all_ind=all_ind,                        3631
    unit_types=UNIT_TYPES,                                              3632
                                type_label=ind_svc.ideal_type_lab       3633
    el,                                                                 3634
                                form=EmptyForm())                       3635
@bp.route("/indicators/<code>/edit", methods=["GET",                    3636
    "POST"])                                                            3637
def indicator_edit(code):                                               3638
    ind = Indicator.query.filter_by(code=code).first_or_404(            3639
    )                                                                   3640
    form = EmptyForm()                                                  3641
    if request.method == "POST" and form.validate_on_submit(            3642
    ):                                                                  3643
        name = request.form.get("name", "").strip()                     3644
        if not name:                                                    3645
            flash("指标名称不能为空", "error")                              3646
            return redirect(url_for("ahp.indicator_edit",               3647
    code=code))                                                         3648
        ind.name = name                                                 3649
        ind.unit = request.form.get("unit", "").strip() or              3650
```

```
    None                                                                3651
        ind.ideal_value = request.form.get("ideal_value",               3652
    "").strip() or None                                                 3653
        ind.ideal_type = request.form.get("ideal_type") or              3654
    ind.ideal_type                                                      3655
        units = request.form.getlist("enabled_units")                   3656
        ind.enabled_units = ",".join(units) if units else               3657
    None                                                                3658
        ind.is_constraint = request.form.get("is_constraint"3659
    ) == "1"                                                            3660
        ind.description = request.form.get("description",               3661
    "").strip() or None                                                 3662
        db.session.commit()                                             3663
        flash("指标「%s」已保存" % ind.name, "ok")                        3664
        return redirect(url_for("ahp.indicators"))                      3665
    return render_template("ahp/indicator_edit.html",                   3666
    ind=ind,                                                            3667
                                unit_types=UNIT_TYPES,                  3668
                                type_label=ind_svc.ideal_type_lab3669
    el, form=form)                                                      3670
@bp.route("/indicators/reset", methods=["POST"])                        3671
def indicators_reset():                                                 3672
    """按论文取值重置指标定义。"""                                        3673
    form = EmptyForm()                                                  3674
    if form.validate_on_submit():                                       3675
        n = ind_svc.seed_indicators()                                   3676
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                  3677
    return redirect(url_for("ahp.indicators"))                          3678
# ------------------------------ 多专家矩阵 ------------------------------3679
    -----                                                               3680
def expert_scope():                                                     3681
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""                3682
    unit_type = request.values.get("unit_type") or                      3683
    UNIT_TYPES[0]                                                       3684
    if unit_type not in DEFAULT_WEIGHTS:                                3685
        unit_type = UNIT_TYPES[0]                                       3686
    level = int(request.values.get("level", 1))                        3687
    l1 = normalize_l1(unit_type, level,                                 3688
    request.values.get("l1"))                                           3689
    return unit_type, level, l1                                         3690
def expert_query(unit_type, level, l1):                                 3691
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,               3692
    level=level)                                                        3693
    if level == 2:                                                      3694
        q = q.filter_by(l1_name=l1)                                     3695
    else:                                                               3696
        q = q.filter(ExpertMatrix.l1_name.is_(None))                    3697
    return q.order_by(ExpertMatrix.id)                                  3698
def default_scheme(unit_type, level, l1=None):                          3699
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""             3700
```

```
    name = "%s · %s%s" % (unit_type,                                  3701
                          "一级指标" if level == 1 else "二级指标",   3702
                          "（%s）" % l1 if l1 else "")                 3703
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ          3704
    e,                                                                3705
                                          level=level,                3706
    name=name).first()                                                3707
    if scheme is None:                                                3708
        scheme = WeightScheme(unit_type=unit_type,                    3709
    level=level, name=name,                                           3710
                              is_default=False)                       3711
        db.session.add(scheme)                                        3712
        db.session.flush()                                            3713
    return scheme                                                     3714
@bp.route("/experts", methods=["GET", "POST"])                        3715
def experts():                                                        3716
    """多专家判断矩阵录入与汇总。"""                                  3717
    unit_type, level, l1 = expert_scope()                             3718
    if unit_type not in DEFAULT_WEIGHTS:                              3719
        flash("无效的微地质单元类型", "error")                        3720
        return redirect(url_for("ahp.index"))                         3721
    names, vec = indicator_list(unit_type, level, l1)                 3722
    n = len(names)                                                    3723
    form = EmptyForm()                                                3724
    if request.method == "POST" and form.validate_on_submit(          3725
    ):                                                                3726
        action = request.form.get("action")                           3727
        if action == "add":                                           3728
            expert_name = request.form.get("expert_name",             3729
    "").strip() or "未具名专家"                                       3730
            try:                                                      3731
                upper = [float(request.form.get("m_%d_%d" %           3732
    (i, j), "1"))                                                     3733
                         for i in range(n) for j in range(i           3734
    + 1, n)]                                                          3735
            except ValueError:                                        3736
                flash("矩阵输入有误，请填入正数", "error")            3737
                return redirect(url_for("ahp.experts",                3738
    unit_type=unit_type,                                              3739
                                        level=level, l1=l1))          3740
            A = ahp.matrix_from_upper(upper)                          3741
            r = ahp.weights_and_consistency(A)                        3742
            scheme = default_scheme(unit_type, level, l1)             3743
            db.session.add(ExpertMatrix(                              3744
                scheme_id=scheme.id,                                  3745
                unit_type=unit_type, l1_name=(l1 if level             3746
    == 2 else None),                                                  3747
                expert_name=expert_name, level=level,                 3748
                matrix_json=json.dumps([list(map(float,               3749
    row)) for row in A]),                                             3750
```

```
                    cr=round(r["cr"], 6)))                                    3751
            db.session.commit()                                               3752
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,          3753
    r["cr"]), "ok")                                                           3754
        elif action == "save_agg":                                            3755
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算                    3756
            weights = _aggregate(unit_type, level, l1)                        3757
            if weights is None:                                               3758
                flash("暂无可汇总的专家矩阵", "error")                          3759
            elif not weights["consistent"]:                                   3760
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"          3761
                      "请先检查或调整专家矩阵" % weights["cr"],                  3762
    "error")                                                                  3763
            else:                                                             3764
                names_, _ = indicator_list(unit_type,                         3765
    level, l1)                                                                3766
                pairs = to_code_pairs(unit_type, level, l1,                   3767
    names_,                                                                   3768
                                      weights["weights"])                     3769
                _, cnt = ws.save_weights(unit_type, pairs)                    3770
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"           3771
                      % (unit_type, cnt, weights["cr"]),                      3772
    "ok")                                                                     3773
        return redirect(url_for("ahp.experts",                                3774
    unit_type=unit_type,                                                      3775
                                level=level, l1=l1))                          3776
    rows = expert_query(unit_type, level, l1).all()                           3777
    default = [[round(vec[i] / vec[j], 4) for j in                            3778
    range(n)] for i in range(n)]                                              3779
    return render_template("ahp/experts.html",                                3780
    unit_type=unit_type, level=level,                                         3781
                           l1=l1, names=names, rows=rows,                     3782
    default=default,                                                          3783
                           n=n, form=form)                                    3784
@bp.route("/experts/<int:mid>/delete", methods=["POST"])                      3785
def expert_delete(mid):                                                       3786
    row = db.get_or_404(ExpertMatrix, mid)                                    3787
    unit_type, level, l1 = row.unit_type, row.level,                          3788
    row.l1_name                                                               3789
    form = EmptyForm()                                                        3790
    if form.validate_on_submit():                                             3791
        db.session.delete(row)                                                3792
        db.session.commit()                                                   3793
        flash("已删除专家矩阵", "ok")                                          3794
    return redirect(url_for("ahp.experts",                                    3795
    unit_type=unit_type,                                                      3796
                            level=level, l1=l1))                              3797
def _aggregate(unit_type, level, l1):                                         3798
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""              3799
    rows = expert_query(unit_type, level, l1).all()                           3800
```

```
    mats = []                                                          3801
    for r in rows:                                                     3802
        try:                                                           3803
            mats.append([list(map(float, line))                        3804
                         for line in json.loads(r.matrix_jso           3805
    n)])                                                               3806
        except Exception:                                              3807
            continue                                                   3808
    if not mats:                                                       3809
        return None                                                    3810
    agg = ahp.aggregate_matrices(mats)                                 3811
    res = ahp.weights_and_consistency(agg)                             3812
    return {                                                           3813
        "count": len(mats),                                            3814
        "weights": res["weights"],                                     3815
        "lambda_max": res["lambda_max"],                               3816
        "ci": res["ci"], "cr": res["cr"],                              3817
        "consistent": res["consistent"],                               3818
        "most_inconsistent": ahp.find_most_inconsistent(agg)           3819
    ,                                                                  3820
        "expert_crs": [r.cr for r in rows if r.cr is not               3821
    None],                                                             3822
    }                                                                  3823
@bp.route("/experts/aggregate")                                        3824
def experts_aggregate():                                               3825
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                                  3826
    unit_type, level, l1 = expert_scope()                              3827
    if unit_type not in DEFAULT_WEIGHTS:                               3828
        flash("无效的微地质单元类型", "error")                                 3829
        return redirect(url_for("ahp.index"))                          3830
    names, vec = indicator_list(unit_type, level, l1)                  3831
    rows = expert_query(unit_type, level, l1).all()                    3832
    result = _aggregate(unit_type, level, l1)                          3833
    if result is not None:                                             3834
        result["weights"] = [round(v, 4) for v in                      3835
    result["weights"]]                                                 3836
        result["lambda_max"] = round(result["lambda_max"],             3837
    4)                                                                 3838
    return render_template("ahp/experts_aggregate.html",               3839
    unit_type=unit_type,                                               3840
                                level=level, l1=l1, names=names,       3841
    rows=rows,                                                         3842
                                result=result, form=EmptyForm())       3843
#  ==== app/views/indicator.py ====                                    3844
#  -*- coding: utf-8 -*-                                              3845
"""指标数据采集与管理 —— 路由与页面逻辑。                                          3846
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标                                    3847
（岩质边坡无土壤指标、平台无地形指标）。                                             3848
"""                                                                    3849
import io                                                              3850
```

```
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))  # True
= 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    return render_template("indicator/index.html",
    patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir
    st()
```

```
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit(
    ):
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
        ind.thickness = f("thickness")
        ind.ph = f("ph")
        ind.organic_matter = f("organic_matter")
        ind.heavy_metals = json.dumps(heavy,
    ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ---------------------------- 批量导入 ----------------------------
    ----
```

```
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit():
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if patch_id else None
        if patch is None:
            flash("请选择矿山图斑", "error")
            return redirect(url_for("indicator.import_data"))
        file = request.files.get("datafile")
        if file is None or not file.filename:
            flash("请选择要导入的文件", "error")
            return redirect(url_for("indicator.import_data"))
        ext = file.filename.rsplit(".", 1)[-1].lower()
        if ext not in ("csv", "xlsx", "xls"):
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")
            return redirect(url_for("indicator.import_data"))
        units = MicroUnit.query.filter_by(patch_id=patch.id).all()
        unit_map = {}
        for u in units:
            unit_map.setdefault(u.unit_type, u)
        if not unit_map:
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
            return redirect(url_for("indicator.import_data"))
        try:
            parsed, errors = imp.parse_upload(file.filename, file.stream, unit_map)
        except Exception as e:
            flash("文件解析失败：%s" % e, "error")
            return redirect(url_for("indicator.import_data"))
        saved = 0
        for rec in parsed:
            unit = unit_map[rec["unit_type"]]
            ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
            if ind is None:
                ind = IndicatorData(unit_id=unit.id)
                db.session.add(ind)
```

```
                f = rec["fields"]                                          4001
                for key in ("soil_ratio", "thickness", "ph",               4002
    "organic_matter",                                                      4003
                            "coverage", "shannon", "slope",                4004
    "aspect"):                                                             4005
                    if key in f:                                           4006
                        setattr(ind, key, f[key])                          4007
                if rec["heavy_metals"]:                                    4008
                    ind.heavy_metals = json.dumps(rec["heavy_met           4009
    als"],                                                                 4010
                                                                           4011
    ensure_ascii=False)                                                    4012
                saved += 1                                                 4013
            db.session.commit()                                            4014
            report = {"patch": patch.name, "saved": saved,                 4015
    "errors": errors,                                                      4016
                      "total": len(parsed) + len(                          4017
                          [e for e in errors if e.startswith("第           4018
    ")])}                                                                  4019
            if saved:                                                      4020
                flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")          4021
            else:                                                          4022
                flash("没有可导入的数据行，请检查文件内容", "error")             4023
    return render_template("indicator/import.html",                        4024
    patches=patches,                                                       4025
                                   form=form, report=report,               4026
    headers=imp.HEADERS)                                                   4027
@bp.route("/import/template")                                              4028
def import_template():                                                     4029
    """下载导入模板（CSV，含一行示例）。"""                                      4030
    csv_text = imp.build_template_csv()                                    4031
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于               4032
    Excel 正确识别中文                                                       4033
    return send_file(io.BytesIO(data), as_attachment=True,                 4034
                     download_name="指标数据导入模板.csv",                   4035
                     mimetype="text/csv")                                  4036
# ==== app/views/evaluate.py ====                                          4037
# -*- coding: utf-8 -*-                                                    4038
"""适宜性定量评价引擎 —— 路由与页面逻辑。                                      4039
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，                      4040
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                          4041
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），    4042
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。                      4043
"""                                                                        4044
import json                                                                4045
from flask import Blueprint, render_template, request,                     4046
    redirect, url_for, flash                                               4047
from app.forms import EmptyForm                                            4048
from app.models import (db, MinePatch, MicroUnit,                          4049
    IndicatorData,                                                         4050
```

```
                                EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).first()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind else None,
            "heavy_metals": ind.heavy_metals_dict() if ind else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else 0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
```

```
    """把评价结果整理为可持久化的 JSON。                                    4101
    完整保留各单元得分、一级指标得分与逐条二级指标明细，                    4102
    使结果页与报告能够复现「已记录的那一次评价」，                          4103
    而不是每次打开时重新推算。                                              4104
    """                                                                     4105
    sources = sources or {}                                                 4106
    out = []                                                                4107
    for u in unit_results:                                                  4108
        out.append({                                                        4109
            "unit_type": u["unit_type"],                                    4110
            "ratio": _round(u["ratio"]),                                    4111
            "unit_score": _round(u["unit_score"]),                          4112
            "weighted": _round(u["weighted"]),                              4113
            "weight_source": sources.get(u["unit_type"]),                   4114
            "l1_scores": {k: _round(v) for k, v in                          4115
u["l1_scores"].items()},                                                    4116
            "details": [{                                                   4117
                "l1": d["l1"], "l2": d["l2"],                               4118
                "raw": None if d["raw"] is None else                        4119
str(d["raw"]),                                                              4120
                "std_value": _round(d["std_value"]),                        4121
                "weight": _round(d["weight"]),                              4122
                "contribution": _round(d["contribution"]),                  4123
            } for d in u["details"]],                                       4124
        })                                                                  4125
    return out                                                              4126
def save_evaluation(patch_id, units):                                       4127
    """执行评价并把结果与逐层明细落库，返回评价记录。                      4128
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用             4129
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、           4130
    一级指标得分与逐条二级指标明细，结果页与报告据此复现                   4131
    记录下来的那次评价，不再重新推算。                                      4132
    """                                                                     4133
    weights, sources = load_weights(units)                                  4134
    res = ev.score_mine(units, weights)                                     4135
    saved = EvaluationResult(                                               4136
        patch_id=patch_id,                                                  4137
        final_score=round(res["final_score"], 4),                           4138
        grade=res["grade"],                                                 4139
        measures=res["grade_info"]["measure"],                              4140
        unit_scores=json.dumps(pack_units(res["unit_results"]               4141
], sources),                                                                4142
                               ensure_ascii=False),                         4143
    )                                                                       4144
    db.session.add(saved)                                                   4145
    db.session.flush()        # 先取得 result.id，再写明细                  4146
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}                4147
    for u in res["unit_results"]:                                           4148
        for d in u["details"]:                                              4149
            db.session.add(ScoreDetail(                                     4150
```

```
                    result_id=saved.id,                                    4151
                    unit_id=unit_ids.get(u["unit_type"]),                  4152
                    l1_name=d["l1"], l2_name=d["l2"],                      4153
                    raw_value=None if d["raw"] is None else                4154
        str(d["raw"]),                                                     4155
                    std_value=d["std_value"],                              4156
        weight=d["weight"],                                                4157
                    contribution=d["contribution"]))                       4158
        db.session.commit()                                                4159
        return saved                                                       4160
def block_reasons(progress):                                               4161
        """把未满足的前置条件转成可读的拦截原因。"""                                  4162
        return ["%s：%s" % (b["label"], b["reason"]) for b in              4163
        progress["blocking"]]                                              4164
@bp.route("/")                                                             4165
def index():                                                               4166
        patches = MinePatch.query.order_by(MinePatch.id.desc()).           4167
        all()                                                              4168
        latest = {}                                                        4169
        progress = {}                                                      4170
        for p in patches:                                                  4171
            latest[p.id] = EvaluationResult.query.filter_by(patc           4172
        h_id=p.id) \                                                       4173
                .order_by(EvaluationResult.id.desc()).first()              4174
            progress[p.id] = wf.patch_progress(p.id)                       4175
        return render_template("evaluate/index.html",                      4176
        patches=patches,                                                   4177
                                    latest=latest, progress=progress)      4178
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                      4179
def run(patch_id):                                                         4180
        patch = db.get_or_404(MinePatch, patch_id)                         4181
        form = EmptyForm()                                                 4182
        progress = wf.patch_progress(patch_id)                             4183
        if request.method == "POST" and form.validate_on_submit(           4184
        ):                                                                 4185
            if not progress["ready"]:                                      4186
                for reason in block_reasons(progress):                     4187
                    flash("无法执行评价 —— " + reason, "error")              4188
                return redirect(url_for("evaluate.run",                    4189
        patch_id=patch_id))                                                4190
            units = load_patch_input(patch_id)                             4191
            saved = save_evaluation(patch.id, units)                       4192
            return redirect(url_for("evaluate.result",                     4193
        result_id=saved.id))                                               4194
        # 展示将采用的权重来源，便于在执行前确认                                       4195
        units_now = load_patch_input(patch_id)                             4196
        _, sources = load_weights(units_now) if units_now else             4197
        ({}, {})                                                           4198
        return render_template("evaluate/run.html",                        4199
        patch=patch, form=form,                                            4200
```

```
                                progress=progress,                    4201
        weight_sources=sources)                                       4202
@bp.route("/result/<int:result_id>")                                  4203
def result(result_id):                                                4204
    r = db.get_or_404(EvaluationResult, result_id)                    4205
    patch = db.session.get(MinePatch, r.patch_id)                     4206
    unit_scores = json.loads(r.unit_scores) if                        4207
    r.unit_scores else []                                             4208
    return render_template("evaluate/result.html", r=r,               4209
    patch=patch,                                                      4210
                                unit_scores=unit_scores)              4211
# ==== app/views/report.py ====                                       4212
# -*- coding: utf-8 -*-                                               4213
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""                4214
import io                                                             4215
import json                                                           4216
from datetime import datetime                                         4217
from flask import Blueprint, render_template, send_file,              4218
    flash, redirect, url_for                                          4219
from app.models import db, MinePatch, ReferenceEcosystem              4220
from app.services import report_service as rs                         4221
from app.services import workflow as wf                               4222
from app.utils import charts                                          4223
from app.utils import browser                                         4224
from app.utils.constants import grade_of                              4225
bp = Blueprint("report", __name__, url_prefix="/report")              4226
def _prepare(patch_id):                                               4227
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。                4228
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，              4229
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与               4230
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查             4231
    结果，由调用方拦截并提示先去执行评价。                                4232
    """                                                               4233
    patch = db.get_or_404(MinePatch, patch_id)                        4234
    prog = wf.report_progress(patch_id)                               4235
    if not prog["ready"]:                                             4236
        return patch, None, None, None, prog                          4237
    rec = prog["result"]                                              4238
    result = {                                                        4239
        "final_score": rec.final_score,                               4240
        "grade": rec.grade,                                           4241
        "grade_info": grade_of(rec.final_score),                      4242
        "unit_results": json.loads(rec.unit_scores) if                4243
    rec.unit_scores else [],                                          4244
    }                                                                 4245
    reference = ReferenceEcosystem.query.filter_by(patch_id=          4246
    patch_id).first()                                                 4247
    return patch, reference, result, rec, prog                        4248
def _deny(progress, patch_id, endpoint="report.detail"):              4249
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""              4250
```

```
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight)
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
```

```
            return _deny(prog, patch_id)                                    4301
        buf = rs.build_report(patch, result)                                4302
        filename = "自然恢复适宜性评价报告_%s.docx" % patch.name              4303
        return send_file(buf, as_attachment=True,                           4304
        download_name=filename,                                             4305
                         mimetype="application/vnd.openxmlformat            4306
        s-officedocument."                                                  4307
                                    "wordprocessingml.document")            4308
@bp.route("/<int:patch_id>/pdf")                                            4309
def pdf(patch_id):                                                          4310
        """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""          4311
        patch, reference, result, record, prog =                            4312
        _prepare(patch_id)                                                  4313
        if result is None:                                                  4314
            return _deny(prog, patch_id)                                    4315
        chart_units, chart_weight = _charts(result)                         4316
        html = render_template(                                             4317
            "report/print.html", patch=patch,                               4318
        reference=reference, result=result,                                 4319
            chart_units=chart_units, chart_weight=chart_weight,             4320
            generated_at=datetime.now().strftime("%Y-%m-%d                  4321
        %H:%M"))                                                            4322
        data, err = browser.html_to_pdf(html)                               4323
        if err:                                                             4324
            flash("PDF 导出失败：%s" % err, "error")                         4325
            return redirect(url_for("report.detail",                        4326
        patch_id=patch_id))                                                 4327
        return send_file(io.BytesIO(data), as_attachment=True,              4328
                         download_name="自然恢复适宜性评价报告_%s.pdf" %      4329
        patch.name,                                                         4330
                         mimetype="application/pdf")                        4331
#  ==== app/views/search.py ====                                            4332
#  -*- coding: utf-8 -*-                                                    4333
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。              4334
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                      4335
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                  4336
"""                                                                         4337
from sqlalchemy import or_                                                  4338
from flask import Blueprint, render_template, request                       4339
from app.models import db, Project, MinePatch, Standard                     4340
bp = Blueprint("search", __name__)                                          4341
LIMIT = 50              #  每类结果上限，避免一次渲染过多                   4342
@bp.route("/search")                                                        4343
def index():                                                                4344
        q = (request.args.get("q") or "").strip()                           4345
        projects, patches, standards = [], [], []                           4346
        if q:                                                               4347
            like = "%" + q + "%"                                            4348
            projects = Project.query.filter(or_(                            4349
                Project.name.like(like),                                    4350
```

```
            Project.code.like(like),                                    4351
            Project.description.like(like),                             4352
        )).order_by(Project.id.desc()).limit(LIMIT).all()               4353
        patches = MinePatch.query.filter(or_(                           4354
            MinePatch.name.like(like),                                  4355
            MinePatch.admin_region.like(like),                          4356
            MinePatch.mineral_type.like(like),                          4357
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()             4358
        standards = Standard.query.filter(or_(                          4359
            Standard.name.like(like),                                   4360
            Standard.doc_number.like(like),                             4361
            Standard.province.like(like),                               4362
            Standard.summary.like(like),                                4363
        )).order_by(Standard.id).limit(LIMIT).all()                     4364
    total = len(projects) + len(patches) + len(standards)               4365
    return render_template("search.html", q=q,                          4366
    projects=projects,                                                  4367
                                patches=patches,                        4368
    standards=standards, total=total)                                   4369
# ==== app/templates/base.html ====                                     4370
{% import "_macros.html" as ui %}<!DOCTYPE html>                        4371
<html lang="zh-CN">                                                     4372
<head>                                                                  4373
  <meta charset="UTF-8">                                                4374
  <meta name="viewport" content="width=device-width,                    4375
    initial-scale=1.0">                                                 4376
  <link rel="icon" href="{{ url_for('static',                           4377
    filename='favicon.ico') }}">                                        4378
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock      4379
    %}</title>                                                          4380
  <style>                                                               4381
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */                4382
    :root {                                                             4383
      --accent: #1f6b45;                  /* 主色：深绿，呼应自然恢复主题 */      4384
      --accent-hover: #185738;                                          4385
      --accent-soft: #e9f4ee;                                           4386
      --side-1: #17503a;                  /* 侧边栏渐变起止色 */              4387
      --side-2: #113c2b;                                                4388
      --danger: #b3261e;                                                4389
      --danger-soft: #fdecea;                                           4390
      --ok: #1e6b3a;                                                    4391
      --ok-soft: #eaf6ef;                                               4392
      --todo: #d98324;                    /* 待办角标 */                  4393
      --ink: #23262b;                                                   4394
      --ink-soft: #6b7280;                                              4395
      --line: #e3e7ea;                                                  4396
      --surface: #ffffff;                                               4397
      --canvas: #f4f7f5;                                                4398
      --radius: 10px;                                                   4399
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);                           4400
```

```
  --header-h: 60px;
  --nav-w: 226px;
}
* { box-sizing: border-box; }
html, body { margin: 0; padding: 0; }
body {
  font-family: "Microsoft YaHei", "PingFang SC",
"SimSun", sans-serif;
  font-size: 14px;
  line-height: 1.6;
  color: var(--ink);
  background: var(--canvas);
}
/* ---------- 跳到主要内容（键盘可达性） ---------- */
.skip-link {
  position: absolute;
  left: 12px;
  top: -60px;
  z-index: 30;
  padding: 8px 14px;
  background: var(--accent);
  color: #fff;
  border-radius: var(--radius);
  text-decoration: none;
  transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------
   固定为一屏高，页面滚动交给右侧内容区，
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */
.app-shell {
  display: flex;
  align-items: stretch;
  height: 100vh;
  height: 100dvh;
  overflow: hidden;
}
/* ---------- 侧边栏 ---------- */
.app-side {
  flex: 0 0 var(--nav-w);
  width: var(--nav-w);
  display: flex;
  flex-direction: column;
  background: linear-gradient(180deg, var(--side-1) 0%,
var(--side-2) 100%);
  color: #fff;
}
.app-brand {
  display: flex;
  align-items: center;
```

```
  gap: 11px;
  padding: 17px 16px 15px;
  border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
  flex: none;
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  border-radius: 10px;
  background: rgba(255, 255, 255, .15);
}
.app-brand__name {
  font-size: 14.5px;
  font-weight: 600;
  line-height: 1.35;
  letter-spacing: .2px;
}
.app-brand__sub {
  font-size: 11.5px;
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:
auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
```

```
    outline-offset: 1px;
  }
  .app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
  }
  .app-nav .ico { flex: none; opacity: .92; }
  .app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
  }
  .app-user {
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
  }
  .app-user__me {
    display: flex;
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
    text-decoration: none;
    transition: background-color .16s ease;
  }
  .app-user__me:hover { background: rgba(255, 255, 255,
  .10); }
  .app-user__me:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
  }
  .app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
```

```
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 13px;
    font-weight: 600;
    object-fit: cover;
}
.app-user__name {
    display: block;
    font-size: 13px;
    line-height: 1.3;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
.app-user__out {
    flex: none;
    padding: 6px;
    border-radius: 6px;
    color: rgba(255, 255, 255, .68);
    text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
    flex: 1;
    min-width: 0;
    display: flex;
    flex-direction: column;
    overflow-y: auto;
    overflow-x: hidden;
    /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
```

```
    scrollbar-width: thin;                                          4601
    scrollbar-color: rgba(31, 107, 69, .28) transparent;            4602
}                                                                   4603
.app-content::-webkit-scrollbar { width: 12px; }                    4604
.app-content::-webkit-scrollbar-track { background:                 4605
transparent; }                                                      4606
.app-content::-webkit-scrollbar-thumb {                             4607
    background: rgba(31, 107, 69, .26);                             4608
    border: 3px solid transparent;                                  4609
    background-clip: content-box;                                   4610
    border-radius: 999px;                                           4611
}                                                                   4612
.app-content::-webkit-scrollbar-thumb:hover {                       4613
    background: rgba(31, 107, 69, .46);                             4614
    background-clip: content-box;                                   4615
}                                                                   4616
/* ---------- 顶栏：面包屑 + 搜索 ----------                        4617
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */                  4618
.app-header {                                                       4619
    position: sticky;                                               4620
    top: 0;                                                         4621
    z-index: 20;                                                    4622
    display: flex;                                                  4623
    align-items: center;                                            4624
    gap: 14px;                                                      4625
    height: var(--header-h);                                        4626
    padding: 0 22px;                                                4627
    background: var(--surface);                                     4628
    border-bottom: 1px solid var(--line);                           4629
}                                                                   4630
.crumb {                                                            4631
    display: flex;                                                  4632
    align-items: center;                                            4633
    gap: 7px;                                                       4634
    font-size: 13px;                                                4635
    color: var(--ink-soft);                                         4636
    white-space: nowrap;                                            4637
}                                                                   4638
.crumb a { color: var(--ink-soft); text-decoration:                 4639
none; }                                                             4640
.crumb a:hover { color: var(--accent); }                            4641
.crumb strong { color: var(--ink); font-weight: 600; }              4642
.crumb .ico { color: #c3c9d0; }                                     4643
.app-header__spacer { flex: 1; }                                    4644
.search-box {                                                       4645
    display: flex;                                                  4646
    align-items: center;                                            4647
    gap: 8px;                                                       4648
    width: 268px;                                                   4649
    padding: 7px 12px;                                              4650
```

```
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
    place-items: center;
    padding: 0;
    border: none;
    background: transparent;
    color: inherit;
    cursor: pointer;
}
.icon-btn {
    display: grid;
    place-items: center;
    width: 34px;
    height: 34px;
    border-radius: 9px;
    border: 1px solid transparent;
    color: var(--ink-soft);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
ease,
                border-color .16s ease;
}
.icon-btn:hover {
    color: var(--accent);
    background: var(--accent-soft);
    border-color: rgba(31, 107, 69, .18);
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 26px;
```

```
}
.container { max-width: 1120px; }
/* ---------- 页脚 ---------- */
.app-foot {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 14px;
  padding: 0 26px 18px;
  font-size: 12px;
  color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
  display: flex;
  flex-direction: column;
  min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 22px 24px;
  margin-bottom: 18px;
}
.card > h2 {
  margin: 0 0 14px;
  font-size: 16px;
  font-weight: 600;
  color: var(--accent);
  text-wrap: balance;
}
.card > h3 {
  margin: 20px 0 10px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
```

```
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
    width: 104px;
    text-align: center;
    font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
    border-color: var(--accent);
}
.checkbox {
```

```
    display: inline-flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;
    user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 8px 18px;
    font-family: inherit;
    font-size: 14px;
    line-height: 1.4;
    color: #fff;
    background: rgba(31, 107, 69, .80);
    border: 1px solid rgba(31, 107, 69, .30);
    border-radius: 8px;
    cursor: pointer;
    text-decoration: none;
    transition: background-color .16s ease, border-color
.16s ease,
                color .16s ease, box-shadow .16s ease,
transform .16s ease;
}
.btn:hover {
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow:
none; }
.btn:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.btn--ghost {
    color: var(--accent);
    background: rgba(31, 107, 69, .06);
    border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
}
.btn--danger {
```

```
  color: var(--danger);                                                    4851
  background: rgba(179, 38, 30, .07);                                      4852
  border-color: rgba(179, 38, 30, .26);                                    4853
}                                                                          4854
.btn--danger:hover {                                                       4855
  color: #fff;                                                             4856
  background: var(--danger);                                               4857
  border-color: var(--danger);                                             4858
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);                           4859
}                                                                          4860
.btn--sm { padding: 5px 13px; font-size: 13px;                             4861
border-radius: 7px; }                                                      4862
.btn:disabled { opacity: .45; cursor: not-allowed; }                       4863
.btn:disabled:hover {                                                      4864
  background: rgba(31, 107, 69, .80);                                      4865
  border-color: rgba(31, 107, 69, .30);                                    4866
  box-shadow: none;                                                        4867
}                                                                          4868
.btn--ghost:disabled:hover { background: rgba(31, 107,                     4869
69, .06); color: var(--accent); }                                          4870
/* ---------- 表格操作列的行内小按钮 ----------                            4871
   取代原来「文字 | 文字 | 文字」的写法：                                  4872
   圆角矩形 + 半透明底色，悬停时变实色。 */                                4873
.row-actions {                                                             4874
  display: inline-flex;                                                    4875
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */       4876
  align-items: center;                                                     4877
  gap: 6px;                                                                4878
  white-space: nowrap;                                                     4879
}                                                                          4880
.row-actions form { display: inline-flex; margin: 0; }                     4881
.link-btn {                                                                4882
  display: inline-flex;                                                    4883
  align-items: center;                                                     4884
  padding: 4px 11px;                                                       4885
  font-family: inherit;                                                    4886
  font-size: 13px;                                                         4887
  line-height: 1.5;                                                        4888
  color: var(--accent);                                                    4889
  background: rgba(31, 107, 69, .07);                                      4890
  border: 1px solid rgba(31, 107, 69, .20);                                4891
  border-radius: 7px;                                                      4892
  cursor: pointer;                                                         4893
  text-decoration: none;                                                   4894
  white-space: nowrap;                                                     4895
  transition: background-color .16s ease, border-color                     4896
.16s ease,                                                                 4897
              color .16s ease, box-shadow .16s ease,                       4898
transform .16s ease;                                                       4899
}                                                                          4900
```

```
.link-btn:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px);
box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items:
center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 居中浮层提示（登录成功、各类操作完成） ----------
   外层 .toast-stack 固定在页面正中心并纵向排列，
   内层每条各自淡入 → 停留 → 淡出，多条提示会依次排开而不重叠。
   半透明底 + 背景模糊做出毛玻璃质感，不做成实心块；
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 50%;
  left: 50%;
```

```
    z-index: 100;                                                    4951
    display: flex;                                                   4952
    flex-direction: column;                                          4953
    align-items: center;                                             4954
    gap: 10px;                                                       4955
    transform: translate(-50%, -50%);                                4956
    pointer-events: none;                                            4957
}                                                                    4958
.toast-center {                                                      4959
   max-width: min(560px, 78vw);                                      4960
    padding: 13px 28px;                                              4961
    border-radius: 999px;                                            4962
    background: rgba(31, 107, 69, .62);              /*              4963
半透明，非实体 */                                                    4964
    color: #fff;                                                     4965
    font-size: 16px;                                                 4966
    font-weight: 600;                                                4967
    line-height: 1.65;                                               4968
    letter-spacing: .03em;                                           4969
    text-align: center;                                              4970
    text-wrap: pretty;                                               4971
    border: 1px solid rgba(255, 255, 255, .28);                      4972
    backdrop-filter: blur(9px) saturate(1.3);                        4973
    -webkit-backdrop-filter: blur(9px) saturate(1.3);                4974
    box-shadow: 0 10px 34px rgba(19, 36, 58, .18);                   4975
    animation: toast-fade 2.6s ease-in-out forwards;                 4976
}                                                                    4977
@keyframes toast-fade {                                              4978
   0%         { opacity: 0; transform: scale(.90); }                 4979
   14%, 74%   { opacity: 1; transform: scale(1); }                   4980
   100%       { opacity: 0; transform: scale(1); }                   4981
}                                                                    4982
/* 系统设置里要求「减少动态效果」时，去掉缩放，只做淡入淡出 */                4983
@media (prefers-reduced-motion: reduce) {                            4984
   .toast-center { animation: toast-fade-plain 2.6s                  4985
linear forwards; }                                                   4986
   @keyframes toast-fade-plain {                                     4987
      0%         { opacity: 0; }                                     4988
      14%, 74%   { opacity: 1; }                                     4989
      100%       { opacity: 0; }                                     4990
   }                                                                 4991
}                                                                    4992
/* 不支持背景模糊时，加深底色保证文字仍然清楚 */                          4993
@supports not (backdrop-filter: blur(1px)) {                         4994
   .toast-center { background: rgba(31, 107, 69, .88); }             4995
}                                                                    4996
/* ---------- 文字工具类 ---------- */                                4997
.muted { color: var(--ink-soft); font-size: 12.5px; }                4998
.mono { font-family: Consolas, "Courier New",                        4999
monospace; font-variant-numeric: tabular-nums; }                     5000
```

```
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%,
                    #edf6f1 0%, rgba(237, 246, 241, 0)
72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%;
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
```

```
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,
                            rgba(11, 48, 33, .68) 50%,
                            rgba(11, 48, 33, .06) 100%),
    url("{{ url_for('static', filename='hero.svg') }}")
    center bottom / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 18px;
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color
.16s ease,
              color .16s ease;
}
```

```
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255,
255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff;
outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px,
1fr));
  gap: 12px;
  margin-bottom: 22px;
}
.stat-card {
  display: block;
  padding: 15px 17px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease;
}
.stat-card:hover {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
  display: block;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.stat-card__value {
  display: block;
  margin: 3px 0 2px;
  font-size: 26px;
  font-weight: 700;
  line-height: 1.25;
```

```
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
    gap: 12px;
}
.step-card {
    position: relative;
    display: block;
    padding: 16px 16px 15px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
ease,
                transform .16s ease;
}
.step-card:hover {
    border-color: rgba(31, 107, 69, .55);
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
```

```
    transform: translateY(-1px);
}
.step-card:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.step-card__no {
    position: absolute;
    top: 13px;
    right: 15px;
    font-size: 17px;
    font-weight: 700;
    color: #d7e3db;
    font-variant-numeric: tabular-nums;
}
.step-card__icon {
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    margin-bottom: 10px;
    border-radius: 9px;
    background: var(--accent-soft);
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 12.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
```

```
  border-color: #eceff0;                                                    5251
  box-shadow: none;                                                         5252
  cursor: default;                                                          5253
}                                                                           5254
.step-card--locked .step-card__no { color: #e6e9e8; }                       5255
.step-card--locked .step-card__icon {                                       5256
  background: #f1f3f2;                                                      5257
  color: #a9b2ae;                                                           5258
}                                                                           5259
.step-card--locked .step-card__title { color: #9aa3a0; }                    5260
.step-card--locked .step-card__desc { color: #b3bab7; }                     5261
.step-card__lock {                                                          5262
  display: inline-block;                                                    5263
  margin-top: 10px;                                                         5264
  padding: 1px 9px;                                                         5265
  font-size: 11.5px;                                                        5266
  border-radius: 9px;                                                       5267
  background: #f1f3f2;                                                      5268
  color: #8b948f;                                                           5269
}                                                                           5270
/* ---------- 工作台：整行入口 ---------- */                                5271
.row-card {                                                                 5272
  display: flex;                                                            5273
  align-items: center;                                                      5274
  gap: 14px;                                                                5275
  padding: 15px 18px;                                                       5276
  background: var(--surface);                                               5277
  border: 1px solid var(--line);                                            5278
  border-radius: var(--radius);                                             5279
  box-shadow: var(--shadow);                                                5280
  color: var(--ink);                                                        5281
  text-decoration: none;                                                    5282
  transition: border-color .16s ease, box-shadow .16s                       5283
ease;                                                                       5284
}                                                                           5285
.row-card:hover {                                                           5286
  border-color: rgba(31, 107, 69, .45);                                     5287
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);                             5288
}                                                                           5289
.row-card__icon {                                                           5290
  display: grid;                                                            5291
  place-items: center;                                                      5292
  width: 36px;                                                              5293
  height: 36px;                                                             5294
  flex: none;                                                               5295
  border-radius: 10px;                                                      5296
  background: var(--accent-soft);                                           5297
  color: var(--accent);                                                     5298
}                                                                           5299
.row-card__main { flex: 1; min-width: 0; }                                  5300
```

```
.row-card__title {
  display: block;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
.row-card__desc {
  display: block;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.row-card__arrow { color: #c3c9d0; flex: none; }
/* ---------- 结果摘要 ---------- */
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
```

```
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
  background: var(--danger-soft);
  border-left-color: var(--danger);
  color: #7a1c17;
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 15px;
  font-weight: 600;
}
.patch-panel__meta {
  font-size: 12.5px;
  color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
```

```
      .patch-panel__tail { margin-left: auto; }                          5401
      .patch-panel .notice { margin: 0; }                                5402
      .patch-panel table { margin: 0; }                                  5403
      /* ---------- 窄屏适配 ---------- */                                5404
      @media (max-width: 960px) {                                        5405
        /* 上下布局时取消「一屏高 + 内容区内部滚动」，                    5406
           否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */            5407
        .app-shell {                                                     5408
          flex-direction: column;                                        5409
          height: auto;                                                  5410
          min-height: 100vh;                                             5411
          min-height: 100dvh;                                            5412
          overflow: visible;                                             5413
        }                                                                5414
        .app-content { overflow: visible; }                              5415
        .app-header { position: static; }                                5416
        .app-side { width: auto; flex: none; }                           5417
        .app-brand { padding: 12px 14px; }                               5418
        .app-nav { padding: 4px 10px 10px; }                             5419
        .app-nav ul { display: flex; flex-wrap: wrap; gap:               5420
      4px; }                                                             5421
        .app-nav a { margin-bottom: 0; }                                 5422
        .app-nav__group { margin: 8px 0 4px; }                           5423
        .app-user { padding: 10px 14px; }                                5424
        .app-header { padding: 0 14px; gap: 10px; }                      5425
        .search-box { width: 180px; }                                    5426
        .app-main { padding: 16px 14px 20px; }                           5427
        .app-foot { padding: 0 14px 14px; }                              5428
      }                                                                  5429
      @media (max-width: 640px) {                                        5430
        .crumb { display: none; }                                        5431
        .search-box { width: auto; flex: 1; }                            5432
      }                                                                  5433
    </style>                                                             5434
</head>                                                                  5435
<body class="{% if not current_user.is_authenticated                     5436
      %}body--login{% endif %}">                                         5437
    <a class="skip-link" href="#main">跳到主要内容</a>                   5438
    {# 取一次即可（同一请求内重复调用返回相同内容）。                   5439
       类别为 toast 的渲染成居中浮层，其余仍是页面顶部的提示条 ——       5440
       错误提示需要停下来细读，不适合做成会消失的浮层。 #}              5441
    {% set flash_msgs = get_flashed_messages(with_categories=t           5442
      rue) %}                                                            5443
    {# 类别为 toast / ok 的渲染成页面正中的浮层（操作完成的即时反馈）；  5444
       其余（尤其 error）仍是页面顶部的提示条 ——                        5445
       错误提示需要停下来细读，不适合做成会消失的浮层。                 5446
       没有浮层时这个容器是空的，不占任何位置。 #}                       5447
    <div class="toast-stack">                                            5448
      {% for category, message in flash_msgs if category in              5449
      ("toast", "ok") %}                                                 5450
```

```
    <div class="toast-center" role="status"
    aria-live="polite">{{ message }}</div>
    {% endfor %}
  </div>
  {% if current_user.is_authenticated %}
  <div class="app-shell">
    <aside class="app-side">
      <div class="app-brand">
        <span class="app-brand__mark">{{
    ui.icon("mountain", 21) }}</span>
        <span>
          <span class="app-brand__name">矿山自然恢复评价</span><br>
          <span class="app-brand__sub">适宜性评价系统 Ｖ１．０</span>
        </span>
      </div>
      <nav class="app-nav" aria-label="功能导航">
        <ul>
          <li>
            <a href="/"{% if home_active %}
    aria-current="page"{% endif %}>
              {{ ui.icon("home") }}<span>工作台</span>
            </a>
          </li>
        </ul>
        {% for g in nav_groups %}
        <p class="app-nav__group">{{ g.title }}</p>
        <ul>
          {% for item in g.links %}
          <li>
            <a href="{{ item.href }}"{% if item.active %}
    aria-current="page"{% endif %}>
              {{ ui.icon(item.icon) }}<span>{{ item.label
    }}</span>
              {% if item.badge %}<span
    class="app-nav__badge">{{ item.badge }}</span>{% endif
    %}
            </a>
          </li>
          {% endfor %}
        </ul>
        {% endfor %}
      </nav>
      <div class="app-user">
        <a class="app-user__me" href="/profile"
    title="个人设置">
          {% if current_user.avatar %}
          <img class="app-user__avatar"
               src="{{ url_for('static',
    filename=current_user.avatar) }}" alt="">
          {% else %}
```

```
          <span class="app-user__avatar"                                5501
    aria-hidden="true">{{ current_user.initial }}</span>                5502
          {% endif %}                                                   5503
          <span style="min-width:0;">                                   5504
            <span class="app-user__name">{{                            5505
    current_user.shown_name }}</span>                                   5506
            <span class="app-user__state">                              5507
              <span class="app-user__dot"                               5508
    aria-hidden="true"></span>                                          5509
              {{ role_labels.get(current_user.role,                     5510
    current_user.role) }}                                               5511
            </span>                                                     5512
          </span>                                                       5513
        </a>                                                            5514
        <a class="app-user__out" href="/logout" title="退出登录"        5515
           aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>        5516
      </div>                                                            5517
    </aside>                                                            5518
    <div class="app-content">                                           5519
      <header class="app-header">                                       5520
        <nav class="crumb" aria-label="当前位置">                       5521
          <a href="/">首页</a>                                          5522
          {{ ui.icon("chevron", 13) }}                                  5523
          <strong>{{ page_title }}</strong>                             5524
        </nav>                                                          5525
        <span class="app-header__spacer"></span>                        5526
        <form class="search-box" method="get"                           5527
    action="/search" role="search">                                     5528
          {{ ui.icon("search", 16) }}                                   5529
          <label for="q" class="visually-hidden">搜索项目、图斑、标准<  5530
    /label>                                                             5531
          <input id="q" type="search" name="q"                          5532
    autocomplete="off"                                                  5533
                 placeholder="搜索项目、图斑、标准" value="{{           5534
    search_q }}">                                                       5535
          <button type="submit" aria-label="搜索">{{                    5536
    ui.icon("chevron", 15) }}</button>                                  5537
        </form>                                                         5538
        {% if current_user.is_admin %}                                  5539
        <a class="icon-btn" href="/profile?tab=logs"                    5540
    title="操作日志"                                                    5541
           aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>          5542
        {% endif %}                                                     5543
      </header>                                                         5544
{% else %}                                                              5545
<div class="login-shell">                                               5546
{% endif %}                                                             5547
      <main id="main" class="app-main">                                 5548
        <div class="container">                                         5549
          {% for category, message in flash_msgs if category            5550
```

```
      not in ("toast", "ok") %}                                                5551
          <p class="flash {{ category }}" role="status">{{                     5552
      message }}</p>                                                           5553
          {% endfor %}                                                         5554
          {% block content %}{% endblock %}                                    5555
        </div>                                                                 5556
      </main>                                                                  5557
    {% if current_user.is_authenticated %}                                     5558
        <footer class="app-foot">                                              5559
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0</span>                  5560
          <span>数据截止 {{ today }}</span>                                    5561
        </footer>                                                              5562
      </div>                                                                   5563
    </div>                                                                     5564
    {% else %}                                                                 5565
    </div>                                                                     5566
    {% endif %}                                                                5567
    {% block scripts %}{% endblock %}                                          5568
</body>                                                                        5569
</html>                                                                        5570
#  ==== app/templates/_macros.html ====                                        5571
{#  共用宏：图标、流程状态面板 #}                                               5572
{#  ============================================================================5573
      ==================                                                       5574
    图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）                       5575
    不引图标字体或外部图片，避免额外请求与资源依赖；                            5576
    stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。                5577
    ============================================================================5578
      ================== #}                                                     5579
{% set ICONS = {                                                               5580
  "dot": '<circle cx="12" cy="12" r="3"/>',                                    5581
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2                     5582
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',                                5583
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2                   5584
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',                        5585
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                                   5586
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4                            5587
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                                        5588
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6                    5589
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',                           5590
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path                       5591
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',                   5592
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20                      5593
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5                           5594
    17h5"/>',                                                                  5595
  "database": '<ellipse cx="12" cy="5.5" rx="8"                                5596
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4                     5597
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4                       5598
    8-3.2"/>',                                                                 5599
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"                    5600
```

```
      cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
    "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
      2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
      2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
    "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
      20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
      3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
      d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
    "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
      21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
      4.4V21"/>',
    "mail": '<rect x="3" y="5" width="18" height="14"
      rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
    "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
      0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
      16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
    "lock": '<rect x="4" y="10.4" width="16" height="10.6"
      rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
    "briefcase": '<rect x="3" y="7.4" width="18"
      height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
      10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
    "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
    "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
    "plus": '<path d="M12 5v14M5 12h14"/>',
    "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
    "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
    "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
    "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
    "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
```

```
    废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。  # }
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
{% endmacro %}
{#  前置条件未满足时的提示块：只列缺什么，并给出处理入口  #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<div class="notice notice--warn">
  <p class="notice__title">尚不能执行本步骤，需要先完成：</p>
  <ol class="notice__list">
    {% for b in progress.blocking %}
    <li>
      <strong>{{ b.label }}</strong>{% if b.reason %}：{{
    b.reason }}{% endif %}
      <a class="link-btn" href="{{ b.fix_url }}">{{
```

```
      b.fix_label }}</a>                                                    5701
      </li>                                                                 5702
      {% endfor %}                                                          5703
   </ol>                                                                    5704
</div>                                                                      5705
{% endif %}                                                                 5706
{% endmacro %}                                                              5707
{# 一行式状态摘要，用于列表页 #}                                             5708
{% macro workflow_badge(progress) %}                                        5709
{% if progress.ready %}                                                     5710
<span class="tag tag--ok">可执行评价</span>                                 5711
{% elif progress.blocking %}                                                5712
<span class="tag tag--warn">{{ progress.blocking[0].label                   5713
      }}</span>                                                             5714
<span class="muted">{{ progress.blocking[0].reason }}</span>                5715
{% endif %}                                                                 5716
{% endmacro %}                                                              5717
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。   5718
   下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就直接列出并给出入口。 #}  5719
{% macro patch_guide(projects_without_patch) %}                             5720
{% if projects_without_patch %}                                             5721
<div class="notice notice--warn">                                           5722
   <p class="notice__title">                                                5723
      以下 {{ projects_without_patch|length }}                              5724
      个项目下还没有图斑，本页不会出现它们：                                5725
   </p>                                                                     5726
   <ul class="notice__list">                                                5727
      {% for p in projects_without_patch %}                                 5728
      <li>                                                                  5729
         {{ p.name }}                                                       5730
         <a class="link-btn" href="/project/{{ p.id                         5731
      }}/patch/add">新建图斑</a>                                            5732
      </li>                                                                 5733
      {% endfor %}                                                          5734
   </ul>                                                                    5735
</div>                                                                      5736
{% endif %}                                                                 5737
{% endmacro %}                                                              5738
# ==== app/templates/_login_badge.svg ====                                  5739
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg                5740
      " viewBox="0 0 240 240" role="img"                                    5741
      aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">               5742
   <title>系统徽标</title>                                                  5743
   <g class="lp-ring" style="--k: 2.7">                                     5744
      <circle cx="162.00" cy="120.00" r="2.60"                              5745
      fill="#c4f5ab"/>                                                      5746
      <circle cx="157.84" cy="138.22" r="2.97"                              5747
      fill="#b8f5ab"/>                                                      5748
      <circle cx="146.19" cy="152.84" r="3.06"                              5749
      fill="#acf5ab"/>                                                      5750
```

```
    <circle cx="129.35" cy="160.95" r="2.80"                    5751
    fill="#abf5b7" />                                           5752
    <circle cx="110.65" cy="160.95" r="2.40"                    5753
    fill="#abf5c2" />                                           5754
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce" />   5755
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da" />   5756
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6" />   5757
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da" />   5758
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce" />    5759
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2" />   5760
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7" />   5761
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab" />   5762
    <circle cx="157.84" cy="101.78" r="2.23"                    5763
    fill="#b8f5ab" />                                           5764
  </g>                                                          5765
  <g class="lp-ring" style="--k: 2.15">                         5766
    <circle cx="177.29" cy="129.07" r="3.92"                    5767
    fill="#a4e28d" />                                           5768
    <circle cx="171.68" cy="146.33" r="4.01"                    5769
    fill="#9be28d" />                                           5770
    <circle cx="161.01" cy="161.01" r="3.87"                    5771
    fill="#91e28d" />                                           5772
    <circle cx="146.33" cy="171.68" r="3.55"                    5773
    fill="#8de292" />                                           5774
    <circle cx="129.07" cy="177.29" r="3.17"                    5775
    fill="#8de29c" />                                           5776
    <circle cx="110.93" cy="177.29" r="2.88"                    5777
    fill="#8de2a6" />                                           5778
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" />   5779
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" />   5780
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" />   5781
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" />   5782
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" />   5783
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" />    5784
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" />    5785
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" />    5786
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" />   5787
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c" />   5788
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" />   5789
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" />   5790
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" />   5791
    <circle cx="177.29" cy="110.93" r="3.63"                    5792
    fill="#a4e28d" />                                           5793
  </g>                                                          5794
  <g class="lp-ring" style="--k: 1.85">                         5795
    <circle cx="194.00" cy="120.00" r="4.94"                    5796
    fill="#91cf72" />                                           5797
    <circle cx="191.85" cy="137.71" r="4.92"                    5798
    fill="#88cf72" />                                           5799
    <circle cx="185.52" cy="154.39" r="4.73"                    5800
```

```
    fill="#80cf72" />                                                    5801
    <circle cx="175.39" cy="169.07" r="4.42"                             5802
    fill="#78cf72" />                                                    5803
    <circle cx="162.04" cy="180.90" r="4.06"                             5804
    fill="#72cf73" />                                                    5805
    <circle cx="146.24" cy="189.19" r="3.73"                             5806
    fill="#72cf7b" />                                                    5807
    <circle cx="128.92" cy="193.46" r="3.51"                             5808
    fill="#72cf83" />                                                    5809
    <circle cx="111.08" cy="193.46" r="3.45"                             5810
    fill="#72cf8b" />                                                    5811
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93" />            5812
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c" />            5813
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4" />            5814
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac" />            5815
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4" />            5816
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc" />            5817
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4" />            5818
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac" />             5819
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4" />             5820
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c" />             5821
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93" />             5822
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b" />            5823
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83" />            5824
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b" />            5825
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73" />            5826
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72" />            5827
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72" />            5828
    <circle cx="191.85" cy="102.29" r="4.80"                             5829
    fill="#88cf72" />                                                    5830
  </g>                                                                   5831
  <g class="lp-ring" style="--k: 1.62">                                  5832
    <circle cx="209.57" cy="128.82" r="5.68"                             5833
    fill="#76bb59" />                                                    5834
    <circle cx="206.12" cy="146.13" r="5.40"                             5835
    fill="#6fbb59" />                                                    5836
    <circle cx="199.37" cy="162.43" r="5.06"                             5837
    fill="#68bb59" />                                                    5838
    <circle cx="189.57" cy="177.10" r="4.71"                             5839
    fill="#61bb59" />                                                    5840
    <circle cx="177.10" cy="189.57" r="4.41"                             5841
    fill="#5abb59" />                                                    5842
    <circle cx="162.43" cy="199.37" r="4.19"                             5843
    fill="#59bb5f" />                                                    5844
    <circle cx="146.13" cy="206.12" r="4.10"                             5845
    fill="#59bb66" />                                                    5846
    <circle cx="128.82" cy="209.57" r="4.15"                             5847
    fill="#59bb6d" />                                                    5848
    <circle cx="111.18" cy="209.57" r="4.32"                             5849
    fill="#59bb73" />                                                    5850
```

```
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a"/>5851
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81"/>5852
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88"/>5853
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f"/>5854
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96"/>5855
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d"/>5856
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4"/>5857
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4"/>5858
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d"/> 5859
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96"/> 5860
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f"/> 5861
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88"/> 5862
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81"/> 5863
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a"/> 5864
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73"/>5865
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d"/>5866
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66"/>5867
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f"/>5868
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59"/>5869
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59"/>5870
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59"/>5871
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59"/>5872
    <circle cx="209.57" cy="111.18" r="5.85"                 5873
    fill="#76bb59"/>                                         5874
  </g>                                                       5875
  <g class="lp-ring" style="--k: 1.46">                      5876
    <circle cx="226.00" cy="120.00" r="5.94"                 5877
    fill="#65a843"/>                                         5878
    <circle cx="224.55" cy="137.45" r="5.62"                 5879
    fill="#5fa843"/>                                         5880
    <circle cx="220.26" cy="154.42" r="5.29"                 5881
    fill="#59a843"/>                                         5882
    <circle cx="213.22" cy="170.45" r="5.00"                 5883
    fill="#53a843"/>                                         5884
    <circle cx="203.65" cy="185.11" r="4.77"                 5885
    fill="#4da843"/>                                         5886
    <circle cx="191.79" cy="197.99" r="4.63"                 5887
    fill="#47a843"/>                                         5888
    <circle cx="177.98" cy="208.74" r="4.59"                 5889
    fill="#43a846"/>                                         5890
    <circle cx="162.58" cy="217.07" r="4.67"                 5891
    fill="#43a84c"/>                                         5892
    <circle cx="146.02" cy="222.76" r="4.85"                 5893
    fill="#43a852"/>                                         5894
    <circle cx="128.75" cy="225.64" r="5.10"                 5895
    fill="#43a858"/>                                         5896
    <circle cx="111.25" cy="225.64" r="5.42"                 5897
    fill="#43a85e"/>                                         5898
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>5899
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>5900
```

```
      <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>
      <circle cx="224.55" cy="102.55" r="6.23"
      fill="#5fa843"/>
   </g>
   <g class="lp-badge__mark">
      <circle cx="120" cy="120" r="27" fill="#14432f"/>
      <g transform="translate(120 120) scale(1.5)
      translate(-12 -12)" fill="none" stroke="#ffffff"
      stroke-width="1.9" stroke-linecap="round"
      stroke-linejoin="round">
         <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>
         <circle cx="17" cy="6" r="2.2"/>
      </g>
   </g>
</svg>
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
   <h2>{{ code }} · {{ message }}</h2>
   <p class="muted">
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
   </p>
   <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
   </p>
```

```
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ==================
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp { color: #1d1d1f; }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
    opacity: calc(1 - var(--p) * .94);
    filter: blur(calc(var(--p) * 2.6px));
  }
  .lp-ring {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
  }
  .lp-badge__mark {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + .18 * var(--p)));
```

```
}                                                                    6001
.lp-title {                                                          6002
  margin: 0 0 18px;                                                  6003
  font-size: clamp(28px, 4.2vw, 42px);                               6004
  font-weight: 700;                                                  6005
  line-height: 1.3;                                                  6006
  letter-spacing: -.4px;                                             6007
  animation: lp-in .7s ease .06s both;                               6008
}                                                                    6009
.lp-cta {                                                            6010
  padding: 13px 52px;                                                6011
  font-family: inherit;                                              6012
  font-size: 17px;                                                   6013
  font-weight: 500;                                                  6014
  color: #fff;                                                       6015
  background: var(--accent);                                         6016
  border: none;                                                      6017
  border-radius: 980px;                                              6018
  cursor: pointer;                                                   6019
  animation: lp-in .7s ease .18s both;                               6020
  transition: background-color .2s ease, box-shadow .2s              6021
  ease,                                                              6022
              transform .2s ease;                                    6023
}                                                                    6024
.lp-cta:hover {                                                      6025
  background: var(--accent-hover);                                   6026
  box-shadow: 0 10px 26px rgba(31, 107, 69, .26);                    6027
}                                                                    6028
.lp-cta:active { transform: translateY(1px); box-shadow:             6029
  none; }                                                            6030
.lp-cta:focus-visible { outline: 3px solid rgba(31, 107,             6031
  69, .40); outline-offset: 3px; }                                   6032
.lp-more {                                                           6033
  margin-top: 38px;                                                  6034
  font-size: 13px;                                                   6035
  color: #86868b;                                                    6036
  text-decoration: none;                                             6037
  animation: lp-in .7s ease .24s both;                               6038
}                                                                    6039
.lp-more:hover { color: var(--accent); }                             6040
@keyframes lp-in {                                                   6041
  from { opacity: 0; transform: translateY(12px); }                  6042
  to   { opacity: 1; transform: none; }                              6043
}                                                                    6044
/* ---------- 滚动进场 ----------                                    6045
   进入视口时按方向弹入，离开视口时弹回，可反复。                    6046
   交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */  6047
.lp-reveal {                                                         6048
  opacity: 0;                                                        6049
  transition: opacity .62s ease,                                     6050
```

```
                transform .62s cubic-bezier(.16, .84, .34,
  1),
                border-color .62s ease, background-color
  .62s ease;
  transition-delay: 0s;
  will-change: opacity, transform;
}
.lp-reveal[data-dir="left"]  { transform:
  translate3d(-48px, 0, 0); }
.lp-reveal[data-dir="right"] { transform:
  translate3d(48px, 0, 0); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0,
  36px, 0); }
.lp-reveal.is-in {
  opacity: 1;
  transform: none;
  transition-delay: calc(var(--i, 0) * 85ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:
  color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: 980px;
  margin: 0 auto;
  padding: 20px 24px 72px;
}
.lp-intro__title {
  margin: 0 0 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
.lp-points {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px,
  1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 24px 22px;
  background: #f5f5f7;
  border-radius: 18px;
}
```

```
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 38px;
  height: 38px;
  margin-bottom: 14px;
  border-radius: 11px;
  background: #fff;
  color: var(--accent);
}
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }
.lp-point p {
  margin: 0;
  font-size: 13.5px;
  line-height: 1.85;
  color: #6e6e73;
  text-wrap: pretty;
}
.lp-sub-title { margin: 0 0 20px; font-size: 17px; }
.lp-mods {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(258px,
  1fr));
  gap: 12px;
  margin-bottom: 46px;
  align-items: start;
}
/*  模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明  */
.lp-mod {
  position: relative;
  padding: 17px 18px 17px 48px;
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  outline: none;
  transition: border-color .3s ease, box-shadow .3s ease;
}
.lp-mod:hover,
.lp-mod:focus-within,
.lp-mod:focus {
  border-color: rgba(31, 107, 69, .42);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.lp-mod__no {
  position: absolute;
  left: 17px;
  top: 17px;
  font-size: 13px;
  font-weight: 700;
  color: #c8d6cd;
```

```
    font-variant-numeric: tabular-nums;
  }
  .lp-mod__name {
    display: block;
    font-size: 14px;
    font-weight: 600;
  }
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
  .lp-mod__body {
    display: grid;
    grid-template-rows: 0fr;
    transition: grid-template-rows .34s ease, margin-top
    .34s ease;
  }
  .lp-mod:hover .lp-mod__body,
  .lp-mod:focus-within .lp-mod__body {
    grid-template-rows: 1fr;
    margin-top: 7px;
  }
  .lp-mod__desc {
    overflow: hidden;
    margin: 0;
    font-size: 12.5px;
    line-height: 1.7;
    color: #6e6e73;
    opacity: 0;
    transition: opacity .3s ease;
    text-wrap: pretty;
  }
  .lp-mod:hover .lp-mod__desc,
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }
  /* 触摸设备没有悬停，直接把说明摊开 */
  @media (hover: none) {
    .lp-mod__body { grid-template-rows: 1fr; margin-top:
    7px; }
    .lp-mod__desc { opacity: 1; }
  }
  .lp-foot {
    margin: 0;
    padding-top: 26px;
    border-top: 1px solid #e6e6e8;
    text-align: center;
    font-size: 12.5px;
    color: #86868b;
  }
  /* ---------- 登录面板：点击后渐变浮现 ---------- */
  .lp-auth {
    position: fixed;
    inset: 0;
    z-index: 60;
```

```
    display: grid;
    place-items: center;
    padding: 24px;
    opacity: 0;
    visibility: hidden;
    transition: opacity .42s ease, visibility 0s linear
    .42s;
  }
  .lp-auth.is-open {
    opacity: 1;
    visibility: visible;
    transition: opacity .42s ease, visibility 0s;
  }
  /* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */
  .lp-auth__scrim {
    position: absolute;
    inset: 0;
    background:
      radial-gradient(1000px 640px at 50% 0%,
                      rgba(31, 107, 69, .22), rgba(31, 107,
    69, 0) 70%),
      linear-gradient(180deg, rgba(255, 255, 255, .84),
                                rgba(246, 250, 247, .95));
    backdrop-filter: blur(16px) saturate(1.25);
    -webkit-backdrop-filter: blur(16px) saturate(1.25);
  }
  .lp-auth__panel {
    position: relative;
    width: 100%;
    max-width: 396px;
    padding: 36px 36px 26px;
    background: #fff;
    border-radius: 20px;
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);
    opacity: 0;
    transform: translateY(18px) scale(.975);
    transition: transform .55s cubic-bezier(.16, .84, .34,
    1) .05s,
                opacity .55s ease .05s;
  }
  .lp-auth.is-open .lp-auth__panel { opacity: 1; transform:
    none; }
  .lp-auth__close {
    position: absolute;
    top: 14px;
    right: 14px;
    display: grid;
    place-items: center;
    width: 30px;
    height: 30px;
```

```
    color: #86868b;                                                    6251
    background: transparent;                                           6252
    border: none;                                                      6253
    border-radius: 50%;                                                6254
    cursor: pointer;                                                   6255
    transition: background-color .18s ease, color .18s ease;           6256
  }                                                                    6257
  .lp-auth__close:hover { background: #f5f5f7; color:                  6258
    #1d1d1f; }                                                         6259
  .lp-auth__title {                                                    6260
    margin: 0 0 6px;                                                   6261
    text-align: center;                                                6262
    font-size: 24px;                                                   6263
    font-weight: 700;                                                  6264
    letter-spacing: -.2px;                                             6265
  }                                                                    6266
  .lp-auth__sub {                                                      6267
    margin: 0 0 26px;                                                  6268
    text-align: center;                                                6269
    font-size: 13px;                                                   6270
    color: #86868b;                                                    6271
  }                                                                    6272
  .lp-field { margin-bottom: 15px; }                                   6273
  .lp-field label {                                                    6274
    display: block;                                                    6275
    margin-bottom: 7px;                                                6276
    font-size: 13px;                                                   6277
    color: #6e6e73;                                                    6278
  }                                                                    6279
  .lp-field input {                                                    6280
    width: 100%;                                                       6281
    padding: 13px 16px;                                                6282
    font-family: inherit;                                              6283
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */  6284
    color: #1d1d1f;                                                    6285
    background: #f5f5f7;                                               6286
    border: 1px solid transparent;                                     6287
    border-radius: 12px;                                               6288
    outline: none;                                                     6289
    transition: background-color .18s ease, border-color               6290
    .18s ease,                                                         6291
                box-shadow .18s ease;                                  6292
  }                                                                    6293
  .lp-field input::placeholder { color: #a1a1a6; }                     6294
  .lp-field input:focus {                                              6295
    background: #fff;                                                  6296
    border-color: rgba(31, 107, 69, .55);                              6297
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                      6298
  }                                                                    6299
  .lp-remember {                                                       6300
```

```
    display: flex;
    align-items: center;
    gap: 8px;
    margin: 4px 0 20px;
    font-size: 13px;
    color: #6e6e73;
    cursor: pointer;
  }
  .lp-remember input {
    width: 16px;
    height: 16px;
    margin: 0;
    accent-color: var(--accent);
    cursor: pointer;
  }
  .lp-submit {
    width: 100%;
    padding: 13px 20px;
    font-family: inherit;
    font-size: 16px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
  }
  .lp-submit:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
  }
  .lp-submit:active { transform: translateY(1px);
    box-shadow: none; }
  .lp-submit:focus-visible { outline: 3px solid rgba(31,
    107, 69, .40); outline-offset: 3px; }
  .lp-auth__hint {
    margin: 18px 0 0;
    text-align: center;
    font-size: 12.5px;
    color: #a1a1a6;
  }
  .lp-auth__reg {
    color: var(--accent);
    font-weight: 500;
    text-decoration: none;
    border-bottom: 1px solid rgba(31, 107, 69, .35);
    transition: border-color .18s ease;
```

```
  }                                                
```

```
    看。'),                                                                    6401
          ('shield', '流程可控',                                               6402
           '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重 6403
    建或人工辅助的结论。'),                                                    6404
          ('file', '结果可复算',                                               6405
           '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')  6406
       ] %}                                                                    6407
       {% for ic, title, text in points %}                                     6408
       <div class="lp-point lp-reveal" data-dir="{{                            6409
    pdirs[loop.index0 % 3] }}"                                                 6410
            style="--i: {{ loop.index0 }}">                                    6411
         <span class="lp-point__icon">{{ icon(ic, 20)                          6412
    }}</span>                                                                  6413
         <h3>{{ title }}</h3>                                                  6414
         <p>{{ text }}</p>                                                     6415
       </div>                                                                  6416
       {% endfor %}                                                            6417
    </div>                                                                     6418
    <h3 class="lp-sub-title lp-reveal"                                         6419
    data-dir="up">九个功能模块</h3>                                            6420
    <div class="lp-mods">                                                      6421
       {% set mdirs = ['left', 'up', 'right'] %}                               6422
       {% for no, name, desc in modules %}                                     6423
       <div class="lp-mod lp-reveal" data-dir="{{                              6424
    mdirs[loop.index0 % 3] }}"                                                 6425
            style="--i: {{ loop.index0 % 3 }}" tabindex="0">                   6426
         <span class="lp-mod__no">{{ no }}</span>                              6427
         <span class="lp-mod__name">{{ name }}</span>                          6428
         <div class="lp-mod__body"><p class="lp-mod__desc">{{                  6429
    desc }}</p></div>                                                          6430
       </div>                                                                  6431
       {% endfor %}                                                            6432
    </div>                                                                     6433
    <p class="lp-foot lp-reveal" data-dir="up">                                6434
       历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０ ·                       6435
       运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库                        6436
    </p>                                                                       6437
</section>                                                                     6438
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}       6439
<div class="lp-auth" id="lp-auth" aria-hidden="true">                          6440
    <div class="lp-auth__scrim" data-lp-close></div>                           6441
    <div class="lp-auth__panel" role="dialog"                                  6442
    aria-modal="true"                                                          6443
          aria-labelledby="lp-auth-title">                                     6444
       <button type="button" class="lp-auth__close"                            6445
    data-lp-close                                                              6446
               aria-label="关闭登录框">                                        6447
         <svg width="16" height="16" viewBox="0 0 24 24"                       6448
    fill="none"                                                                6449
              stroke="currentColor" stroke-width="2"                           6450
```

```
      stroke-linecap="round"                                                6451
                aria-hidden="true"><path d="M6 6l12 12M18 6L6               6452
      18"/></svg>                                                           6453
        </button>                                                           6454
        <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>             6455
        <p class="lp-auth__sub">使用系统账号登录</p>                        6456
        <form method="post" novalidate>                                     6457
          {{ form.hidden_tag() }}                                           6458
          <div class="lp-field">                                            6459
            <label for="{{ form.username.id }}">账户名称</label>            6460
            {{ form.username(id=form.username.id,                           6461
      autocomplete="username",                                              6462
                             placeholder="请输入账户名称") }}               6463
          </div>                                                            6464
          <div class="lp-field">                                            6465
            <label for="{{ form.password.id }}">密码</label>                6466
            {{ form.password(id=form.password.id,                           6467
      autocomplete="current-password",                                      6468
                             placeholder="请输入密码") }}                   6469
          </div>                                                            6470
          <label class="lp-remember" for="{{ form.remember.id               6471
      }}">                                                                  6472
            {{ form.remember(id=form.remember.id) }}                        6473
            <span>记住我（下次自动登录，一个月内有效）</span>               6474
          </label>                                                          6475
          {{ form.submit(class_="lp-submit") }}                             6476
        </form>                                                             6477
        <p class="lp-auth__hint">                                           6478
          还没有账号？<a class="lp-auth__reg"                               6479
                         href="{{ url_for('auth.register')                  6480
      }}">点我注册</a>                                                      6481
        </p>                                                                6482
        <p class="lp-auth__hint" style="margin-top:10px;">                  6483
          演示账号：admin 密码：admin123                                    6484
        </p>                                                                6485
      </div>                                                                6486
    </div>                                                                  6487
</div>                                                                      6488
{% endblock %}                                                              6489
{% block scripts %}                                                         6490
<script>                                                                    6491
(function () {                                                              6492
  var reduce = window.matchMedia &&                                         6493
    window.matchMedia('(prefers-reduced-motion:                             6494
    reduce)').matches;                                                      6495
  /* ---------- 登录面板 ---------- */                                       6496
  var btn = document.getElementById('lp-open');                             6497
  var auth = document.getElementById('lp-auth');                            6498
  var lastFocus = null;                                                     6499
  function openPanel() {                                                    6500
```

```
    lastFocus = document.activeElement;                                  6501
    auth.classList.add('is-open');                                       6502
    auth.setAttribute('aria-hidden', 'false');                           6503
    btn.setAttribute('aria-expanded', 'true');                           6504
    document.body.style.overflow = 'hidden';                             6505
    //  记到地址栏，直接访问  /login#login  也能展开登录框                  6506
    if (location.hash !== '#login') {                                    6507
      history.replaceState(null, '', location.pathname +                 6508
    '#login');                                                           6509
    }                                                                    6510
    var first = auth.querySelector('input');                             6511
    if (first) window.setTimeout(function () {                           6512
    first.focus(); }, 300);                                              6513
  }                                                                      6514
  function closePanel() {                                                6515
    auth.classList.remove('is-open');                                    6516
    auth.setAttribute('aria-hidden', 'true');                            6517
    btn.setAttribute('aria-expanded', 'false');                          6518
    document.body.style.overflow = '';                                   6519
    if (location.hash === '#login') {                                    6520
      history.replaceState(null, '', location.pathname);                 6521
    }                                                                    6522
    if (lastFocus && lastFocus.focus) lastFocus.focus();                 6523
  }                                                                      6524
  if (btn && auth) {                                                     6525
    btn.addEventListener('click', openPanel);                            6526
    Array.prototype.forEach.call(                                        6527
      auth.querySelectorAll('[data-lp-close]'),                          6528
      function (el) { el.addEventListener('click',                       6529
    closePanel); }                                                       6530
    );                                                                   6531
    document.addEventListener('keydown', function (e) {                 6532
      if (e.key === 'Escape' && auth.classList.contains('is-             6533
    open')) closePanel();                                                6534
    });                                                                  6535
    //  登录失败回到本页、或直接带  #login  访问时，直接展开登录框          6536
    if (location.hash === '#login' ||                                    6537
    document.querySelector('.flash.error')) {                            6538
      openPanel();                                                       6539
    }                                                                    6540
  }                                                                      6541
  /*  ----------  内容进场  +  徽标散开：同一套滚动计算  ----------        6542
      原先用  IntersectionObserver，实测在滚动到位置后回调并未生效，      6543
      内容始终停在  opacity：0。改为直接按元素与视口的相对位置判断，      6544
      与徽标用同一次滚动回调，行为可预期、也便于排查。  */                6545
  var items = document.querySelectorAll('.lp-reveal');                   6546
  var badge = document.getElementById('lp-badge');                       6547
  var ticking = false;                                                   6548
  function applyScroll() {                                               6549
    ticking = false;                                                     6550
```

```
      var vh = window.innerHeight || 800;                                  6551
      /* 徽标：滚动前约三分之一屏内完成散开淡出 */                              6552
      if (badge) {                                                         6553
        var range = vh * 0.34;                                             6554
        var p = range > 0 ? window.scrollY / range : 0;                    6555
        badge.style.setProperty('--p', Math.min(1,                         6556
      Math.max(0, p)).toFixed(3));                                         6557
      }                                                                    6558
      /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */                  6559
      if (reduce || !items.length) return;                                 6560
      var box = [];                                                        6561
      for (var i = 0; i < items.length; i++) {                             6562
        box.push(items[i].getBoundingClientRect());                        6563
      }                                                                    6564
      for (var j = 0; j < items.length; j++) {                             6565
        var r = box[j];                                                    6566
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内                6567
        var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;             6568
        if (shown) items[j].classList.add('is-in');                        6569
        else items[j].classList.remove('is-in');                           6570
      }                                                                    6571
    }                                                                      6572
    function onScroll() {                                                  6573
      if (!ticking) {                                                      6574
        ticking = true;                                                    6575
        window.requestAnimationFrame(applyScroll);                         6576
      }                                                                    6577
    }                                                                      6578
    // 减少动态效果时不做滚动特效，内容直接显示                              6579
    if (reduce) {                                                          6580
      Array.prototype.forEach.call(items, function (el) {                  6581
        el.classList.add('is-in');                                         6582
      });                                                                  6583
      if (badge) badge.style.setProperty('--p', '0');                      6584
    } else {                                                               6585
      window.addEventListener('scroll', onScroll, { passive:               6586
      true });                                                             6587
      window.addEventListener('resize', onScroll);                         6588
      applyScroll();                                                       6589
    }                                                                      6590
})();                                                                      6591
</script>                                                                  6592
{% endblock %}                                                             6593
# ==== app/templates/auth/register.html ====                               6594
{% extends "base.html" %}                                                  6595
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}     6596
{% block content %}                                                        6597
<style>                                                                    6598
  /* ================== 注册页 ================== */                        6599
     与登录页同一套观感：浅绿渐变底 + 白色卡片。                             6600
```

```
      必填只有账户名称与密码，其余收在「选填」分隔线以下，                                    6601
      这样首次注册不会被一堆输入框挡住。  */                                                6602
.rg {                                                                                      6603
  min-height: 100dvh;                                                                      6604
  display: grid;                                                                           6605
  place-items: center;                                                                     6606
  padding: 40px 20px 56px;                                                                 6607
  background:                                                                              6608
    radial-gradient(900px 520px at 50% 0%,                                                 6609
                    rgba(31, 107, 69, .14), rgba(31, 107,                                  6610
  69, 0) 72%),                                                                             6611
    #f6faf7;                                                                               6612
}                                                                                          6613
.rg-card {                                                                                 6614
  width: 100%;                                                                             6615
  max-width: 470px;                                                                        6616
  padding: 38px 40px 30px;                                                                 6617
  background: #fff;                                                                        6618
  border-radius: 20px;                                                                     6619
  box-shadow: 0 26px 70px rgba(14, 46, 32, .13);                                           6620
  animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)                                      6621
  both;                                                                                    6622
}                                                                                          6623
@keyframes rg-in {                                                                         6624
  from { opacity: 0; transform: translateY(16px)                                           6625
  scale(.98); }                                                                            6626
  to   { opacity: 1; transform: none; }                                                    6627
}                                                                                          6628
.rg-title {                                                                                6629
  margin: 0 0 6px;                                                                         6630
  text-align: center;                                                                      6631
  font-size: 25px;                                                                         6632
  font-weight: 700;                                                                        6633
  letter-spacing: -.2px;                                                                   6634
}                                                                                          6635
.rg-sub {                                                                                  6636
  margin: 0 0 28px;                                                                        6637
  text-align: center;                                                                      6638
  font-size: 13px;                                                                         6639
  color: #86868b;                                                                          6640
}                                                                                          6641
.rg-field { margin-bottom: 15px; }                                                         6642
.rg-field > label {                                                                        6643
  display: block;                                                                          6644
  margin-bottom: 7px;                                                                      6645
  font-size: 13px;                                                                         6646
  color: #6e6e73;                                                                          6647
}                                                                                          6648
.rg-field > label .rg-opt {                                                                6649
  margin-left: 5px;                                                                        6650
```

```
    font-size: 12px;                                                    6651
    color: #a1a1a6;                                                     6652
  }                                                                     6653
  .rg-field input[type="text"],                                         6654
  .rg-field input[type="password"] {                                    6655
    width: 100%;                                                        6656
    padding: 12px 15px;                                                 6657
    font-family: inherit;                                               6658
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    6659
    color: #1d1d1f;                                                     6660
    background: #f5f5f7;                                                6661
    border: 1px solid transparent;                                      6662
    border-radius: 12px;                                                6663
    outline: none;                                                      6664
    transition: background-color .18s ease, border-color                6665
    .18s ease,                                                          6666
                box-shadow .18s ease;                                   6667
  }                                                                     6668
  .rg-field input::placeholder { color: #a1a1a6; }                      6669
  .rg-field input:focus {                                               6670
    background: #fff;                                                   6671
    border-color: rgba(31, 107, 69, .55);                               6672
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       6673
  }                                                                     6674
  .rg-field input[aria-invalid="true"] {                                6675
    border-color: rgba(180, 60, 50, .55);                               6676
    background: #fff;                                                   6677
  }                                                                     6678
  .rg-field select {                                                    6679
    width: 100%;                                                        6680
    padding: 12px 15px;                                                 6681
    font-family: inherit;                                               6682
    font-size: 15px;                                                    6683
    color: #1d1d1f;                                                     6684
    background: #f5f5f7;                                                6685
    border: 1px solid transparent;                                      6686
    border-radius: 12px;                                                6687
    outline: none;                                                      6688
  }                                                                     6689
  .rg-field select:focus {                                              6690
    background: #fff;                                                   6691
    border-color: rgba(31, 107, 69, .55);                               6692
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       6693
  }                                                                     6694
  .rg-err {                                                             6695
    margin: 6px 0 0;                                                    6696
    font-size: 12.5px;                                                  6697
    color: var(--danger);                                               6698
  }                                                                     6699
  /* 「选填」分隔线 */                                                   6700
```

```
.rg-split {
  display: flex;
  align-items: center;
  gap: 12px;
  margin: 24px 0 18px;
  font-size: 12.5px;
  color: #a1a1a6;
}
.rg-split::before,
.rg-split::after {
  content: "";
  flex: 1;
  height: 1px;
  background: #ececee;
}
.rg-pair {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0 14px;
}
@media (max-width: 480px) { .rg-pair {
  grid-template-columns: 1fr; } }
/* 头像：圆形预览 + 选择文件 */
.rg-avatar {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 15px;
}
.rg-avatar__face {
  display: grid;
  place-items: center;
  flex: none;
  width: 54px;
  height: 54px;
  border-radius: 50%;
  background: #eef2f0;
  color: #9aa8a1;
  font-size: 12px;
  overflow: hidden;
  background-size: cover;
  background-position: center;
}
.rg-avatar__pick { min-width: 0; }
.rg-avatar__pick input[type="file"] {
  max-width: 100%;
  font-family: inherit;
  font-size: 12.5px;
  color: #6e6e73;
}
```

```
  .rg-avatar__tip {                                                    6751
    margin: 4px 0 0;                                                   6752
    font-size: 12px;                                                   6753
    color: #a1a1a6;                                                    6754
  }                                                                    6755
  .rg-submit {                                                         6756
    width: 100%;                                                       6757
    margin-top: 24px;                                                  6758
    padding: 13px 20px;                                                6759
    font-family: inherit;                                              6760
    font-size: 16px;                                                   6761
    font-weight: 500;                                                  6762
    color: #fff;                                                       6763
    background: var(--accent);                                         6764
    border: none;                                                      6765
    border-radius: 980px;                                              6766
    cursor: pointer;                                                   6767
    transition: background-color .2s ease, box-shadow .2s              6768
    ease,                                                              6769
                transform .2s ease;                                    6770
  }                                                                    6771
  .rg-submit:hover {                                                   6772
    background: var(--accent-hover);                                   6773
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                    6774
  }                                                                    6775
  .rg-submit:active { transform: translateY(1px);                      6776
    box-shadow: none; }                                                6777
  .rg-submit:focus-visible { outline: 3px solid rgba(31,               6778
    107, 69, .40); outline-offset: 3px; }                              6779
  .rg-foot {                                                           6780
    margin: 20px 0 0;                                                  6781
    text-align: center;                                                6782
    font-size: 13px;                                                   6783
    color: #86868b;                                                    6784
  }                                                                    6785
  .rg-foot a { color: var(--accent); text-decoration: none;            6786
    }                                                                  6787
  .rg-foot a:hover { text-decoration: underline; }                     6788
  @media (prefers-reduced-motion: reduce) {                            6789
    .rg-card { animation: none; }                                      6790
  }                                                                    6791
</style>                                                               6792
<div class="rg">                                                       6793
  <div class="rg-card">                                                6794
    <h1 class="rg-title">创建账号</h1>                                  6795
    <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>          6796
    <form method="post" enctype="multipart/form-data"                  6797
    novalidate>                                                        6798
      {{ form.hidden_tag() }}                                          6799
      <div class="rg-field">                                           6800
```

```
      <label for="{{ form.username.id }}">账户名称</label>
      {{ form.username(id=form.username.id,
autocomplete="username",
                       placeholder="2–64 个字符",
                       **({'aria-invalid': 'true'} if
form.username.errors else {})) }}
      {% for e in form.username.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.password.id }}">密码</label>
      {{ form.password(id=form.password.id,
autocomplete="new-password",
                       placeholder="至少 6 位",
                       **({'aria-invalid': 'true'} if
form.password.errors else {})) }}
      {% for e in form.password.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.confirm.id }}">确认密码</label>
      {{ form.confirm(id=form.confirm.id,
autocomplete="new-password",
                      placeholder="再次输入密码",
                      **({'aria-invalid': 'true'} if
form.confirm.errors else {})) }}
      {% for e in form.confirm.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <p class="rg-split">以下均为选填</p>
    <div class="rg-avatar">
      <span class="rg-avatar__face" id="rg-face"
aria-hidden="true">头像</span>
      <div class="rg-avatar__pick">
        <label class="visually-hidden"
for="rg-avatar">选择头像图片</label>
        <input id="rg-avatar" type="file" name="avatar"
               accept=".png,.jpg,.jpeg,.gif,.webp">
        <p class="rg-avatar__tip">PNG / JPG / GIF /
WebP，不超过 4 MB</p>
      </div>
    </div>
    <div class="rg-pair">
      <div class="rg-field">
        <label for="{{ form.display_name.id }}">昵称</label>
        {{ form.display_name(id=form.display_name.id,
                             placeholder="留空则显示账户名称") }}
      </div>
      <div class="rg-field">
        <label for="{{ form.gender.id }}">性别</label>
```

```
            {{ form.gender(id=form.gender.id) }}
          </div>
        </div>
        <div class="rg-field">
          <label for="{{ form.email.id }}">邮箱</label>
          {{ form.email(id=form.email.id,
      autocomplete="email",
                        placeholder="用于联系，可留空",
                        **({'aria-invalid': 'true'} if
      form.email.errors else {})) }}
          {% for e in form.email.errors %}<p
      class="rg-err">{{ e }}</p>{% endfor %}
        </div>
        <div class="rg-field" style="margin-bottom:0;">
          <label for="{{ form.phone.id }}">手机号</label>
          {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
// 选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
```

```
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {% if dash.progress.ready %}其中 {{ dash.progress.ready
    }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/unit/">
    <span class="stat-card__label">微地质单元</span>
    <span class="stat-card__value">{{ dash.stats.unit_types
```

```
      }}<span class="stat-card__unit">类</span></span>
      <span class="stat-card__note">
        {% if dash.progress.units %}共划分 {{
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
      </span>
    </a>
    <a class="stat-card" href="/standard/">
      <span class="stat-card__label">标准与政策</span>
      <span class="stat-card__value">{{ dash.stats.standards
      }}<span class="stat-card__unit">项</span></span>
      <span class="stat-card__note">地方标准与政策文件</span>
    </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
    <div class="section__head">
      <h2 class="section__title">评价流程</h2>
      <p class="section__desc">
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
      </p>
    </div>
    <div class="step-grid">
      {% for s in dash.steps %}
      {% if s.available %}
      <a class="step-card" href="{{ s.href }}">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        {% if s.pending %}
        <span class="step-card__badge">待处理 {{ s.pending
      }}</span>
        {% endif %}
      </a>
      {% else %}
      <div class="step-card step-card--locked"
      aria-disabled="true">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        <span class="step-card__lock">{{ s.note }}</span>
      </div>
      {% endif %}
      {% endfor %}
    </div>
</section>
{# ---------- 系统管理 ---------- #}
```

```
<section class="section">
  <div class="section__head">
    <h2 class="section__title">系统管理</h2>
  </div>
  <a class="row-card" href="/profile">
    <span class="row-card__icon">{{ icon("user", 19) }}</span>
    <span class="row-card__main">
      <span class="row-card__title">个人设置</span>
      <span class="row-card__desc">维护昵称、性别、邮箱、手机号与登录密码</span>
    </span>
    <span class="row-card__arrow">{{ icon("chevron", 17) }}</span>
  </a>
  {% if current_user.is_admin %}
  <a class="row-card" href="/profile?tab=users">
    <span class="row-card__icon">{{ icon("users", 19) }}</span>
    <span class="row-card__main">
      <span class="row-card__title">账号与操作日志</span>
      <span class="row-card__desc">新增账号并设置职务；全部写操作自动留痕，可查询追溯</span>
    </span>
    <span class="row-card__arrow">{{ icon("chevron", 17) }}</span>
  </a>
  {% endif %}
</section>
{% endblock %}
# ==== app/templates/auth/add_user.html ====
{% extends "base.html" %}
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label>
      <input id="username" type="text" name="username" required
             maxlength="64" autocomplete="off" style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password" required
```

```
            autocomplete="new-password"                                  7051
    style="width:280px;">                                                7052
    </div>                                                               7053
    <div class="field">                                                  7054
      <label class="field__label" for="role">职务</label>                7055
      <select id="role" name="role">                                     7056
        {% for code, label in role_labels.items() %}                     7057
        <option value="{{ code }}" {% if code ==                         7058
    default_role %}selected{% endif %}>{{ label }}</option>              7059
        {% endfor %}                                                     7060
      </select>                                                          7061
      <p class="muted" style="margin:6px 0 0;">                          7062
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。          7063
      </p>                                                               7064
    </div>                                                               7065
    <hr style="margin:22px 0;border:none;border-top:1px                  7066
    solid var(--line);">                                                 7067
    <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3     7068
    >                                                                    7069
    <div class="field">                                                  7070
      <label class="field__label" for="display_name">昵称</lab           7071
    el>                                                                  7072
      <input id="display_name" type="text"                               7073
    name="display_name" maxlength="64"                                   7074
            placeholder="留空则显示账户名称">                                7075
    </div>                                                               7076
    <div class="field">                                                  7077
      <label class="field__label" for="gender">性别</label>              7078
      <select id="gender" name="gender">                                 7079
        <option value="">不填</option>                                   7080
        {% for g in genders %}<option value="{{ g }}">{{ g               7081
    }}</option>{% endfor %}                                              7082
      </select>                                                          7083
    </div>                                                               7084
    <div class="field">                                                  7085
      <label class="field__label" for="email">邮箱</label>               7086
      <input id="email" type="text" name="email"                         7087
    maxlength="128">                                                     7088
    </div>                                                               7089
    <div class="field">                                                  7090
      <label class="field__label" for="phone">手机号</label>             7091
      <input id="phone" type="text" name="phone"                         7092
    maxlength="32">                                                      7093
    </div>                                                               7094
    <p class="actions" style="margin-top:18px;">                         7095
      <button type="submit" class="btn">保存</button>                    7096
      <a class="btn btn--ghost" href="{{                                 7097
    url_for('auth.profile', tab='users') }}">返回</a>                    7098
    </p>                                                                 7099
  </form>                                                                7100
```

```
</div>                                                                  7101
{% endblock %}                                                          7102
# ==== app/templates/auth/edit_user.html ====                           7103
{% extends "base.html" %}                                               7104
{% block title %}编辑账号 · {{ user.username }}{% endblock %}           7105
{% block content %}                                                     7106
<div class="card" style="max-width:560px;">                             7107
  <h2>编辑账号 · {{ user.username }}</h2>                               7108
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>             7109
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"            7110
    >                                                                   7111
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。       7112
  </p>                                                                  7113
  <form method="post">                                                  7114
    {{ form.hidden_tag() }}                                             7115
    <div class="field">                                                 7116
      <label class="field__label" for="display_name">昵称</lab          7117
    el>                                                                 7118
      <input id="display_name" type="text"                              7119
    name="display_name" maxlength="64"                                  7120
             value="{{ user.display_name or '' }}"                      7121
             placeholder="留空则显示账户名称">                          7122
    </div>                                                              7123
    <div class="field">                                                 7124
      <label class="field__label" for="gender">性别</label>             7125
      <select id="gender" name="gender">                                7126
        <option value="">不填</option>                                  7127
        {% for g in genders %}                                          7128
        <option value="{{ g }}" {% if user.gender == g                  7129
    %}selected{% endif %}>{{ g }}</option>                              7130
        {% endfor %}                                                    7131
      </select>                                                         7132
    </div>                                                              7133
    <div class="field">                                                 7134
      <label class="field__label" for="email">邮箱</label>              7135
      <input id="email" type="text" name="email"                        7136
    maxlength="128"                                                     7137
             value="{{ user.email or '' }}">                            7138
    </div>                                                              7139
    <div class="field">                                                 7140
      <label class="field__label" for="phone">手机号</label>            7141
      <input id="phone" type="text" name="phone"                        7142
    maxlength="32"                                                      7143
             value="{{ user.phone or '' }}">                            7144
    </div>                                                              7145
    <hr style="margin:22px 0;border:none;border-top:1px                 7146
    solid var(--line);">                                                7147
    <h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>         7148
    <p class="muted" style="margin:0 0                                  7149
    14px;font-size:12.5px;">                                            7150
```

```
        职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
        操作员可使用全部业务功能。
      </p>
      <div class="field">
        <label class="field__label" for="role">职务</label>
        <select id="role" name="role">
          {% for code, label in role_labels.items() %}
          <option value="{{ code }}" {% if user.role == code
      %}selected{% endif %}>{{ label }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="checkbox" for="active">
          <input id="active" type="checkbox" name="active"
      value="1"
                 {% if user.active %}checked{% endif %}>
          <span>启用该账号</span>
        </label>
        <p class="muted" style="margin:6px 0 0;">
          停用后该账号无法登录，但其历史操作记录会保留。
          系统至少需保留一名启用状态的管理员。
        </p>
      </div>
      <div class="field">
        <label class="field__label" for="password">重置密码（留空则不修改
      ）</label>
        <input id="password" type="password" name="password"
               autocomplete="new-password"
      style="width:280px;"
               placeholder="留空表示保持原密码">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="{{
      url_for('auth.profile', tab='users') }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 个人设置（仿 Apple 账户页） ==================
     左栏是身份与分类，右栏是该分类下的功能卡。
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——
     不点就不会误改，页面也不会一上来堆满输入框。 */
```

```
.acct {
  display: grid;
  grid-template-columns: 244px 1fr;
  gap: 40px;
  align-items: start;
}
@media (max-width: 900px) {
  .acct { grid-template-columns: 1fr; gap: 24px; }
}
/* ---------- 左栏 ---------- */
.acct-rail__avatar {
  display: grid;
  place-items: center;
  width: 88px;
  height: 88px;
  border-radius: 50%;
  background: #e9edeb;
  color: #97a09b;
  font-size: 32px;
  font-weight: 500;
  object-fit: cover;
}
.acct-rail__name {
  margin: 16px 0 2px;
  font-size: 16px;
  font-weight: 600;
}
.acct-rail__sub {
  margin: 0 0 24px;
  font-size: 13px;
  color: #86868b;
  word-break: break-all;
}
.acct-tabs {
  display: flex;
  flex-direction: column;
  gap: 2px;
  margin-bottom: 24px;
}
.acct-tabs a {
  padding: 7px 12px;
  border: 1px solid transparent;
  border-radius: 7px;
  font-size: 15px;
  color: #1d1d1f;
  text-decoration: none;
  transition: background-color .16s ease, border-color
  .16s ease;
}
.acct-tabs a:hover { background: #f5f5f7; }
```

```
.acct-tabs a.is-active {
  font-weight: 600;
  border-color: #bcc9c1;
  background: #fff;
}
/* 更换头像：折叠在左栏，点开才出现文件选择 */
.acct-avatar-edit > summary {
  display: inline-block;
  padding: 5px 12px;
  font-size: 13px;
  color: var(--accent);
  background: var(--accent-soft);
  border-radius: 7px;
  cursor: pointer;
  list-style: none;
}
.acct-avatar-edit > summary::-webkit-details-marker { display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%; margin-bottom: 8px; }
.acct-avatar-edit__file {
  margin: 0 0 10px;
  font-size: 12px;
  color: var(--ink-soft);
  word-break: break-all;
}
/* ---------- 右栏 ---------- */
.acct-title {
  margin: 0 0 8px;
  font-size: 26px;
  font-weight: 700;
  letter-spacing: -.3px;
}
.acct-lead {
  margin: 0 0 26px;
  font-size: 14px;
  color: #6e6e73;
}
/* 固定两列，与 Apple 账户页的 2×2 一致；
   卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
.acct-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 16px;
  align-items: start;
}
@media (max-width: 760px) {
  .acct-grid { grid-template-columns: 1fr; }
```

```
}
/* 功能卡：details/summary 实现，点开才展开表单 */
.acct-card {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  transition: border-color .2s ease, box-shadow .2s ease;
}
.acct-card[open] {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.acct-card > summary {
  position: relative;
  display: block;
  padding: 18px 56px 18px 20px;
  border-radius: 14px;
  cursor: pointer;
  list-style: none;
}
.acct-card > summary::-webkit-details-marker { display:
  none; }
.acct-card > summary:hover { background: #fafbfa; }
.acct-card > summary:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}
.acct-card__title {
  display: block;
  font-size: 15px;
  font-weight: 600;
}
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
```

```
}                                                                    7351
.acct-card__body {                                                   7352
  padding: 4px 20px 20px;                                            7353
  border-top: 1px solid #f0f1f2;                                     7354
  margin-top: 2px;                                                   7355
}                                                                    7356
.acct-card__body .field:first-child { margin-top: 16px; }            7357
.acct-card__body .btn { width: 100%; }                               7358
/* 只读信息列表 */                                                    7359
.acct-kv {                                                           7360
  display: grid;                                                     7361
  grid-template-columns: 88px 1fr;                                   7362
  gap: 7px 12px;                                                     7363
  margin: 16px 0 0;                                                  7364
  font-size: 13.5px;                                                 7365
}                                                                    7366
.acct-kv dt { color: var(--ink-soft); }                              7367
.acct-kv dd { margin: 0; }                                           7368
.field__error {                                                      7369
  margin: 5px 0 0;                                                   7370
  font-size: 12.5px;                                                 7371
  color: var(--danger);                                              7372
}                                                                    7373
/* 管理员分类（账号 / 操作日志）里的表格面板 */                        7374
.acct-panel {                                                        7375
  background: #fff;                                                  7376
  border: 1px solid #e6e6e8;                                         7377
  border-radius: 14px;                                               7378
  overflow: hidden;                                                  7379
}                                                                    7380
.acct-panel__head {                                                  7381
  display: flex;                                                     7382
  align-items: center;                                               7383
  justify-content: space-between;                                    7384
  gap: 12px;                                                         7385
  padding: 13px 20px;                                                7386
  border-bottom: 1px solid #f0f1f2;                                  7387
}                                                                    7388
.acct-panel__count {                                                 7389
  margin: 0;                                                         7390
  font-size: 13.5px;                                                 7391
  color: #6e6e73;                                                    7392
}                                                                    7393
.acct-panel table { margin: 0; }                                     7394
.acct-panel th,                                                      7395
.acct-panel td { border: 0; border-bottom: 1px solid                 7396
  #f0f1f2; }                                                         7397
.acct-panel thead th { background: #f7f9f8; }                        7398
.acct-panel tbody tr:last-child td { border-bottom: 0; }             7399
.acct-panel tbody tr:hover { background: #fafbfa; }                  7400
```

```
</style>                                                                    7401
<div class="acct">                                                          7402
  {# ---------------- 左栏：身份 + 分类 ---------------- #}                  7403
  <aside class="acct-rail">                                                 7404
    {% if current_user.avatar %}                                            7405
    <img class="acct-rail__avatar"                                          7406
         src="{{ url_for('static', filename=current_user.ava                7407
    tar) }}"                                                                7408
         alt="{{ current_user.shown_name }} 的头像">                        7409
    {% else %}                                                              7410
    <span class="acct-rail__avatar" aria-hidden="true">{{                   7411
    current_user.initial }}</span>                                          7412
    {% endif %}                                                             7413
    <p class="acct-rail__name">{{ current_user.shown_name                   7414
    }}</p>                                                                  7415
    <p class="acct-rail__sub">                                              7416
      {{ role_labels.get(current_user.role,                                 7417
    current_user.role) }}                                                   7418
      {% if current_user.email %} · {{ current_user.email                   7419
    }}{% endif %}                                                           7420
    </p>                                                                    7421
    <nav class="acct-tabs" aria-label="个人设置分类">                       7422
      <a href="{{ url_for('auth.profile', tab='info') }}"                   7423
         {% if tab == 'info' %}class="is-active"                            7424
    aria-current="page"{% endif %}>个人信息</a>                             7425
      <a href="{{ url_for('auth.profile', tab='security')                   7426
    }}"                                                                     7427
         {% if tab == 'security' %}class="is-active"                        7428
    aria-current="page"{% endif %}>登录与安全性</a>                         7429
      {% if current_user.is_admin %}                                        7430
      <a href="{{ url_for('auth.profile', tab='users') }}"                  7431
         {% if tab == 'users' %}class="is-active"                           7432
    aria-current="page"{% endif %}>账号</a>                                 7433
      <a href="{{ url_for('auth.profile', tab='logs') }}"                   7434
         {% if tab == 'logs' %}class="is-active"                            7435
    aria-current="page"{% endif %}>操作日志</a>                             7436
      {% endif %}                                                           7437
    </nav>                                                                  7438
    <details class="acct-avatar-edit">                                      7439
      <summary>更换头像</summary>                                           7440
      <div class="acct-avatar-edit__body">                                  7441
        <form method="post" action="/profile/avatar"                        7442
              enctype="multipart/form-data">                                7443
          {{ avatar_form.hidden_tag() }}                                    7444
          <input type="hidden" name="tab" value="{{ tab }}">                7445
          <label class="visually-hidden"                                    7446
    for="avatar-file">选择头像图片</label>                                  7447
          <input id="avatar-file" type="file" name="avatar"                 7448
                 accept=".png,.jpg,.jpeg,.gif,.webp">                       7449
          <p class="acct-avatar-edit__file">PNG / JPG / GIF                 7450
```

```
    /  WebP，不超过  4  MB</p>                                                  7451
          <button  type="submit"  class="btn                                   7452
    btn--sm">上传头像</button>                                                 7453
        </form>                                                                7454
        {%  if  current_user.avatar  %}                                        7455
        <form  method="post"  action="/profile/avatar/delete"                  7456
              onsubmit="return  confirm('确定移除当前头像吗？');">             7457
          {{  avatar_form.hidden_tag()  }}                                     7458
          <input  type="hidden"  name="tab"  value="{{  tab  }}">              7459
          <button  type="submit"  class="btn  btn--sm                          7460
    btn--ghost">移除头像</button>                                              7461
        </form>                                                                7462
        {%  endif  %}                                                          7463
      </div>                                                                   7464
    </details>                                                                 7465
  </aside>                                                                     7466
  {#  ----------------  右栏：功能卡  ----------------  #}                     7467
  <div  class="acct-main">                                                     7468
    {%  if  tab  ==  'security'  %}                                            7469
    <h1  class="acct-title">登录与安全性</h1>                                  7470
    <p  class="acct-lead">管理与登录账号、账号安全有关的设置。</p>            7471
    <div  class="acct-grid">                                                   7472
      <details  class="acct-card"                                              7473
                {%  if  pwd_form.old_password.errors  or                       7474
    pwd_form.new_password.errors  or  pwd_form.confirm.errors                  7475
    %}open{%  endif  %}>                                                       7476
        <summary>                                                              7477
          <span  class="acct-card__title">密码</span>                          7478
          <span  class="acct-card__value">修改登录密码</span>                  7479
          <span  class="acct-card__icon">{{  icon("lock",  15)                 7480
    }}</span>                                                                  7481
        </summary>                                                             7482
        <div  class="acct-card__body">                                         7483
          <form  method="post">                                                7484
            {{  pwd_form.hidden_tag()  }}                                      7485
            <input  type="hidden"  name="action"                               7486
    value="password">                                                          7487
            <input  type="hidden"  name="tab"                                  7488
    value="security">                                                          7489
            <div  class="field">                                               7490
              <label  class="field__label"  for="{{                            7491
    pwd_form.old_password.id  }}">当前密码</label>                             7492
              {{  pwd_form.old_password(id=pwd_form.old_passw                  7493
    ord.id,                                                                    7494
                                                                               7495
    autocomplete="current-password")  }}                                       7496
              {%  for  e  in  pwd_form.old_password.errors  %}                 7497
              <p  class="field__error">{{  e  }}</p>                           7498
              {%  endfor  %}                                                   7499
            </div>                                                             7500
```

```
          <div class="field">
            <label class="field__label" for="{{
pwd_form.new_password.id }}">新密码</label>
            {{ pwd_form.new_password(id=pwd_form.new_passw
ord.id,

autocomplete="new-password") }}
            {% for e in pwd_form.new_password.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <div class="field">
            <label class="field__label" for="{{
pwd_form.confirm.id }}">确认新密码</label>
            {{ pwd_form.confirm(id=pwd_form.confirm.id,
                                autocomplete="new-password
") }}
            {% for e in pwd_form.confirm.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <p class="actions">{{ pwd_form.submit_pwd(class_
="btn") }}</p>
        </form>
      </div>
    </details>
    <details class="acct-card">
      <summary>
        <span class="acct-card__title">登录账号</span>
        <span class="acct-card__value">{{
current_user.username }}</span>
        <span class="acct-card__icon">{{ icon("user", 15)
}}</span>
      </summary>
      <div class="acct-card__body">
        <dl class="acct-kv">
          <dt>登录账号</dt><dd class="mono">{{
current_user.username }}</dd>
          <dt>职务</dt>
          <dd>{{ role_labels.get(current_user.role,
current_user.role) }}</dd>
          <dt>状态</dt>
          <dd>
            {% if current_user.active %}
            <span class="tag tag--ok">启用</span>
            {% else %}
            <span class="tag tag--warn">停用</span>
            {% endif %}
          </dd>
          <dt>创建时间</dt>
```

```
            <dd class="mono">
              {{ current_user.created_at.strftime("%Y-%m-%d %H:%M")
                 if current_user.created_at else "—" }}
            </dd>
          </dl>
        </div>
      </details>
    </div>
    {% elif tab == 'users' %}
    <h1 class="acct-title">账号</h1>
    <p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p>
    {% include "auth/_users.html" %}
    {% elif tab == 'logs' %}
    <h1 class="acct-title">操作日志</h1>
    <p class="acct-lead">记录各账号在本系统中的关键操作。</p>
    {% include "auth/_logs.html" %}
    {% else %}
    <h1 class="acct-title">个人信息</h1>
    <p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
    <form method="post">
      {{ info_form.hidden_tag() }}
      <input type="hidden" name="action" value="info">
      <input type="hidden" name="tab" value="info">
      <div class="acct-grid">
        <details class="acct-card" {% if info_form.display_name.errors %}open{% endif %}>
          <summary>
            <span class="acct-card__title">昵称</span>
            <span class="acct-card__value">
              {{ current_user.display_name or "未设置" }}
            </span>
            <span class="acct-card__icon">{{ icon("user", 15) }}</span>
          </summary>
          <div class="acct-card__body">
            <div class="field">
              <label class="field__label" for="{{ info_form.display_name.id }}">
                昵称
              </label>
              {{ info_form.display_name(id=info_form.display_name.id,
                                        placeholder="留空则显示账户名称") }}
              {% for e in info_form.display_name.errors %}
              <p class="field__error">{{ e }}</p>
              {% endfor %}
            </div>
```

```
          <p class="actions">{{ info_form.submit_info(clas 7601
s_="btn") }}</p>                                          7602
        </div>                                            7603
      </details>                                          7604
      <details class="acct-card" {% if                    7605
info_form.gender.errors %}open{% endif %}>                7606
        <summary>                                         7607
          <span class="acct-card__title">性别</span>      7608
          <span class="acct-card__value">{{               7609
current_user.gender or "未设置" }}</span>                 7610
          <span class="acct-card__icon">{{ icon("user",   7611
15) }}</span>                                             7612
        </summary>                                        7613
        <div class="acct-card__body">                     7614
          <div class="field">                             7615
            <label class="field__label" for="{{           7616
info_form.gender.id }}">性别</label>                      7617
            {{ info_form.gender(id=info_form.gender.id) }}7618
            {% for e in info_form.gender.errors %}        7619
            <p class="field__error">{{ e }}</p>           7620
            {% endfor %}                                  7621
          </div>                                          7622
          <p class="actions">{{ info_form.submit_info(clas7623
s_="btn") }}</p>                                          7624
        </div>                                            7625
      </details>                                          7626
      <details class="acct-card" {% if                    7627
info_form.email.errors %}open{% endif %}>                 7628
        <summary>                                         7629
          <span class="acct-card__title">邮箱</span>      7630
          <span class="acct-card__value">{{               7631
current_user.email or "未设置" }}</span>                  7632
          <span class="acct-card__icon">{{ icon("mail",   7633
15) }}</span>                                             7634
        </summary>                                        7635
        <div class="acct-card__body">                     7636
          <div class="field">                             7637
            <label class="field__label" for="{{           7638
info_form.email.id }}">邮箱</label>                       7639
            {{ info_form.email(id=info_form.email.id) }}  7640
            {% for e in info_form.email.errors %}         7641
            <p class="field__error">{{ e }}</p>           7642
            {% endfor %}                                  7643
          </div>                                          7644
          <p class="actions">{{ info_form.submit_info(clas7645
s_="btn") }}</p>                                          7646
        </div>                                            7647
      </details>                                          7648
      <details class="acct-card" {% if                    7649
info_form.phone.errors %}open{% endif %}>                 7650
```

```
            <summary>                                                    7651
              <span class="acct-card__title">手机号</span>               7652
              <span class="acct-card__value">{{                          7653
    current_user.phone or "未设置" }}</span>                             7654
              <span class="acct-card__icon">{{ icon("phone",             7655
    15) }}</span>                                                        7656
            </summary>                                                   7657
            <div class="acct-card__body">                                7658
              <div class="field">                                        7659
                <label class="field__label" for="{{                      7660
    info_form.phone.id }}">手机号</label>                                7661
                {{ info_form.phone(id=info_form.phone.id) }}             7662
                {% for e in info_form.phone.errors %}                    7663
                <p class="field__error">{{ e }}</p>                      7664
                {% endfor %}                                             7665
              </div>                                                     7666
              <p class="actions">{{ info_form.submit_info(clas           7667
    s_="btn") }}</p>                                                     7668
            </div>                                                       7669
          </details>                                                     7670
        </div>                                                           7671
    </form>                                                              7672
    {% endif %}                                                          7673
  </div>                                                                 7674
</div>                                                                   7675
{% endblock %}                                                           7676
{% block scripts %}                                                      7677
<script>                                                                 7678
// 同时只展开一张卡，避免点开多张后页面被撑得很长                                  7679
(function () {                                                           7680
  var cards = document.querySelectorAll('.acct-card');                   7681
  Array.prototype.forEach.call(cards, function (card) {                  7682
    card.addEventListener('toggle', function () {                        7683
      if (!card.open) return;                                            7684
      Array.prototype.forEach.call(cards, function (other) {             7685
        if (other !== card && other.open) other.open =                   7686
    false;                                                               7687
      });                                                                7688
    });                                                                  7689
  });                                                                    7690
})();                                                                    7691
</script>                                                                7692
{% endblock %}                                                           7693
# ==== app/templates/auth/_users.html ====                               7694
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}                7695
<div class="acct-panel">                                                 7696
  <div class="acct-panel__head">                                         7697
    <p class="acct-panel__count">共 {{ users|length }}                   7698
    个账号</p>                                                           7699
    <a class="btn btn--sm" href="{{ url_for('auth.add_user')             7700
```

```
    }}">新增账号</a>                                                    7701
</div>                                                                  7702
<table>                                                                 7703
  <thead>                                                               7704
    <tr>                                                                7705
      <th scope="col">ID</th>                                           7706
      <th scope="col">账户名称</th>                                     7707
      <th scope="col">昵称</th>                                         7708
      <th scope="col">职务</th>                                         7709
      <th scope="col">状态</th>                                         7710
      <th scope="col">操作</th>                                         7711
    </tr>                                                               7712
  </thead>                                                              7713
  <tbody>                                                               7714
    {% for u in users %}                                                7715
    <tr>                                                                7716
      <td class="num">{{ u.id }}</td>                                   7717
      <td class="mono">                                                 7718
        {{ u.username }}                                                7719
        {% if u.id == current_user.id %}<span                           7720
  class="muted">（当前登录）</span>{% endif %}                          7721
      </td>                                                             7722
      <td style="text-align:left;">                                     7723
        {% if u.display_name %}{{ u.display_name }}                     7724
        {% else %}<span class="muted">—</span>{% endif %}               7725
      </td>                                                             7726
      <td>                                                              7727
        <span class="tag {{ 'tag--ok' if u.role ==                      7728
  'admin' else 'tag--muted' }}">                                        7729
          {{ role_labels.get(u.role, u.role) }}                         7730
        </span>                                                         7731
      </td>                                                             7732
      <td>                                                              7733
        {% if u.active %}<span class="tag                               7734
  tag--ok">启用</span>                                                  7735
        {% else %}<span class="tag tag--muted">停用</span>{%            7736
  endif %}                                                              7737
      </td>                                                             7738
      <td class="nowrap">                                               7739
        <span class="row-actions">                                      7740
          <a class="link-btn" href="{{                                  7741
  url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                  7742
          <form method="post"                                           7743
                action="{{ url_for('auth.delete_user',                  7744
  user_id=u.id) }}"                                                     7745
                onsubmit="return confirm('确定删除账号「{{              7746
  u.username }}」吗？\n该操作无法恢复。');">                            7747
            {{ form.hidden_tag() }}                                     7748
            <button type="submit" class="link-btn                       7749
  link-btn--danger"                                                     7750
```

```
                                {% if u.id == current_user.id
    %}disabled
                                title="不能删除当前登录的账号"{% endif
    %}>删除</button>
                </form>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
</div>
# ==== app/templates/auth/_logs.html ====
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">
      共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页
    </p>
    {% if pages > 1 %}
    <nav class="row-actions" aria-label="日志分页">
      {% if page > 1 %}
      <a class="link-btn"
         href="{{ url_for('auth.profile', tab='logs',
    page=page - 1) }}">上一页</a>
      {% endif %}
      {% if page < pages %}
      <a class="link-btn"
         href="{{ url_for('auth.profile', tab='logs',
    page=page + 1) }}">下一页</a>
      {% endif %}
    </nav>
    {% endif %}
  </div>
  {% if not records %}
  <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">时间</th>
        <th scope="col">操作人</th>
        <th scope="col">操作</th>
        <th scope="col">对象</th>
        <th scope="col">说明</th>
      </tr>
    </thead>
    <tbody>
      {% for r in records %}
      <tr>
```

```
            <td class="mono nowrap">{{ r.created_at.strftime("%Y
      -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>
            <td>{{ users_map.get(r.user_id, "—") }}</td>
            <td style="text-align:left;">{{ r.action }}</td>
            <td>
              {% if r.target_type %}{{ r.target_type }}{% if
      r.target_id %}#{{ r.target_id }}{% endif %}
              {% else %}—{% endif %}
            </td>
            <td style="text-align:left;">{{ r.detail or "—"
      }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
  <h2>项目与矿山档案</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/add">新建项目</a>
  </p>
  {% if not projects %}
  <p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  <table>
    <caption class="muted">共 {{ projects|length }}
    个项目</caption>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
```

```
            <td><span class="tag tag--muted">{{ p.status                    7851
        }}</span></td>                                                      7852
            <td class="num">                                                7853
              {% if p.patches.count() %}                                    7854
                {{ p.patches.count() }}                                     7855
              {% else %}                                                    7856
                <span class="tag tag--warn">0</span>                        7857
              {% endif %}                                                   7858
            </td>                                                           7859
            <td class="nowrap">                                             7860
              <span class="row-actions">                                    7861
                <a class="link-btn" href="/project/{{ p.id                  7862
        }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+           7863
        新建图斑{% endif %}</a>                                             7864
                <a class="link-btn" href="/project/{{ p.id                  7865
        }}/edit">编辑</a>                                                   7866
                <form method="post" action="/project/{{ p.id                7867
        }}/delete"                                                          7868
                      onsubmit="return confirm('确定删除项目「{{            7869
        p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  7870
                  {{ form.hidden_tag() }}                                   7871
                  <button type="submit" class="link-btn                     7872
        link-btn--danger">删除</button>                                     7873
                </form>                                                     7874
              </span>                                                       7875
            </td>                                                           7876
          </tr>                                                             7877
          {% endfor %}                                                      7878
        </tbody>                                                            7879
      </table>                                                              7880
      {% endif %}                                                           7881
</div>                                                                      7882
{% endblock %}                                                              7883
# ==== app/templates/project/edit.html ====                                 7884
{% extends "base.html" %}                                                   7885
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%              7886
      endblock %}                                                           7887
{% block content %}                                                         7888
<div class="card" style="max-width:640px;">                                 7889
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                      7890
  <form method="post">                                                      7891
    {{ form.hidden_tag() }}                                                 7892
    <div class="field">                                                     7893
      <label class="field__label" for="name">项目名称</label>               7894
      <input id="name" type="text" name="name"                              7895
    style="width:100%;" required                                            7896
             value="{{ project.name if project else '' }}">                 7897
    </div>                                                                  7898
    <div class="field">                                                     7899
      <label class="field__label" for="code">项目编号</label>               7900
```

```
        <input id="code" type="text" name="code"                    7901
      style="width:240px;"                                          7902
             value="{{ project.code or '' if project else           7903
      '' }}">                                                       7904
      </div>                                                        7905
      {% if project %}                                              7906
      <div class="field">                                           7907
        <label class="field__label" for="status">状态</label>       7908
        <select id="status" name="status">                          7909
          {% for s in ["进行中", "已完成", "已归档"] %}             7910
          <option value="{{ s }}" {% if project.status == s         7911
      %}selected{% endif %}>{{ s }}</option>                        7912
          {% endfor %}                                              7913
        </select>                                                   7914
      </div>                                                        7915
      {% endif %}                                                   7916
      <div class="field">                                           7917
        <label class="field__label" for="description">项目说明</la 7918
      bel>                                                          7919
        <textarea id="description" name="description">{{          7920
      project.description or '' if project else ''                  7921
      }}</textarea>                                                 7922
      </div>                                                        7923
      <p class="actions" style="margin-top:18px;">                  7924
        <button type="submit" class="btn">保存</button>             7925
        <a class="btn btn--ghost" href="/project/">返回</a>         7926
      </p>                                                          7927
    </form>                                                         7928
</div>                                                              7929
{% endblock %}                                                      7930
# ==== app/templates/project/patches.html ====                      7931
{% extends "base.html" %}                                           7932
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}        7933
{% block content %}                                                 7934
<div class="card">                                                  7935
  <h2>矿山图斑 · {{ project.name }}</h2>                            7936
  <p class="actions" style="margin-bottom:12px;">                   7937
    <a class="btn btn--sm" href="/project/{{ project.id             7938
    }}/patch/add">新建图斑</a>                                      7939
    <a class="btn btn--sm btn--ghost"                               7940
    href="/project/">返回项目列表</a>                               7941
  </p>                                                              7942
  {% if not project.patches.count() %}                              7943
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>  7944
  {% else %}                                                        7945
  <table>                                                           7946
    <caption class="muted">共 {{ project.patches.count() }}         7947
    个图斑</caption>                                                7948
    <thead>                                                         7949
      <tr>                                                          7950
```

```
            <th scope="col">图斑名称</th>
            <th scope="col">行政区</th>
            <th scope="col">面积（公顷）</th>
            <th scope="col">开采矿种</th>
            <th scope="col">微地质单元</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
          {% for p in project.patches %}
          <tr>
            <td style="text-align:left;">{{ p.name }}</td>
            <td>{{ p.admin_region or "—" }}</td>
            <td class="num">{{ p.area_ha if p.area_ha is not none else "—" }}</td>
            <td>{{ p.mineral_type or "—" }}</td>
            <td class="num">{{ p.units.count() }}</td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/unit/{{ p.id }}">微地质单元</a>
                <a class="link-btn" href="/constraint/{{ p.id }}">限制性判定</a>
                <a class="link-btn" href="/project/patch/{{ p.id }}/edit">编辑</a>
                <form method="post" action="/project/patch/{{ p.id }}/delete"
                      onsubmit="return confirm('确定删除图斑「{{ p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
                  {{ form.hidden_tag() }}
                  <button type="submit" class="link-btn link-btn--danger">删除</button>
                </form>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
    </div>
    {% endblock %}
    # ==== app/templates/project/patch_edit.html ====
    {% extends "base.html" %}
    {% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{% endblock %}
    {% block content %}
    <div class="card" style="max-width:720px;">
      <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name }}</h2>
```

```
<form method="post">
  {{ form.hidden_tag() }}
  <div class="field">
    <label class="field__label" for="patch-name">图斑名称</lab
  el>
    <input id="patch-name" type="text" name="name"
  style="width:100%;" required
           value="{{ patch.name if patch else '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="admin-region">所在行政区</
  label>
    <input id="admin-region" type="text"
  name="admin_region" style="width:100%;"
           value="{{ patch.admin_region or '' if patch
  else '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="area-ha">图斑面积（公顷）</la
  bel>
    <input id="area-ha" type="number" step="any"
  name="area_ha"
           value="{{ patch.area_ha if patch and
  patch.area_ha is not none else '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="mineral-type">开采矿种</l
  abel>
    <input id="mineral-type" type="text"
  name="mineral_type" style="width:240px;"
           value="{{ patch.mineral_type or '' if patch
  else '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="mining-method">开采方式</
  label>
    <input id="mining-method" type="text"
  name="mining_method" style="width:240px;"
           value="{{ patch.mining_method or '' if patch
  else '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="stop-years">停产年限</lab
  el>
    <input id="stop-years" type="number" step="any"
  name="stop_years"
           value="{{ patch.stop_years if patch and
  patch.stop_years is not none else '' }}">
  </div>
  <div class="field">
```

```
        <label class="checkbox" for="responsible-missing">
          <input id="responsible-missing" type="checkbox"
    name="responsible_entity_missing" value="1"
                 {% if not patch or patch.responsible_entity_m
    issing %}checked{% endif %}>
          <span>责任主体缺失</span>
        </label>
      </div>
      <h3>参照生态系统（指标标准化基准）</h3>
      <p class="muted">
        论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。
      </p>
      <div class="field">
        <label class="field__label" for="coverage-benchmark">植
    被覆盖度基准（%）</label>
        <input id="coverage-benchmark" type="number"
    step="any" name="coverage_benchmark"
               value="{{ reference.coverage_benchmark if
    reference else 75 }}">
      </div>
      <div class="field">
        <label class="field__label" for="shannon-benchmark">Sh
    annon-Wiener 多样性指数基准</label>
        <input id="shannon-benchmark" type="number"
    step="any" name="shannon_benchmark"
               value="{{ reference.shannon_benchmark if
    reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种
    </label>
        <input id="dominant-species" type="text"
    name="dominant_species" style="width:100%;"
               value="{{ reference.dominant_species or '' if
    reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
    project.id }}/patches">返回</a>
      </p>
    </form>
  </div>
  {% if patch %}
  <div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
```

```
</p>
{% if patch.image_path %}
<figure style="margin:0 0 16px;">
  <img src="{{ url_for('static', filename=patch.image_path) }}"
       alt="图斑「{{ patch.name }}」已挂接的遥感影像"
       style="max-width:100%;border:1px solid var(--line);border-radius:6px;">
  <figcaption class="muted" style="margin-top:6px;">
    {{ patch.image_note or "遥感影像" }}
    {% if patch.image_uploaded_at %}
    · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d %H:%M") }}
    {% endif %}
  </figcaption>
</figure>
<form method="post" action="/project/patch/{{ patch.id }}/image/delete"
      onsubmit="return confirm('确定移除该遥感影像吗？');">
  {{ form.hidden_tag() }}
  <button type="submit" class="btn btn--sm btn--danger">移除影像</button>
</form>
<hr style="margin:16px 0;border:none;border-top:1px solid var(--line);">
{% endif %}
<form method="post" action="/project/patch/{{ patch.id }}/image"
      enctype="multipart/form-data">
  {{ form.hidden_tag() }}
  <div class="field">
    <label class="field__label" for="image-file">
      {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
    </label>
    <input id="image-file" type="file" name="image"
           accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.tiff" required>
  </div>
  <div class="field">
    <label class="field__label" for="image-note">影像说明（如来源、时相）</label>
    <input id="image-note" type="text" name="image_note"
           style="width:100%;"
           placeholder="如：2025 年 6 月无人机正射影像"
           value="{{ patch.image_note or '' }}">
  </div>
  <p class="actions">
    <button type="submit" class="btn">{{ "替换影像" if patch.image_path else "上传并挂接" }}</button>
  </p>
```

```
  </form>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else
"—" }}</td>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
          {% if r.count %}
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}
            {% if r.ratio_sum < 0.999 or r.ratio_sum >
1.001 %}
              <span class="warn">（不等于 100%）</span>
            {% endif %}
          {% else %}—{% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ r.patch.id
}}">划分与维护</a>
            <a class="link-btn" href="/indicator/">录入指标数据</a
>
          </span>
```

```
          </td>                                                                    8201
        </tr>                                                                      8202
        {% endfor %}                                                               8203
      </tbody>                                                                     8204
    </table>                                                                       8205
    {% endif %}                                                                    8206
</div>                                                                             8207
{% endblock %}                                                                     8208
# ==== app/templates/unit/index.html ====                                          8209
{% extends "base.html" %}                                                          8210
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}                       8211
{% block content %}                                                                8212
<div class="card">                                                                 8213
  <h2>微地质单元 · {{ patch.name }}</h2>                                           8214
  {% if units %}                                                                   8215
  <table>                                                                          8216
    <caption class="muted">                                                        8217
      共 {{ units|length }} 个单元；                                               8218
      面积合计 <span class="num">{{ "%.4f"|format(total_area)                      8219
    }}</span>；                                                                    8220
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *                    8221
    100) }}</span>                                                                 8222
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >                        8223
    1.001) %}                                                                      8224
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>            8225
      {% endif %}                                                                  8226
    </caption>                                                                     8227
    <thead>                                                                        8228
      <tr>                                                                         8229
        <th scope="col">微地质单元</th>                                            8230
        <th scope="col">面积</th>                                                  8231
        <th scope="col">面积占比</th>                                              8232
        <th scope="col">说明</th>                                                  8233
        <th scope="col">操作</th>                                                  8234
      </tr>                                                                        8235
    </thead>                                                                       8236
    <tbody>                                                                        8237
      {% for u in units %}                                                         8238
      <tr>                                                                         8239
        <td>{{ u.unit_type }}</td>                                                 8240
        <td class="num">{{ u.area if u.area is not none                            8241
    else "—" }}</td>                                                               8242
        <td class="num">                                                           8243
          {% if u.ratio is not none %}{{                                           8244
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                        8245
        </td>                                                                      8246
        <td style="text-align:left;">{{ u.description or                           8247
    "—" }}</td>                                                                    8248
        <td class="nowrap">                                                        8249
          <span class="row-actions">                                               8250
```

```
                <a class="link-btn" href="/unit/edit/{{ u.id }}">编辑</a>
                <form method="post" action="/unit/delete/{{ u.id }}"
                      onsubmit="return confirm('确定删除微地质单元「{{ u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
                  {{ form.hidden_tag() }}
                  <button type="submit" class="link-btn link-btn--danger">删除</button>
                </form>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% else %}
      <p class="muted">尚未划分微地质单元。请在下方添加。</p>
      {% endif %}
      <h3>添加微地质单元</h3>
      <form method="post" action="/unit/{{ patch.id }}/add">
        {{ form.hidden_tag() }}
        <div class="actions">
          <label class="field__label" for="add-unit-type" style="margin:0;">类型</label>
          <select id="add-unit-type" name="unit_type">
            {% for t in unit_types %}<option value="{{ t }}">{{ t }}</option>{% endfor %}
          </select>
          <label class="field__label" for="add-unit-area" style="margin:0;">面积</label>
          <input id="add-unit-area" type="number" step="any" name="area">
          <label class="field__label" for="add-unit-desc" style="margin:0;">说明</label>
          <input id="add-unit-desc" type="text" name="description" style="width:220px;">
          <button type="submit" class="btn btn--sm">添加</button>
        </div>
      </form>
      <p class="actions" style="margin-top:20px;">
        <a class="btn btn--sm btn--ghost" href="/project/{{ patch.project_id }}/patches">返回图斑列表</a>
        <a class="btn btn--sm btn--ghost" href="/indicator/">去录入指标数据</a>
      </p>
    </div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
```

```
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>编辑微地质单元 · {{ patch.name }}</h2>
  <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type">
        {% for t in unit_types %}
        <option value="{{ t }}" {% if unit.unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="unit-area">面积</label>
      <input id="unit-area" type="number" step="any" name="area"
             value="{{ unit.area if unit.area is not none else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit-desc">说明</label>
      <input id="unit-desc" type="text" name="description" style="width:420px;"
             value="{{ unit.description or '' }}">
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/unit/{{ unit.patch_id }}">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost" href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
```

```
{% if local %}
<table>
  <thead>
    <tr>
      <th scope="col">省份</th>
      <th scope="col">名称</th>
      <th scope="col">编号</th>
      <th scope="col">规则数</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for s in local %}
    <tr>
      <td>{{ s.province }}</td>
      <td style="text-align:left;">{{ s.name }}</td>
      <td class="mono">{{ s.doc_number or "—" }}</td>
      <td class="num">{{ s.rules|length }}</td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/standard/{{ s.id }}/rules">规则</a>
          <a class="link-btn" href="/standard/{{ s.id }}/edit">编辑</a>
          <form method="post" action="/standard/{{ s.id }}/delete"
                onsubmit="return confirm('确定删除「{{ s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length }} 条规则将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="link-btn link-btn--danger">删除</button>
          </form>
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% else %}
<p class="muted">暂无地方标准。运行 <span class="mono">python seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
{% endif %}
<h3>政策文件</h3>
{% if policy %}
<table>
  <thead>
    <tr>
      <th scope="col">省份</th>
      <th scope="col">名称</th>
```

```
            <th scope="col">文号</th>                                    8401
            <th scope="col">规则数</th>                                  8402
            <th scope="col">操作</th>                                    8403
          </tr>                                                          8404
        </thead>                                                         8405
        <tbody>                                                          8406
          {% for s in policy %}                                          8407
          <tr>                                                           8408
            <td>{{ s.province }}</td>                                    8409
            <td style="text-align:left;">{{ s.name }}</td>               8410
            <td class="mono">{{ s.doc_number or "—" }}</td>              8411
            <td class="num">{{ s.rules|length }}</td>                    8412
            <td class="nowrap">                                          8413
              <span class="row-actions">                                 8414
                <a class="link-btn" href="/standard/{{ s.id              8415
        }}/rules">规则</a>                                               8416
                <a class="link-btn" href="/standard/{{ s.id              8417
        }}/edit">编辑</a>                                               8418
                <form method="post" action="/standard/{{ s.id            8419
        }}/delete"                                                       8420
                      onsubmit="return confirm('确定删除「{{             8421
        s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length   8422
        }} 条规则将一并删除，且无法恢复。');">                           8423
                  {{ form.hidden_tag() }}                                8424
                  <button type="submit" class="link-btn                  8425
        link-btn--danger">删除</button>                                  8426
                </form>                                                  8427
              </span>                                                    8428
            </td>                                                        8429
          </tr>                                                          8430
          {% endfor %}                                                   8431
        </tbody>                                                         8432
      </table>                                                           8433
      {% else %}                                                         8434
      <p class="muted">暂无政策文件。</p>                                8435
      {% endif %}                                                        8436
</div>                                                                   8437
{% endblock %}                                                           8438
# ==== app/templates/standard/edit.html ====                             8439
{% extends "base.html" %}                                                8440
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%          8441
    endblock %}                                                          8442
{% block content %}                                                      8443
<div class="card" style="max-width:800px;">                              8444
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                  8445
  <form method="post">                                                   8446
    {{ form.hidden_tag() }}                                              8447
    <div class="field">                                                  8448
      <label class="field__label" for="province">省份</label>            8449
      <input id="province" type="text" name="province"                   8450
```

```
style="width:160px;" required                                    8451
          value="{{ standard.province if standard else           8452
'' }}">                                                          8453
</div>                                                           8454
<div class="field">                                              8455
  <label class="field__label" for="std-name">名称</label>        8456
  <input id="std-name" type="text" name="name"                   8457
style="width:100%;" required                                     8458
          value="{{ standard.name if standard else ''            8459
}}">                                                             8460
</div>                                                           8461
<div class="field">                                              8462
  <label class="field__label" for="doc-number">标准编号 /        8463
文号</label>                                                     8464
  <input id="doc-number" type="text" name="doc_number"           8465
style="width:320px;"                                             8466
          value="{{ standard.doc_number or '' if                 8467
standard else '' }}">                                            8468
</div>                                                           8469
<div class="field">                                              8470
  <label class="field__label" for="category">类别</label>        8471
  <select id="category" name="category">                         8472
    {% for c in ["地方标准", "政策文件"] %}                      8473
    <option value="{{ c }}" {% if standard and                   8474
standard.category == c %}selected{% endif %}>{{ c                8475
}}</option>                                                      8476
    {% endfor %}                                                 8477
  </select>                                                      8478
</div>                                                           8479
<div class="field">                                              8480
  <label class="field__label" for="publish-date">发布时间</l     8481
abel>                                                            8482
  <input id="publish-date" type="text"                           8483
name="publish_date" style="width:200px;"                         8484
          placeholder="如 2023-03-30"                            8485
          value="{{ standard.publish_date or '' if               8486
standard else '' }}">                                            8487
</div>                                                           8488
<div class="field">                                              8489
  <label class="field__label" for="implement-date">实施时间<     8490
/label>                                                          8491
  <input id="implement-date" type="text"                         8492
name="implement_date" style="width:200px;"                       8493
          placeholder="如 2023-04-30（未明确可留空）"            8494
          value="{{ standard.implement_date or '' if             8495
standard else '' }}">                                            8496
</div>                                                           8497
<div class="field">                                              8498
  <label class="field__label" for="summary">特点 /               8499
判定标准</label>                                                 8500
```

```
        <textarea id="summary" name="summary"                          8501
      style="min-height:96px;">{{ standard.summary or '' if            8502
      standard else '' }}</textarea>                                   8503
      </div>                                                           8504
      <p class="actions" style="margin-top:18px;">                     8505
        <button type="submit" class="btn">保存</button>                8506
        {% if standard %}                                              8507
        <form method="post" action="/standard/{{ standard.id           8508
      }}/delete" style="display:inline;"                               8509
              onsubmit="return confirm('确定删除「{{                    8510
      standard.province }} {{ standard.name                            8511
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">                  8512
          {{ form.hidden_tag() }}                                      8513
          <button type="submit" class="btn                             8514
      btn--danger">删除该标准</button>                                   8515
        </form>                                                        8516
        {% endif %}                                                    8517
        <a class="btn btn--ghost" href="/standard/">返回</a>           8518
      </p>                                                             8519
    </form>                                                            8520
</div>                                                                 8521
{% endblock %}                                                         8522
# ==== app/templates/standard/rules.html ====                          8523
{% extends "base.html" %}                                              8524
{% block title %}结构化规则 · {{ standard.province }}{% endblock       8525
    %}                                                                 8526
{% block content %}                                                    8527
<div class="card">                                                     8528
    <h2>结构化规则 · {{ standard.province }} {{ standard.name          8529
      }}</h2>                                                          8530
    {% if rules %}                                                     8531
    <table>                                                            8532
      <caption class="muted">共 {{ rules|length }}                     8533
      条规则</caption>                                                 8534
      <thead>                                                          8535
        <tr>                                                           8536
          <th scope="col">判定项</th>                                  8537
          <th scope="col">条件</th>                                    8538
          <th scope="col">条件类型</th>                                8539
          <th scope="col">适用分区</th>                                8540
          <th scope="col">备注</th>                                    8541
          <th scope="col">操作</th>                                    8542
        </tr>                                                          8543
      </thead>                                                         8544
      <tbody>                                                          8545
        {% for r in rules %}                                           8546
        <tr>                                                           8547
          <td style="text-align:left;">{{ r.item }}</td>               8548
          <td class="mono nowrap">{{ r.operator }} {{                  8549
      r.threshold or "" }}</td>                                        8550
```

```
        <td>{{ r.condition_type }}</td>                                    8551
        <td>{{ r.region or "—" }}</td>                                     8552
        <td style="text-align:left;">{{ r.note or "—"                      8553
  }}</td>                                                                  8554
        <td>                                                               8555
          <form method="post" action="/standard/rule/{{                    8556
  r.id }}/delete"                                                          8557
                onsubmit="return confirm('确定删除规则「{{ r.item          8558
  }} {{ r.operator }} {{ r.threshold or '' }}」吗?');">                    8559
            {{ form.hidden_tag() }}                                        8560
            <button type="submit" class="btn btn--sm                       8561
  btn--danger">删除</button>                                               8562
          </form>                                                          8563
        </td>                                                              8564
      </tr>                                                                8565
      {% endfor %}                                                         8566
    </tbody>                                                               8567
  </table>                                                                 8568
  {% else %}                                                               8569
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>                8570
  {% endif %}                                                              8571
  <h3>添加规则</h3>                                                        8572
  <form method="post">                                                     8573
    {{ form.hidden_tag() }}                                                8574
    <div class="actions">                                                  8575
      <div>                                                                8576
        <label class="field__label" for="r-item">判定项</label>            8577
        <input id="r-item" type="text" name="item"                         8578
  placeholder="如：植被覆盖度" required>                                   8579
      </div>                                                               8580
      <div>                                                                8581
        <label class="field__label" for="r-op">比较符</label>              8582
        <select id="r-op" name="operator">                                 8583
          {% for o in [">=", ">", "<=", "<", "=="]                         8584
  %}<option value="{{ o }}">{{ o }}</option>{% endfor %}                   8585
        </select>                                                          8586
      </div>                                                               8587
      <div>                                                                8588
        <label class="field__label" for="r-thr">阈值</label>               8589
        <input id="r-thr" type="text" name="threshold"                     8590
  placeholder="如：50" style="width:110px;">                               8591
      </div>                                                               8592
      <div>                                                                8593
        <label class="field__label" for="r-cond">条件类型</label           8594
  >                                                                        8595
        <select id="r-cond" name="condition_type">                         8596
          <option value="必要条件">必要条件</option>                       8597
          <option value="充分条件">充分条件</option>                       8598
        </select>                                                          8599
      </div>                                                               8600
```

```
        <div>
          <label class="field__label" for="r-region">适用分区</lab
    el>
          <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
    style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
    <h2>省际口径比对</h2>
    {% if not groups %}
    <p class="muted">
      暂无规则数据。请先在「标准知识库」中为标准录入规则，
      或运行 <span class="mono">python seed_demo.py</span> 载入论文表
      1 数据。
    </p>
    {% else %}
    {% for item, rs in groups.items() %}
    <h3>{{ item }}</h3>
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">标准 / 文件</th>
          <th scope="col">条件</th>
          <th scope="col">条件类型</th>
          <th scope="col">备注</th>
        </tr>
      </thead>
      <tbody>
```

```
        {% for r in rs %}                                                    8651
        <tr>                                                                 8652
          <td>{{ r.standard.province }}</td>                                 8653
          <td style="text-align:left;">{{ r.standard.name                    8654
      }}</td>                                                                8655
          <td class="mono nowrap">{{ r.operator }} {{                        8656
      r.threshold or "" }}</td>                                              8657
          <td>{{ r.condition_type }}</td>                                    8658
          <td style="text-align:left;">{{ r.note or "—"                      8659
      }}</td>                                                                8660
        </tr>                                                                8661
        {% endfor %}                                                         8662
      </tbody>                                                               8663
    </table>                                                                 8664
    {% endfor %}                                                             8665
    {% endif %}                                                              8666
    <p class="actions" style="margin-top:16px;">                             8667
      <a class="btn btn--ghost" href="/standard/">返回标准列表</a>             8668
    </p>                                                                     8669
</div>                                                                       8670
{% endblock %}                                                               8671
# ==== app/templates/constraint/index.html ====                              8672
{% extends "base.html" %}                                                    8673
{% from "_macros.html" import patch_guide %}                                 8674
{% block title %}限制性指标判定{% endblock %}                                8675
{% block content %}                                                          8676
<div class="card">                                                           8677
    <h2>限制性指标判定</h2>                                                  8678
    {{ patch_guide(projects_without_patch) }}                                8679
    {% if not patches %}                                                     8680
    <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>  8681
    {% else %}                                                               8682
    <table>                                                                  8683
      <thead>                                                                8684
        <tr>                                                                 8685
          <th scope="col">所属项目</th>                                      8686
          <th scope="col">矿山图斑</th>                                      8687
          <th scope="col">判定结果</th>                                      8688
          <th scope="col">操作</th>                                          8689
        </tr>                                                                8690
      </thead>                                                               8691
      <tbody>                                                                8692
        {% for p in patches %}                                               8693
        {% set r = latest.get(p.id) %}                                       8694
        <tr>                                                                 8695
          <td>{{ p.project.name if p.project else "—" }}</td>                8696
          <td style="text-align:left;">{{ p.name }}</td>                     8697
          <td style="text-align:left;">                                      8698
            {% if r %}                                                       8699
              {% if r.passed %}                                              8700
```

```
                <span class="tag tag--ok">通过</span>                    8701
                {% else %}                                               8702
                <span class="tag tag--warn">不通过</span>                8703
                {% endif %}                                              8704
                <span class="muted">{{ r.reason }}</span>                8705
              {% else %}                                                 8706
              <span class="tag tag--muted">未判定</span>                 8707
              {% endif %}                                                8708
            </td>                                                        8709
            <td class="nowrap">                                          8710
              <span class="row-actions">                                 8711
                <a class="link-btn" href="/constraint/{{ p.id            8712
    }}">执行判定</a>                                                     8713
              </span>                                                    8714
            </td>                                                        8715
          </tr>                                                          8716
          {% endfor %}                                                   8717
        </tbody>                                                         8718
      </table>                                                           8719
      {% endif %}                                                        8720
</div>                                                                   8721
{% endblock %}                                                           8722
# ==== app/templates/constraint/check.html ====                          8723
{% extends "base.html" %}                                                8724
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}          8725
{% block content %}                                                      8726
<div class="card" style="max-width:760px;">                              8727
  <h2>限制性条件判定 · {{ patch.name }}</h2>                             8728
  <form method="post">                                                   8729
    {{ form.hidden_tag() }}                                              8730
    {% for field, name, val in field_states %}                           8731
    <fieldset style="border:1px solid                                    8732
    var(--line);border-radius:6px;padding:10px                           8733
    14px;margin:0 0 10px;">                                              8734
      <legend style="padding:0 6px;font-weight:600;">{{                  8735
    name }}</legend>                                                     8736
      <div class="actions">                                              8737
        <label class="checkbox">                                         8738
          <input type="radio" name="{{ field }}" value="1"               8739
                 {% if val is none or val %}checked{% endif              8740
    %}>                                                                  8741
          <span>通过</span>                                              8742
        </label>                                                         8743
        <label class="checkbox">                                         8744
          <input type="radio" name="{{ field }}" value="0"               8745
                 {% if val is not none and not val                       8746
    %}checked{% endif %}>                                                8747
          <span>不通过</span>                                            8748
        </label>                                                         8749
      </div>                                                             8750
```

```
      </fieldset>                                                    8751
      {% endfor %}                                                   8752
      <p class="actions" style="margin-top:16px;">                   8753
        <button type="submit" class="btn">保存判定</button>          8754
        <a class="btn btn--ghost" href="/constraint/">返回</a>       8755
      </p>                                                           8756
    </form>                                                          8757
    {% if record %}                                                  8758
    <div style="margin-top:18px;padding:14px;border-radius:6px 8759
      ;border-left:4px solid;                                        8760
                  {% if record.passed %}background:var(--ok-soft 8761
      );border-left-color:var(--ok);                                 8762
                  {% else %}background:var(--danger-soft);border 8763
      -left-color:var(--danger);{% endif %}">                        8764
      <p style="margin:0 0 4px;font-weight:600;">                    8765
        判定结论：                                                   8766
        {% if record.passed %}                                       8767
        <span class="ok">通过，可进入定量评价</span>                 8768
        {% else %}                                                   8769
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>   8770
        {% endif %}                                                  8771
      </p>                                                           8772
      <p class="muted" style="margin:0;">{{ record.reason            8773
      }}</p>                                                         8774
    </div>                                                           8775
    {% endif %}                                                      8776
</div>                                                               8777
{% endblock %}                                                       8778
# ==== app/templates/ahp/index.html ====                             8779
{% extends "base.html" %}                                            8780
{% block title %}AHP 权重计算{% endblock %}                          8781
{% block content %}                                                  8782
<div class="card" style="max-width:720px;">                          8783
    <h2>AHP 权重计算 · 选择指标层级</h2>                            8784
    <form method="post">                                             8785
      {{ form.hidden_tag() }}                                        8786
      <div class="field">                                            8787
        <label class="field__label" for="{{ form.unit_type.id 8788
      }}">微地质单元类型</label>                                     8789
        {{ form.unit_type(id=form.unit_type.id) }}                   8790
      </div>                                                         8791
      <div class="field">                                            8792
        <label class="field__label" for="{{ form.level.id            8793
      }}">指标层级</label>                                           8794
        {{ form.level(id=form.level.id) }}                           8795
      </div>                                                         8796
      <div class="field">                                            8797
        <label class="field__label" for="{{ form.l1.id }}">          8798
          一级指标（层级选「二级指标」时需选择）                     8799
        </label>                                                     8800
```

```
        {{ form.l1(id=form.l1.id) }}                                    8801
      </div>                                                            8802
      <p class="actions" style="margin-top:16px;">                      8803
        {{ form.submit(class_="btn") }}                                 8804
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维 8805
      护</a>                                                            8806
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a 8807
      >                                                                 8808
      </p>                                                              8809
    </form>                                                             8810
</div>                                                                  8811
<div class="card">                                                      8812
    <h2>各微地质单元当前生效权重</h2>                                   8813
    <table>                                                             8814
      <thead>                                                           8815
        <tr>                                                            8816
          <th scope="col">微地质单元</th>                               8817
          <th scope="col">当前权重来源</th>                             8818
          <th scope="col">操作</th>                                     8819
        </tr>                                                           8820
      </thead>                                                          8821
      <tbody>                                                           8822
        {% for r in weight_rows %}                                      8823
        <tr>                                                            8824
          <td style="text-align:left;">{{ r.unit_type }}</td>           8825
          <td style="text-align:left;">                                 8826
            {% if r.custom %}                                           8827
            <span class="tag tag--warn">{{ r.label }}</span>            8828
            {% else %}                                                  8829
            <span class="tag tag--muted">{{ r.label }}</span>           8830
            {% endif %}                                                 8831
          </td>                                                         8832
          <td class="nowrap">                                           8833
            <span class="row-actions">                                  8834
              <a class="link-btn" href="/ahp/matrix?unit_type=          8835
      {{ r.unit_type }}&level=1">一级指标权重</a>                       8836
              <a class="link-btn" href="/ahp/matrix?unit_type=          8837
      {{ r.unit_type }}&level=1&calc=1">查看一致性</a>                  8838
              {% if r.custom %}                                         8839
              <form method="post" action="/ahp/weights/reset"           8840
                    onsubmit="return confirm('确定把「{{                8841
      r.unit_type }}」恢复为论文表 3 的默认权重吗？');">                8842
                <input type="hidden" name="csrf_token"                  8843
      value="{{ csrf_token() }}">                                       8844
                <input type="hidden" name="unit_type"                   8845
      value="{{ r.unit_type }}">                                        8846
                <button type="submit" class="link-btn                   8847
      link-btn--danger">恢复默认</button>                               8848
              </form>                                                   8849
              {% endif %}                                               8850
```

```
            </span>                                                    8851
          </td>                                                        8852
        </tr>                                                          8853
        {% endfor %}                                                   8854
      </tbody>                                                         8855
    </table>                                                           8856
</div>                                                                 8857
{% endblock %}                                                         8858
# ==== app/templates/ahp/matrix.html ====                              8859
{% extends "base.html" %}                                              8860
{% block title %}判断矩阵录入{% endblock %}                              8861
{% block content %}                                                    8862
<div class="card">                                                     8863
  <h2>                                                                 8864
    判断矩阵录入                                                       8865
    · {{ unit_type }}                                                  8866
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%      8867
    endif %}                                                           8868
  </h2>                                                                8869
  <p class="muted">                                                    8870
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，  8871
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。  8872
    矩阵已按论文表 3 的权重预填。                                      8873
  </p>                                                                 8874
  <form method="post">                                                 8875
    {{ form.hidden_tag() }}                                            8876
    <table style="width:auto;">                                        8877
      <thead>                                                          8878
        <tr>                                                           8879
          <th></th>                                                    8880
          {% for name in names %}<th scope="col">{{ name               8881
    }}</th>{% endfor %}                                                8882
        </tr>                                                          8883
      </thead>                                                         8884
      <tbody>                                                          8885
        {% for i in range(names|length) %}                             8886
        <tr>                                                           8887
          <th scope="row">{{ names[i] }}</th>                          8888
          {% for j in range(names|length) %}                           8889
            {% if i == j %}                                            8890
            <td class="muted num">1</td>                               8891
            {% elif i < j %}                                           8892
            <td>                                                       8893
              <label class="muted" style="position:absolute;           8894
    left:-9999px;"                                                     8895
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}            8896
    相对 {{ names[j] }} 的重要性</label>                               8897
              <input id="m-{{ i }}-{{ j }}" type="number"               8898
    step="0.01" min="0.01"                                             8899
                     name="m_{{ i }}_{{ j }}" value="{{                8900
```

```
    default[i][j] }}">                                                    8901
            </td>                                                         8902
            {% else %}                                                    8903
            <td class="muted num">{{ default[i][j] }}</td>                8904
            {% endif %}                                                   8905
          {% endfor %}                                                    8906
        </tr>                                                             8907
        {% endfor %}                                                      8908
      </tbody>                                                            8909
    </table>                                                              8910
    <p class="actions">                                                   8911
      <button type="submit" class="btn" name="action"                     8912
    value="calc">计算权重</button>                                        8913
      {% if result and result.consistent %}                               8914
      <button type="submit" class="btn" name="action"                     8915
    value="save">保存并生效</button>                                      8916
      {% endif %}                                                         8917
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                 8918
    </p>                                                                  8919
  </form>                                                                 8920
  <p class="muted" style="margin-top:10px;">                              8921
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label              8922
    }}</strong>。                                                         8923
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；  8924
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。       8925
  </p>                                                                    8926
  {% if result %}                                                         8927
  <hr style="margin:22px 0;border:none;border-top:1px solid               8928
    var(--line);">                                                        8929
  <h3>计算结果</h3>                                                       8930
  <table style="width:auto;">                                             8931
    <thead>                                                               8932
      <tr>                                                                8933
        <th scope="col">指标</th>                                         8934
        {% for name in names %}<th scope="col">{{ name                    8935
    }}</th>{% endfor %}                                                   8936
      </tr>                                                               8937
    </thead>                                                              8938
    <tbody>                                                               8939
      <tr>                                                                8940
        <th scope="row">权重</th>                                         8941
        {% for w in result.weights %}<td class="mono">{{ w                8942
    }}</td>{% endfor %}                                                   8943
      </tr>                                                               8944
    </tbody>                                                              8945
  </table>                                                                8946
  <div class="stat-row">                                                  8947
    <div>                                                                 8948
      <div class="stat__label">最大特征值 λmax</div>                      8949
      <div class="stat__value mono">{{ result.lambda_max                  8950
```

```
      }}</div>                                                          8951
      </div>                                                            8952
      <div>                                                             8953
        <div class="stat__label">一致性指标 CI</div>                    8954
        <div class="stat__value mono">{{                               8955
      "%.4f"|format(result.ci) }}</div>                                 8956
      </div>                                                            8957
      <div>                                                             8958
        <div class="stat__label">一致性比例 CR</div>                    8959
        <div class="stat__value mono">{{                               8960
      "%.4f"|format(result.cr) }}</div>                                 8961
      </div>                                                            8962
      <div>                                                             8963
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>   8964
        <div class="stat__value">                                       8965
          {% if result.consistent %}<span class="ok">通过</span>        8966
          {% else %}<span class="warn">不通过</span>{% endif %}         8967
        </div>                                                          8968
      </div>                                                            8969
    </div>                                                              8970
    {% if not result.consistent and result.most_inconsistent            8971
      %}                                                                8972
    <p style="margin-top:14px;padding:12px;border-radius:6px;b          8973
      ackground:var(--danger-soft);                                     8974
                border-left:4px solid var(--danger);">                  8975
      <strong class="warn">建议优先调整：</strong>                      8976
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                 8977
      result.most_inconsistent.j + 1 }} 列的比较                        8978
      （当前 <span class="mono num">{{ result.most_inconsistent.v       8979
      alue|round(3) }}</span>，                                         8980
      理想约 <span class="mono num">{{ result.most_inconsistent.e       8981
      xpected|round(3) }}</span>）                                      8982
    </p>                                                                8983
    {% endif %}                                                         8984
    {% endif %}                                                         8985
</div>                                                                  8986
{% endblock %}                                                          8987
# ==== app/templates/ahp/indicators.html ====                           8988
{% extends "base.html" %}                                               8989
{% block title %}指标定义维护{% endblock %}                             8990
{% block content %}                                                     8991
<div class="card">                                                      8992
  <h2>指标体系维护</h2>                                                 8993
  <p class="actions" style="margin-bottom:14px;">                       8994
    <a class="btn btn--sm btn--ghost"                                   8995
    href="/ahp/">返回权重计算</a>                                       8996
    <a class="btn btn--sm btn--ghost"                                   8997
    href="/ahp/experts">多专家矩阵</a>                                  8998
    <form method="post" action="/ahp/indicators/reset"                  8999
    style="display:inline;"                                             9000
```

```
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 9001
    确定继续？');"> 9002
      {{ form.hidden_tag() }} 9003
      <button type="submit" class="btn btn--sm 9004
    btn--danger">按论文取值重置</button> 9005
    </form> 9006
  </p> 9007
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3, 9008
    "三级指标")] %} 9009
  {% set level_items = all_ind|selectattr("level", 9010
    "equalto", level)|list %} 9011
  {% if level_items %} 9012
  <h3>{{ title }}（{{ level_items|length }} 项）</h3> 9013
  <table> 9014
    <thead> 9015
      <tr> 9016
        <th scope="col">指标</th> 9017
        <th scope="col">上级指标</th> 9018
        <th scope="col">单位</th> 9019
        <th scope="col">标准化类型</th> 9020
        <th scope="col">理想取值</th> 9021
        <th scope="col">启用单元</th> 9022
        <th scope="col">限制性</th> 9023
        <th scope="col">操作</th> 9024
      </tr> 9025
    </thead> 9026
    <tbody> 9027
      {% for i in level_items %} 9028
      <tr> 9029
        <td style="text-align:left;"> 9030
          <span class="mono muted">{{ i.code 9031
    }}</span><br>{{ i.name }} 9032
        </td> 9033
        <td>{{ (all_ind|selectattr("code", "equalto", 9034
    i.parent_code)|list|first).name if i.parent_code else 9035
    "—" }}</td> 9036
        <td>{{ i.unit or "—" }}</td> 9037
        <td>{{ type_label(i.ideal_type) }}</td> 9038
        <td class="mono">{{ i.ideal_value or "—" }}</td> 9039
        <td style="text-align:left;">{{ i.enabled_units or 9040
    "—" }}</td> 9041
        <td> 9042
          {% if i.is_constraint %}<span class="tag 9043
    tag--warn">是</span> 9044
          {% else %}<span class="tag tag--muted">否</span>{% 9045
    endif %} 9046
        </td> 9047
        <td class="nowrap"> 9048
          <span class="row-actions"> 9049
            <a class="link-btn" href="/ahp/indicators/{{ 9050
```

```
      i.code }}/edit">编辑</a>                                          9051
            </span>                                                     9052
          </td>                                                         9053
        </tr>                                                           9054
        {% endfor %}                                                    9055
      </tbody>                                                          9056
    </table>                                                            9057
    {% endif %}                                                         9058
    {% endfor %}                                                        9059
</div>                                                                  9060
{% endblock %}                                                          9061
# ==== app/templates/ahp/indicator_edit.html ====                       9062
{% extends "base.html" %}                                               9063
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                9064
{% block content %}                                                     9065
<div class="card" style="max-width:680px;">                             9066
  <h2>编辑指标 · {{ ind.name }}</h2>                                    9067
  <p class="muted">                                                     9068
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{          9069
    ind.level }} 级，                                                   9070
    由论文指标体系确定，不可修改。                                      9071
  </p>                                                                  9072
  <form method="post">                                                  9073
    {{ form.hidden_tag() }}                                             9074
    <div class="field">                                                 9075
      <label class="field__label" for="name">指标名称</label>           9076
      <input id="name" type="text" name="name"                          9077
    style="width:100%;" required                                        9078
             value="{{ ind.name }}">                                    9079
    </div>                                                              9080
    <div class="field">                                                 9081
      <label class="field__label" for="unit">计量单位</label>           9082
      <input id="unit" type="text" name="unit"                          9083
    style="width:180px;"                                                9084
             value="{{ ind.unit or '' }}">                              9085
    </div>                                                              9086
    <div class="field">                                                 9087
      <label class="field__label" for="ideal-type">标准化类型</la9088
    bel>                                                                9089
      <select id="ideal-type" name="ideal_type">                        9090
        {% for code, label in [("benefit","越大越优"),("cost","越9091
    小越优"),("mid","适中型"),("discrete","离散赋值")] %}               9092
        <option value="{{ code }}" {% if ind.ideal_type ==              9093
    code %}selected{% endif %}>{{ label }}</option>                     9094
        {% endfor %}                                                    9095
      </select>                                                         9096
    </div>                                                              9097
    <div class="field">                                                 9098
      <label class="field__label" for="ideal-value">理想取值</la9099
    bel>                                                                9100
```

```
        <input id="ideal-value" type="text"
      name="ideal_value" style="width:100%;"
                value="{{ ind.ideal_value or '' }}">
        <p class="muted" style="margin:4px 0 0;">
          如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
        </p>
      </div>
      <fieldset style="border:1px solid
      var(--line);border-radius:6px;padding:10px
      14px;margin:0 0 12px;">
        <legend style="padding:0 6px;font-weight:600;">启用该指标的微
      地质单元</legend>
        {% for t in unit_types %}
        <label class="checkbox" style="margin-right:18px;">
          <input type="checkbox" name="enabled_units"
      value="{{ t }}"
                 {% if t in ind.units_list() %}checked{%
      endif %}>
          <span>{{ t }}</span>
        </label>
        {% endfor %}
        <p class="muted" style="margin:8px 0 0;">
          论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
        </p>
      </fieldset>
      <div class="field">
        <label class="checkbox" for="is-constraint">
          <input id="is-constraint" type="checkbox"
      name="is_constraint" value="1"
                 {% if ind.is_constraint %}checked{% endif %}>
          <span>同时作为限制性指标（参与一票否决判定）</span>
        </label>
      </div>
      <div class="field">
        <label class="field__label" for="description">取值依据说明</
      label>
        <textarea id="description" name="description"
      style="min-height:88px;">{{ ind.description or ''
      }}</textarea>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列
      表</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
```

```
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家判断矩阵 · 录入与汇总</h2>
  <form method="get" class="actions" style="margin-bottom:14px;">
    <div>
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type" onchange="this.form.submit()">
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}
        <option value="{{ t }}" {% if unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div>
      <label class="field__label" for="level">指标层级</label>
      <select id="level" name="level" onchange="this.form.submit()">
        <option value="1" {% if level == 1 %}selected{% endif %}>一级指标</option>
        <option value="2" {% if level == 2 %}selected{% endif %}>二级指标</option>
      </select>
    </div>
  </form>
  <p class="muted">
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
  {% if rows %}
  <table>
    <caption class="muted">已录入 {{ rows|length }} 位专家的矩阵</caption>
    <thead>
      <tr>
        <th scope="col">专家</th>
        <th scope="col">矩阵规模</th>
        <th scope="col">该矩阵 CR</th>
        <th scope="col">一致性</th>
        <th scope="col">录入时间</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
```

```
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="num">{{ n }} × {{ n }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
    not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span
    class="tag tag--ok">通过</span>
          {% else %}<span class="tag tag--warn">不通过</span>{%
    endif %}
        </td>
        <td class="mono nowrap">{{ r.created_at.strftime("%Y
    -%m-%d %H:%M") if r.created_at else "—" }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id
    }}/delete"
                onsubmit="return confirm('确定删除专家「{{
    r.expert_name }}」的矩阵吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-bottom:10px;">
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%
    endif %}">
      汇总计算（几何平均）
    </a>
  </p>
  {% else %}
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
  {% endif %}
  <h3>录入一位专家的判断矩阵</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <input type="hidden" name="action" value="add">
    <input type="hidden" name="unit_type" value="{{
    unit_type }}">
    <input type="hidden" name="level" value="{{ level }}">
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1
    }}">{% endif %}
    <div class="field">
      <label class="field__label" for="expert-name">专家姓名</la
    bel>
```

```
        <input id="expert-name" type="text" name="expert_name"
                style="width:220px;" placeholder="如：张工">
      </div>
      <table style="width:auto;">
        <thead>
          <tr>
            <th></th>
            {% for name in names %}<th scope="col">{{ name
      }}</th>{% endfor %}
          </tr>
        </thead>
        <tbody>
          {% for i in range(n) %}
          <tr>
            <th scope="row">{{ names[i] }}</th>
            {% for j in range(n) %}
              {% if i == j %}
              <td class="muted num">1</td>
              {% elif i < j %}
              <td>
                <label class="muted" style="position:absolute;
      left:-9999px;"
                       for="e-{{ i }}-{{ j }}">{{ names[i] }}
      相对 {{ names[j] }} 的重要性</label>
                <input id="e-{{ i }}-{{ j }}" type="number"
      step="0.01" min="0.01"
                       name="m_{{ i }}_{{ j }}" value="{{
      default[i][j] }}">
              </td>
              {% else %}
              <td class="muted num">{{ default[i][j] }}</td>
              {% endif %}
            {% endfor %}
          </tr>
          {% endfor %}
        </tbody>
      </table>
      <p class="actions" style="margin-top:12px;">
        <button type="submit" class="btn">录入该专家矩阵</button>
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护<
      /a>
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts_aggregate.html ====
{% extends "base.html" %}
{% block title %}多专家矩阵汇总{% endblock %}
{% block content %}
```

```
<div class="card">
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}</h2>
  {% if not result %}
  <p class="muted">
    该范围内尚未录入专家矩阵。
    <a href="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">去录入</a>
  </p>
  {% else %}
  <div class="stat-row">
    <div>
      <div class="stat__label">参与汇总的专家</div>
      <div class="stat__value">{{ result.count }} 位</div>
    </div>
    <div>
      <div class="stat__label">λmax</div>
      <div class="stat__value mono">{{ result.lambda_max }}</div>
    </div>
    <div>
      <div class="stat__label">CI</div>
      <div class="stat__value mono">{{ "%.4f"|format(result.ci) }}</div>
    </div>
    <div>
      <div class="stat__label">CR（要求 &lt; 0.1）</div>
      <div class="stat__value mono">{{ "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>{% else %}<span class="warn">不通过</span>{% endif %}
      </div>
    </div>
  </div>
  <h3>汇总后权重</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
```

```
          <th scope="row">权重</th>                                          9351
          {% for w in result.weights %}<td class="mono">{{ w                 9352
    }}</td>{% endfor %}                                                      9353
        </tr>                                                                9354
    </tbody>                                                                 9355
</table>                                                                     9356
<h3>各专家矩阵的一致性</h3>                                                  9357
<table style="width:auto;">                                                  9358
    <thead>                                                                  9359
        <tr><th scope="col">专家</th><th scope="col">该矩阵                  9360
    CR</th><th scope="col">判定</th></tr>                                    9361
    </thead>                                                                 9362
    <tbody>                                                                  9363
        {% for r in rows %}                                                  9364
        <tr>                                                                 9365
            <td>{{ r.expert_name }}</td>                                     9366
            <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is               9367
    not none else "—" }}</td>                                                9368
            <td>                                                             9369
                {% if r.cr is not none and r.cr < 0.1 %}<span                9370
    class="ok">通过</span>                                                   9371
                {% else %}<span class="warn">不通过</span>{% endif %}        9372
            </td>                                                            9373
        </tr>                                                                9374
        {% endfor %}                                                         9375
    </tbody>                                                                 9376
</table>                                                                     9377
{% if not result.consistent and result.most_inconsistent                     9378
  %}                                                                         9379
<p style="padding:12px;border-radius:6px;background:var(--                   9380
    danger-soft);                                                            9381
              border-left:4px solid var(--danger);">                         9382
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>       9383
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                        9384
    result.most_inconsistent.j + 1 }} 列的比较                               9385
    （当前 <span class="mono">{{ result.most_inconsistent.value              9386
    |round(3) }}</span>，                                                    9387
    理想约 <span class="mono">{{ result.most_inconsistent.expec              9388
    ted|round(3) }}</span>）。                                               9389
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。              9390
</p>                                                                         9391
{% endif %}                                                                  9392
{% endif %}                                                                  9393
<p class="actions" style="margin-top:16px;">                                 9394
    <form method="post"                                                      9395
          action="/ahp/experts?unit_type={{ unit_type                        9396
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"                  9397
          style="display:inline;"                                            9398
          onsubmit="return confirm('保存后，「{{ unit_type                   9399
    }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">               9400
```

```
        {{ form.hidden_tag() }}                                          9401
        <input type="hidden" name="action" value="save_agg">             9402
        <button type="submit" class="btn" {% if not                      9403
      result.consistent %}disabled{% endif %}>                           9404
          保存为生效权重                                                 9405
        </button>                                                        9406
      </form>                                                            9407
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={           9408
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1               9409
      }}{% endif %}">返回录入</a>                                        9410
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>            9411
    </p>                                                                 9412
</div>                                                                   9413
{% endblock %}                                                           9414
# ==== app/templates/indicator/index.html ====                           9415
{% extends "base.html" %}                                                9416
{% from "_macros.html" import patch_guide %}                             9417
{% block title %}指标数据采集{% endblock %}                              9418
{% block content %}                                                      9419
<div class="card">                                                       9420
  <h2>指标数据采集与管理</h2>                                            9421
  <p class="actions" style="margin-bottom:12px;">                        9422
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV        9423
    / Excel）</a>                                                        9424
  </p>                                                                   9425
  {{ patch_guide(projects_without_patch) }}                              9426
  {% if not patches %}                                                   9427
  <p class="muted">                                                      9428
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，                  9429
    或运行 <span class="mono">python seed_demo.py</span>                9430
    载入论文案例。                                                       9431
  </p>                                                                   9432
  {% else %}                                                             9433
  {% for p in patches %}                                                 9434
  <div class="patch-panel">                                              9435
    <div class="patch-panel__head">                                      9436
      <h3 class="patch-panel__name">{{ p.name }}</h3>                    9437
      <span class="patch-panel__meta">                                   9438
        所属项目：{{ p.project.name if p.project else "—" }}             9439
      </span>                                                            9440
      <span class="patch-panel__meta">                                   9441
        微地质单元：{{ p.units.count() }} 个                             9442
      </span>                                                            9443
      {% if p.units.count() %}                                           9444
      <span class="patch-panel__meta patch-panel__tail">                 9445
        <a class="link-btn" href="/indicator/import">批量导入本图斑      9446
    数据</a>                                                             9447
      </span>                                                            9448
      {% endif %}                                                        9449
    </div>                                                               9450
```

```
    {% if not p.units.count() %}
    <div class="notice notice--warn">
      <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
      <p style="margin:0;">
        指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
        请先完成划分，再回到本页录入。
      </p>
      <p style="margin:10px 0 0;">
        <a class="btn btn--sm" href="/unit/{{ p.id
    }}">去划分微地质单元</a>
      </p>
    </div>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">微地质单元</th>
          <th scope="col">面积占比</th>
          <th scope="col">数据状态</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for u in p.units %}
        <tr>
          <td>{{ u.unit_type }}</td>
          <td class="num">
            {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
          </td>
          <td>
            {% if u.indicator_data %}
            <span class="tag tag--ok">已录入</span>
            {% else %}
            <span class="tag tag--muted">未录入</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/indicator/{{ u.id
    }}">录入 / 编辑</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
  </div>
  {% endfor %}
```

```
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/indicator/edit.html ====
{% extends "base.html" %}
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}
{% block content %}
<div class="card" style="max-width:820px;">
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>
  <p class="muted">
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for group_name, fields, has_hm in groups %}
    <h3>{{ group_name }}</h3>
    {% for fname, cname, unit_ in fields %}
    <div class="field">
      <label class="field__label" for="f-{{ fname }}">
        {{ cname }}{% if unit_ %} ({{ unit_ }}){% endif %}
      </label>
      {% if fname == "aspect" %}
      <select id="f-{{ fname }}" name="aspect">
        <option value="">— 不适用 —</option>
        {% for a in aspect_types %}
        <option value="{{ a }}" {% if current.get("aspect")
    == a %}selected{% endif %}>{{ a }}</option>
        {% endfor %}
      </select>
      {% else %}
      <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
             value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
      {% endif %}
    </div>
    {% endfor %}
    {% endfor %}
    {% for group_name, fields, has_hm in groups %}
    {% if has_hm %}
    <h3>重金属含量（mg/kg）</h3>
    <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
    <table style="width:auto;">
      <thead>
        <tr>
          {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
        </tr>
      </thead>
```

```
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number" step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy', {}).get(m) if current.get('heavy', {}).get(m) is not none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
       href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
       href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</label>
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
```

```
          {% for p in patches %}                                    9601
          <option value="{{ p.id }}">{{ p.name }} ({{              9602
      p.units.count() }} 个单元）</option>                          9603
          {% endfor %}                                              9604
        </select>                                                   9605
      </div>                                                        9606
      <div class="field">                                           9607
        <label class="field__label" for="datafile">数据文件</label 9608
      >                                                             9609
        <input id="datafile" type="file" name="datafile"            9610
               accept=".csv,.xlsx,.xls" required>                   9611
      </div>                                                        9612
      <p class="actions" style="margin-top:16px;">                  9613
        <button type="submit" class="btn">开始导入</button>         9614
      </p>                                                          9615
    </form>                                                         9616
    <h3>模板列说明</h3>                                             9617
    <p class="muted">                                               9618
      共 {{ headers|length }} 列。留空表示该指标「不适用」。        9619
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可  9620
      （按论文注释视为未受污染）。                                  9621
    </p>                                                            9622
    <p class="mono muted" style="line-height:1.9;">{{               9623
      headers|join(" , ") }}</p>                                    9624
    <h3>导入规则</h3>                                               9625
    <ul class="muted" style="padding-left:20px;line-height:1.9 9626
      ;">                                                           9627
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>  9628
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>     9629
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>  9630
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非  9631
      <span class="mono">10cm</span>）。</li>                       9632
    </ul>                                                           9633
</div>                                                              9634
{% if report %}                                                     9635
<div class="card" style="max-width:860px;">                         9636
    <h2>导入结果</h2>                                               9637
    <div class="stat-row">                                          9638
      <div>                                                         9639
        <div class="stat__label">目标图斑</div>                     9640
        <div style="font-size:15px;">{{ report.patch }}</div>       9641
      </div>                                                        9642
      <div>                                                         9643
        <div class="stat__label">成功导入</div>                     9644
        <div class="stat__value ok">{{ report.saved }}              9645
      个单元</div>                                                  9646
      </div>                                                        9647
      <div>                                                         9648
        <div class="stat__label">问题行</div>                       9649
        <div class="stat__value {% if report.errors %}warn{%       9650
```

```
      endif %}">{{ report.errors|length }}</div>
      </div>
   </div>
   {% if report.errors %}
   <h3>未导入的行</h3>
   <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
   </ul>
   {% else %}
   <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
   {% endif %}
   <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
   </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
   <h2>适宜性定量评价</h2>
   {{ patch_guide(projects_without_patch) }}
   {% if not patches %}
   <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
   {% else %}
   <table>
      <thead>
         <tr>
            <th scope="col">所属项目</th>
            <th scope="col">矿山图斑</th>
            <th scope="col">前置条件</th>
            <th scope="col">最近评价结果</th>
            <th scope="col">操作</th>
         </tr>
      </thead>
      <tbody>
         {% for p in patches %}
         {% set pr = progress.get(p.id) %}
         <tr>
            <td>{{ p.project.name if p.project else "—" }}</td>
            <td style="text-align:left;">{{ p.name }}</td>
            <td style="text-align:left;">
               {% if pr and pr.ready %}
               <span class="tag tag--ok">已具备</span>
               {% elif pr %}
               <span class="tag tag--warn">缺：{{
```

```
      pr.blocking[0].label }}</span>                                    9701
            <span class="muted">{{ pr.blocking[0].reason               9702
      }}</span>                                                        9703
            {% endif %}                                                9704
          </td>                                                        9705
          <td>                                                         9706
            {% if latest.get(p.id) %}                                  9707
            F = <span class="mono num">{{                              9708
      latest[p.id].final_score }}</span>,                              9709
            <span class="tag tag--ok">{{ latest[p.id].grade            9710
      }}</span>                                                        9711
            {% else %}                                                 9712
            <span class="tag tag--muted">未评价</span>                  9713
            {% endif %}                                                9714
          </td>                                                        9715
          <td class="nowrap">                                          9716
            <span class="row-actions">                                 9717
              <a class="link-btn" href="/evaluate/{{ p.id              9718
      }}">执行评价</a>                                                 9719
              {% if latest.get(p.id) %}                                9720
              <a class="link-btn" href="/evaluate/result/{{            9721
      latest[p.id].id }}">查看结果</a>                                 9722
              {% endif %}                                              9723
            </span>                                                    9724
          </td>                                                        9725
        </tr>                                                          9726
        {% endfor %}                                                   9727
      </tbody>                                                         9728
    </table>                                                           9729
    {% endif %}                                                        9730
</div>                                                                 9731
{% endblock %}                                                         9732
# ==== app/templates/evaluate/run.html ====                            9733
{% extends "base.html" %}                                              9734
{% import "_macros.html" as ui %}                                      9735
{% block title %}执行评价 · {{ patch.name }}{% endblock %}             9736
{% block content %}                                                    9737
<div class="card" style="max-width:880px;">                            9738
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                           9739
  {{ ui.workflow_steps(progress) }}                                    9740
  {{ ui.workflow_blocking(progress) }}                                 9741
  {% if progress.ready %}                                              9742
  <form method="post">                                                 9743
    {{ form.hidden_tag() }}                                            9744
    <p class="actions">                                                9745
      <button type="submit" class="btn">开始评价</button>              9746
      <a class="btn btn--ghost" href="/evaluate/">返回</a>             9747
    </p>                                                               9748
  </form>                                                              9749
  {% else %}                                                           9750
```

```
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
        <td>
          {% if progress.constraint.passed %}<span
    class="tag tag--ok">已通过</span>
          {% elif progress.constraint.done %}<span
    class="tag tag--warn">未通过</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.constraint.detail }}</td>
      </tr>
      <tr>
        <td>划分微地质单元</td>
        <td>
          {% if progress.units.done %}<span class="tag
    tag--ok">已完成</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.units.detail }}</td>
      </tr>
      <tr>
        <td>录入指标数据</td>
        <td>
          {% if progress.data.ready %}<span class="tag
    tag--ok">已完成</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.data.detail }}</td>
      </tr>
    </tbody>
```

```
  </table>
</div>
{% if weight_sources %}
<div class="card" style="max-width:880px;">
  <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">
    <thead>
      <tr><th scope="col">微地质单元</th><th
    scope="col">权重来源</th></tr>
    </thead>
    <tbody>
      {% for ut, label in weight_sources.items() %}
      <tr>
        <td style="text-align:left;">{{ ut }}</td>
        <td style="text-align:left;">
          {% if "自定义" in label %}
          <span class="tag tag--warn">{{ label }}</span>
          {% else %}
          <span class="tag tag--muted">{{ label }}</span>
          {% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>
{% endif %}
{% endblock %}
#  ==== app/templates/evaluate/result.html ====
{% extends "base.html" %}
{% block title %}评价结果{% endblock %}
{% block content %}
<div class="card">
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{ r.final_score
    }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
```

```
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F′</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ u.unit_score }}</td>
        <td class="mono">{{ "%.4f"|format(u.ratio *
    u.unit_score) }}</td>
        <td class="muted">{{ u.weight_source or "论文表 3
    默认权重" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:18px;">
    <a class="btn btn--ghost" href="/report/{{ r.patch_id
    }}">查看得分明细</a>
    <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/report/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
  <h2>报告与结果输出</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">可生成报告</th>
```

```
            <th scope="col">操作</th>                                                    9901
          </tr>                                                                          9902
        </thead>                                                                         9903
        <tbody>                                                                          9904
          {% for p in patches %}                                                         9905
          {% set pr = progress.get(p.id) %}                                              9906
          <tr>                                                                           9907
            <td>{{ p.project.name if p.project else "—" }}</td>                          9908
            <td style="text-align:left;">{{ p.name }}</td>                               9909
            <td style="text-align:left;">                                                9910
              {% if pr and pr.ready %}                                                   9911
              <span class="tag tag--ok">可以</span>                                      9912
              {% elif pr %}                                                              9913
              <span class="tag tag--warn">暂不可</span>                                  9914
              <span class="muted">{{ pr.blocking[0].reason                               9915
        }}</span>                                                                        9916
              {% endif %}                                                                9917
            </td>                                                                        9918
            <td class="nowrap">                                                          9919
              {% if pr and pr.ready %}                                                   9920
              <span class="row-actions">                                                 9921
                <a class="link-btn" href="/report/{{ p.id                                9922
        }}">查看得分明细</a>                                                             9923
                <a class="link-btn" href="/report/{{ p.id                                9924
        }}/pdf">导出 PDF</a>                                                             9925
                <a class="link-btn" href="/report/{{ p.id                                9926
        }}/word">导出 Word</a>                                                           9927
              </span>                                                                    9928
              {% else %}                                                                 9929
              <span class="muted">—</span>                                               9930
              {% endif %}                                                                9931
            </td>                                                                        9932
          </tr>                                                                          9933
          {% endfor %}                                                                   9934
        </tbody>                                                                         9935
      </table>                                                                           9936
      {% endif %}                                                                        9937
</div>                                                                                   9938
{% endblock %}                                                                           9939
# ==== app/templates/report/detail.html ====                                             9940
{% extends "base.html" %}                                                                9941
{% block title %}得分明细 · {{ patch.name }}{% endblock %}                               9942
{% block content %}                                                                      9943
<div class="card">                                                                       9944
  <h2>得分明细 · {{ patch.name }}</h2>                                                   9945
  <div class="stat-row">                                                                 9946
    <div>                                                                                9947
      <div class="stat__label">最终评价因子 F</div>                                      9948
      <div class="stat__value mono">{{                                                   9949
    "%.4f"|format(result.final_score) }}</div>                                           9950
```

```
      </div>                                                                    9951
      <div>                                                                     9952
        <div class="stat__label">适宜性等级</div>                               9953
        <div class="stat__value ok">{{ result.grade }}</div>                    9954
      </div>                                                                    9955
      <div style="flex:1;min-width:240px;">                                     9956
        <div class="stat__label">推荐措施</div>                                 9957
        <div style="font-size:15px;">{{                                         9958
      result.grade_info.measure }}</div>                                        9959
      </div>                                                                    9960
    </div>                                                                      9961
    <p class="muted" style="margin-top:12px;">                                  9962
      本页与导出报告均取自                                                      9963
      <strong>{% if record %}{{ record.created_at.strftime("%Y                  9964
      -%m-%d %H:%M:%S") }}{% endif %}                                           9965
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else    9966
      "—" }}），                                                                9967
      不会因事后修改指标数据而变化。                                            9968
    </p>                                                                        9969
    <p class="actions" style="margin-top:16px;">                                9970
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF                 9971
      报告</a>                                                                  9972
      <a class="btn btn--ghost" href="/report/{{ patch.id                       9973
      }}/word">导出 Word 报告</a>                                               9974
      <a class="btn btn--ghost" href="/report/">返回</a>                        9975
    </p>                                                                        9976
</div>                                                                          9977
<div class="card">                                                              9978
    <h2>结果图表</h2>                                                          9979
    <h3>各微地质单元得分对比</h3>                                              9980
    <div>{{ chart_units|safe }}</div>                                           9981
    <h3>各一级指标平均得分</h3>                                                9982
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>9983
    <div>{{ chart_weight|safe }}</div>                                          9984
</div>                                                                          9985
{% for u in result.unit_results %}                                              9986
<div class="card">                                                              9987
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *                   9988
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>                 9989
    {% if u.weight_source %}                                                    9990
    <p class="muted" style="margin-top:-6px;">权重来源：{{                     9991
      u.weight_source }}</p>                                                    9992
    {% endif %}                                                                 9993
    <table>                                                                     9994
      <thead>                                                                   9995
        <tr>                                                                    9996
          <th scope="col">一级指标</th>                                         9997
          <th scope="col">二级指标</th>                                         9998
          <th scope="col">实测值</th>                                           9999
          <th scope="col">标准化值</th>                                        10000
```

```
          <th scope="col">权重</th>                                    10001
          <th scope="col">贡献</th>                                    10002
        </tr>                                                          10003
      </thead>                                                         10004
      <tbody>                                                          10005
        {% for d in u.details %}                                       10006
        <tr>                                                           10007
          <td>{{ d.l1 }}</td>                                          10008
          <td style="text-align:left;">{{ d.l2 }}</td>                 10009
          <td class="mono" style="text-align:left;">{{ "—" if 10010
      d.raw is none else d.raw }}</td>                                 10011
          <td class="mono">{{ "—" if d.std_value is none else 10012
      "%.4f"|format(d.std_value) }}</td>                               10013
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td> 10014
          <td class="mono">{{ "—" if d.contribution is none  10015
      else "%.4f"|format(d.contribution) }}</td>                       10016
        </tr>                                                          10017
        {% endfor %}                                                   10018
      </tbody>                                                         10019
    </table>                                                           10020
    <p class="muted" style="margin:0;">                                10021
      一级指标得分：                                                   10022
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span     10023
      class="num">{{ "%.4f"|format(v) }}</span>{% if not       10024
      loop.last %};{% endif %}{% endfor %}                             10025
    </p>                                                               10026
</div>                                                                 10027
{% endfor %}                                                           10028
{% endblock %}                                                         10029
# ==== app/templates/report/print.html ====                            10030
<!DOCTYPE html>                                                        10031
<html lang="zh-CN">                                                    10032
<head>                                                                 10033
<meta charset="UTF-8">                                                 10034
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>               10035
<style>                                                                10036
  @page { size: A4; margin: 18mm 16mm; }                               10037
  * { box-sizing: border-box; }                                        10038
  body {                                                               10039
    margin: 0; color: #23262b; background: #fff;                       10040
    font-family: "Microsoft YaHei", "SimSun", sans-serif;   10041
    font-size: 10.5pt; line-height: 1.65;                              10042
  }                                                                    10043
  h1 { font-size: 17pt; text-align: center; margin: 0 0      10044
    6px; }                                                             10045
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;10046
       border-bottom: 1px solid #dfe3e8; padding-bottom:     10047
    4px; }                                                             10048
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0    10049
    6px; }                                                             10050
```

```
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace;
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 V1.0 ·
    生成时间 {{ generated_at }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
  </table>
```

```
<h2>二、评价结论</h2>
<table class="kv">
  <tr><th>最终评价因子 F</th><td class="mono"><strong>{{ "%.4f"|format(result.final_score) }}</strong></td></tr>
  <tr><th>适宜性等级</th><td><strong>{{ result.grade }}</strong></td></tr>
  <tr><th>判定依据</th><td>{{ result.grade_info.meaning }}</td></tr>
  <tr><th>推荐措施</th><td>{{ result.grade_info.measure }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
  <thead>
    <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子 F'</th><th>加权贡献</th></tr>
  </thead>
  <tbody>
    {% for u in result.unit_results %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="mono">{{ "%.2f%%"|format(u.ratio * 100) }}</td>
      <td class="mono">{{ "%.4f"|format(u.unit_score) }}</td>
      <td class="mono">{{ "%.4f"|format(u.weighted) }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio * 100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
  <thead>
    <tr><th>一级指标</th><th>二级指标</th><th>实测值</th><th>标准化值</th><th>权重</th><th>贡献</th></tr>
  </thead>
  <tbody>
    {% for d in u.details %}
    <tr>
      <td>{{ d.l1 }}</td>
      <td class="left">{{ d.l2 }}</td>
```

```
            <td class="mono left">{{ "—" if d.raw is none else
      d.raw }}</td>
            <td class="mono">{{ "—" if d.std_value is none else
      "%.4f"|format(d.std_value) }}</td>
            <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
            <td class="mono">{{ "—" if d.contribution is none
      else "%.4f"|format(d.contribution) }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      <p class="note">
        一级指标得分：
        {% for k, v in u.l1_scores.items() %}{{ k }} = {{
        "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
        endfor %}
      </p>
      {% endfor %}
      <h2>六、评价依据说明</h2>
      <ul class="note">
        <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、
            适中型与离散赋值四类。</li>
        <li>单元评价因子按论文式（9） 由二级指标权重加权求得；
            整矿评价因子由各微地质单元按面积占比加权合成。</li>
        <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7
        适宜。</li>
        <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
      </ul>
      <p class="sig">
        评价人（签字）：_____　　日期：_____
      </p>
    </body>
    </html>
    # ==== app/templates/search.html ====
    {% extends "base.html" %}
    {% from "_macros.html" import icon %}
    {% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
    {% block content %}
    <div class="card">
      <h2>搜索结果</h2>
      {% if not q %}
      <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
        标准名称与文号。</p>
      {% else %}
      <p class="muted">
        关键词「<strong>{{ q }}</strong>」共找到
        <span class="num">{{ total }}</span> 条结果。
      </p>
      {% if not total %}
      <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
```

```
  {% endif %}
  {% endif %}
</div>
{% if projects %}
<div class="card">
  <h2>评价项目（{{ projects|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
        <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
        <td class="num">{{ p.patches.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/project/{{ p.id
    }}/patches">矿山图斑</a>
            <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>
{% endif %}
{% if patches %}
<div class="card">
  <h2>矿山图斑（{{ patches|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">矿种</th>
        <th scope="col">操作</th>
      </tr>
```

```
      </thead>                                                                  10251
      <tbody>                                                                   10252
        {% for m in patches %}                                                  10253
        <tr>                                                                    10254
          <td>{{ m.project.name if m.project else "—" }}</td>                   10255
          <td style="text-align:left;">{{ m.name }}</td>                        10256
          <td>{{ m.admin_region or "—" }}</td>                                  10257
          <td>{{ m.mineral_type or "—" }}</td>                                  10258
          <td class="nowrap">                                                   10259
            <span class="row-actions">                                          10260
              <a class="link-btn" href="/constraint/{{ m.id                     10261
      }}">限制性判定</a>                                                        10262
              <a class="link-btn" href="/unit/{{ m.id                           10263
      }}">微地质单元</a>                                                        10264
              <a class="link-btn" href="/evaluate/{{ m.id                       10265
      }}">执行评价</a>                                                          10266
            </span>                                                             10267
          </td>                                                                 10268
        </tr>                                                                   10269
        {% endfor %}                                                            10270
      </tbody>                                                                  10271
    </table>                                                                    10272
</div>                                                                          10273
{% endif %}                                                                     10274
{% if standards %}                                                              10275
<div class="card">                                                              10276
    <h2>标准与政策文件（{{ standards|length }}）</h2>                           10277
    <table>                                                                     10278
      <thead>                                                                   10279
        <tr>                                                                    10280
          <th scope="col">类别</th>                                             10281
          <th scope="col">省份</th>                                             10282
          <th scope="col">名称</th>                                             10283
          <th scope="col">文号</th>                                             10284
          <th scope="col">操作</th>                                             10285
        </tr>                                                                   10286
      </thead>                                                                  10287
      <tbody>                                                                   10288
        {% for s in standards %}                                                10289
        <tr>                                                                    10290
          <td>{{ s.category }}</td>                                             10291
          <td>{{ s.province }}</td>                                             10292
          <td style="text-align:left;">{{ s.name }}</td>                        10293
          <td class="mono">{{ s.doc_number or "—" }}</td>                       10294
          <td class="nowrap">                                                   10295
            <span class="row-actions">                                          10296
              <a class="link-btn" href="/standard/{{ s.id                       10297
      }}/rules">规则</a>                                                        10298
            </span>                                                             10299
          </td>                                                                 10300
```

```
        </tr>                                                                  10301
        {% endfor %}                                                           10302
      </tbody>                                                                 10303
    </table>                                                                   10304
</div>                                                                         10305
{% endif %}                                                                    10306
{% endblock %}                                                                 10307
#  ==== seed_demo.py ====                                                     10308
#  -*- coding: utf-8 -*-                                                      10309
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                          10310
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。                     10311
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，      10312
                                               不会删除用户自建的项目与图斑  10313
      python seed_demo.py --reset          连同标准库与指标定义一并重建      10314
"""                                                                            10315
import json                                                                    10316
import sys                                                                     10317
sys.stdout.reconfigure(encoding="utf-8")                                       10318
from app import create_app                                                     10319
from app.models import (db, User, Project, MinePatch,                          10320
    ReferenceEcosystem,                                                        10321
                            MicroUnit, IndicatorData, Standard,                10322
    StandardRule,                                                              10323
                            ConstraintCheck, EvaluationResult,                 10324
    ScoreDetail)                                                               10325
app = create_app()                                                             10326
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」             10327
DEMO_CODE = "DEMO-001"                                                         10328
#  论文表 1：11 项地方标准与政策文件                                          10329
STANDARDS = [                                                                  10330
    dict(category="地方标准", province="辽宁",                                 10331
    name="矿山废弃地植被自然恢复技术规程",                                     10332
        doc_number="DB21/T 3702—2023",                                         10333
    publish_date="2023-03-30",                                                 10334
        implement_date="2023-04-30",                                           10335
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",  10336
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),                   10337
               ("无地质灾害", "==", "是", "必要条件", None),                   10338
               ("无水土污染", "==", "是", "必要条件", None),                   10339
               ("与参照系差距", "<", "20", "充分条件", "%")]),                 10340
    dict(category="地方标准", province="湖北",                                 10341
    name="废弃露天矿山生态环境自然恢复标准",                                   10342
        doc_number="DB42/T 2421—2025",                                         10343
    publish_date="2025-08-20",                                                 10344
        implement_date="2025-10-20",                                           10345
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",                  10346
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),         10347
    dict(category="地方标准", province="内蒙古",                               10348
    name="历史遗留废弃矿山自然恢复判定规范",                                   10349
        doc_number="DB15/T 4280—2026",                                         10350
```

```
        publish_date="2026-01-30",                                          10351
        implement_date="2026-03-01",                                        10352
        summary="调查内容包含参照生态系统调查",                                  10353
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),          10354
               ("地形地貌与周边协调", "==", "是", "必要条件", None),            10355
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),      10356
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),          10357
    dict(category="地方标准", province="江西",                                10358
    name="废弃矿山自然恢复技术规范",                                            10359
        doc_number="DB36/T 2221—2026",                                      10360
    publish_date="2026-02-02",                                              10361
        implement_date="2026-08-01",                                        10362
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",                 10363
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),              10364
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),        10365
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",10366
        doc_number=None, publish_date="2026-06-18",                         10367
    implement_date=None,                                                    10368
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",              10369
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),              10370
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),      10371
               ("与参照系差距", "<", "20", "充分条件", "%")]),                 10372
    dict(category="政策文件", province="安徽",                                10373
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",10374
        doc_number="皖自然资修函[2020]34号",                                 10375
    publish_date="2020-03-03", implement_date=None,                         10376
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",10377
        rules=[("责任主体", "==", "政府", "必要条件", None),                  10378
               ("无地质灾害", "==", "是", "必要条件", None),                  10379
               ("无水土污染", "==", "是", "必要条件", None)]),                10380
    dict(category="政策文件", province="山东",                                10381
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",10382
        doc_number="鲁自然资字[2022]104号",                                  10383
    publish_date="2022-08-02", implement_date=None,                         10384
        summary="条件与安徽省相同，另增加自然恢复认定标准",                      10385
        rules=[("无地质灾害", "==", "是", "必要条件", None),                  10386
               ("无水土污染", "==", "是", "必要条件", None)]),                10387
    dict(category="政策文件", province="湖南",                                10388
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",10389
        doc_number="湘自资办发[2023]58号",                                   10390
    publish_date="2023-07-20", implement_date=None,                         10391
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度10392
    >75%、连续3年正向增长",                                                  10393
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),                    10394
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),10395
               ("无地质灾害", "==", "是", "必要条件", None),                  10396
               ("符合国土空间规划", "==", "是", "必要条件", None)]),          10397
    dict(category="政策文件", province="新疆",                                10398
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",      10399
        doc_number="新自然资源办函〔2024〕219号",                            10400
```

```
        publish_date="2024-12-20", implement_date=None,
            summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
            rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="广西",
            name="广西壮族自治区历史遗留矿山自然恢复技术指南",
            doc_number="桂自然资办〔2024〕250号",
        publish_date="2024-12-30", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
```

```
                        "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                                "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                        "coverage": 58.42, "shannon": 0.72},
            },
    ],
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE).
    all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
```

```
            synchronize_session=False)                                  10501
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_    10502
    ids)).delete(                                                       10503
            synchronize_session=False)                                  10504
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).    10505
    delete(                                                             10506
            synchronize_session=False)                                  10507
    if demo_ids:                                                        10508
        Project.query.filter(Project.id.in_(demo_ids)).delet    10509
    e(                                                                  10510
            synchronize_session=False)                                  10511
    if reset:                                                           10512
        StandardRule.query.delete()                                     10513
        Standard.query.delete()                                         10514
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                  10515
    #  避免长期运行后 score_details 表堆积无效数据。                   10516
    _purge_orphan_details()                                             10517
    db.session.commit()                                                 10518
    return len(demo_ids)                                                10519
def demo_patch():                                                       10520
    """取内置演示案例的图斑。                                          10521
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），               10522
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。       10523
    没有演示案例时才退回第一个图斑。                                   10524
    """                                                                 10525
    proj = Project.query.filter_by(code=DEMO_CODE).first()              10526
    if proj is not None:                                                10527
        m = MinePatch.query.filter_by(project_id=proj.id).fi    10528
    rst()                                                               10529
        if m is not None:                                               10530
            return m                                                    10531
    return MinePatch.query.first()                                      10532
def _purge_orphan_details():                                            10533
    """清理指向已不存在评价记录的孤儿得分明细。                        10534
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。    10535
    """                                                                 10536
    live_ids = [r.id for r in EvaluationResult.query.all()]             10537
    if live_ids:                                                        10538
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(    10539
    live_ids)).delete(                                                  10540
            synchronize_session=False)                                  10541
    else:                                                               10542
        ScoreDetail.query.delete()                                      10543
    db.session.commit()                                                 10544
def seed(reset=False):                                                  10545
    with app.app_context():                                             10546
        db.create_all()          #  全新环境下先建表，保证本脚本可独立运行  10547
        _purge_orphan_details()                                         10548
        admin = User.query.filter_by(username="admin").first    10549
    ()                                                                  10550
```

```
    if not admin:                                                    10551
        from werkzeug.security import                                10552
generate_password_hash                                               10553
        admin = User(username="admin",                               10554
                     password_hash=generate_password_has             10555
h("admin123"),                                                       10556
                     role="admin")                                   10557
        db.session.add(admin)                                        10558
        db.session.commit()                                          10559
    #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目                  10560
    existing = Project.query.filter_by(code=DEMO_CODE).f             10561
irst()                                                               10562
    if existing and not reset:                                       10563
        print("演示案例已存在，跳过写入（用户自建项目不受影响）")                  10564
        _report()                                                    10565
        _seed_reference_data(reset=False)                            10566
        return                                                       10567
    clean_case(reset=reset)                                          10568
    project = Project(name=CASE["name"], code=DEMO_CODE,             10569
                      description="论文第 5 节典型案例",                  10570
created_by=admin.id)                                                 10571
    db.session.add(project)                                          10572
    db.session.flush()                                               10573
    patch = MinePatch(project_id=project.id,                         10574
name=CASE["patch_name"],                                             10575
                      admin_region="河北省唐山市玉田县",                   10576
                      area_ha=CASE["area_ha"],                       10577
                      mineral_type=CASE["mineral_type"],             10578
                      mining_method="露天开采",                        10579
stop_years=None,                                                     10580
                      responsible_entity_missing=True)               10581
    db.session.add(patch)                                            10582
    db.session.flush()                                               10583
    ref = ReferenceEcosystem(patch_id=patch.id,                      10584
                             coverage_benchmark=75.0,                10585
shannon_benchmark=0.7)                                               10586
    db.session.add(ref)                                              10587
    for u in CASE["units"]:                                          10588
        unit = MicroUnit(patch_id=patch.id,                          10589
unit_type=u["unit_type"],                                            10590
                         ratio=u["ratio"], area=None)                10591
        db.session.add(unit)                                         10592
        db.session.flush()                                           10593
        d = u["data"]                                                10594
        ind = IndicatorData(                                         10595
            unit_id=unit.id,                                         10596
            soil_ratio=d.get("soil_ratio"),                          10597
            thickness=d.get("thickness"),                            10598
            ph=d.get("ph"),                                          10599
            organic_matter=d.get("organic_matter"),                  10600
```

```
                heavy_metals=json.dumps(d["heavy_metals"],  10601
    ensure_ascii=False)                                     10602
                                    if d.get("heavy_metals") else10603
    None,                                                   10604
                coverage=d.get("coverage"),                 10605
                shannon=d.get("shannon"),                   10606
                slope=d.get("slope"),                       10607
                aspect=d.get("aspect"),                     10608
            )                                               10609
            db.session.add(ind)                             10610
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、  10611
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。  10612
        db.session.add(ConstraintCheck(                     10613
            patch_id=patch.id,                              10614
            policy_pass=True, location_pass=True,           10615
    pollution_pass=True,                                    10616
            geo_pass=True, farmland_pass=True,              10617
            passed=True, reason="全部通过"))                10618
        db.session.commit()                                 10619
        print("已写入案例：", CASE["name"])                 10620
        print("   图斑：", CASE["patch_name"])              10621
        print("   微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台  10622
    23.19%")                                                10623
        print("   管理员账号：admin / admin123")            10624
        _seed_reference_data(reset=reset)                   10625
        _report()                                           10626
def _seed_reference_data(reset=False):                      10627
    """写入标准库与指标定义。                               10628
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护  10629
    的标准条目与指标定义。                                  10630
    """                                                     10631
    if reset or Standard.query.count() == 0:                10632
        seed_standards()                                    10633
    from app.services.indicator_service import              10634
    seed_indicators                                         10635
    n = seed_indicators()                                   10636
    print("指标定义：%d 项" % n)                            10637
def _report():                                              10638
    total_p = Project.query.count()                         10639
    total_patch = MinePatch.query.count()                   10640
    print()                                                 10641
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))  10642
    if total_p > total_patch:                               10643
        print("   提示：项目是归档单位，评价针对的是图斑。")  10644
        print("         新项目需先建立图斑，才会出现在后续各模块中。")  10645
def seed_standards():                                       10646
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""  10647
    StandardRule.query.delete()                             10648
    Standard.query.delete()                                 10649
    db.session.commit()                                     10650
```

```
    for s in STANDARDS:                                                    10651
        std = Standard(category=s["category"],                             10652
    province=s["province"],                                                10653
                           name=s["name"],                                 10654
    doc_number=s["doc_number"],                                            10655
                           publish_date=s["publish_date"],                 10656
                           implement_date=s["implement_date"],             10657
                           summary=s["summary"])                           10658
        db.session.add(std)                                                10659
        db.session.flush()                                                 10660
        for item, op, thr, ctype, note in s["rules"]:                      10661
            db.session.add(StandardRule(standard_id=std.id,                10662
    item=item,                                                             10663
                                        operator=op,                       10664
    threshold=thr,                                                         10665
                                                                           10666
    condition_type=ctype, note=note))                                      10667
    db.session.commit()                                                    10668
    print("已写入标准/政策文件：%d 项，规则 %d 条"                                  10669
          % (len(STANDARDS), sum(len(s["rules"]) for s in                  10670
    STANDARDS)))                                                           10671
if __name__ == "__main__":                                                 10672
    # --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。                      10673
    # 无论是否 --reset，都不会删除用户自建的项目与图斑。                                 10674
    seed(reset="--reset" in sys.argv[1:])                                  10675
#  ==== 测试/test_ahp.py ====                                             10676
#  -*- coding: utf-8 -*-                                                  10677
"""地基自检：验证 AHP 引擎与应用可启动。"""                                          10678
import sys                                                                 10679
sys.stdout.reconfigure(encoding="utf-8")                                   10680
import numpy as np                                                         10681
import os                                                                  10682
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                             10683
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                  10684
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil10685
    e__)))                                                                 10686
if ROOT not in sys.path:                                                   10687
    sys.path.insert(0, ROOT)                                               10688
from app.services import ahp_service as ahp                                10689
LINE = "=" * 66                                                            10690
#  ------------------------------------------------------------------10691
    ------------------                                                     10692
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                                        10693
#  ------------------------------------------------------------------10694
    ------------------                                                     10695
print(LINE)                                                                10696
print("测试 1  完全一致判断矩阵的权重还原与一致性")                                    10697
print(LINE)                                                                10698
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                                    10699
w_true = np.array([0.44, 0.31, 0.25])                                      10700
```

```
A = np.outer(w_true, 1.0 / w_true)                                    10701
r = ahp.weights_and_consistency(A)                                    10702
print("  判断矩阵 A：")                                               10703
for row in A:                                                         10704
    print("    " + "  ".join(f"{v:8.4f}" for v in row))               10705
print()                                                               10706
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                   10707
    r["weights"]))                                                    10708
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))          10709
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"              10710
      % (r["lambda_max"], r["ci"], r["cr"],                           10711
    r["consistent"]))                                                 10712
assert np.allclose(r["weights"], w_true, atol=1e-6),                  10713
    "权重未精确还原"                                                  10714
assert r["consistent"], "完全一致矩阵应通过检验"                      10715
# ------------------------------------------------------------------10716
    ----------------                                                  10717
# 2） 上三角输入 → 完整矩阵 → 一致性定位                             10718
# ------------------------------------------------------------------10719
    ----------------                                                  10720
print()                                                               10721
print(LINE)                                                           10722
print("测试 2  上三角录入与最不一致项定位")                          10723
print(LINE)                                                           10724
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）   10725
M = ahp.matrix_from_upper([2, 5, 3])                                  10726
r2 = ahp.weights_and_consistency(M)                                   10727
bad = ahp.find_most_inconsistent(M)                                   10728
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")                       10729
for row in M:                                                         10730
    print("    " + "  ".join(f"{v:8.4f}" for v in row))               10731
print("  权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),        10732
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))         10733
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"          10734
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],                    10735
    bad["expected"]))                                                 10736
# ------------------------------------------------------------------10737
    ----------------                                                  10738
# 3） 多专家聚合（几何平均）                                         10739
# ------------------------------------------------------------------10740
    ----------------                                                  10741
print()                                                               10742
print(LINE)                                                           10743
print("测试 3  多专家判断矩阵聚合（几何平均）")                      10744
print(LINE)                                                           10745
m1 = ahp.matrix_from_upper([2, 4, 3])                                 10746
m2 = ahp.matrix_from_upper([3, 5, 2])                                 10747
agg = ahp.aggregate_matrices([m1, m2])                                10748
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                10749
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"                   10750
```

```
        %（agg[0，1]，agg[0，2]，agg[1，2]））
assert abs(agg[0，1] - np.sqrt(2 * 3)) < 1e-9，"几何平均错误"
# ----------------------------------------------------------------
    ----------------
# 4） 论文表 3 权重可还原性
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 （0.44， 0.31， 0.25）")
print(LINE)
j = ahp.judge_from_weights([0.44， 0.31， 0.25])
print("  λmax = %.6f    CR = %.3e    一致： %s" %
    (j["lambda_max"]， j["cr"]， j["consistent"]))
assert j["consistent"]， "论文权重应能作为完全一致矩阵主特征向量"
# ----------------------------------------------------------------
    ----------------
# 5） 应用可启动 + 建表
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
    )
if ROOT not in sys.path:
    sys.path.insert(0， ROOT)
from app.services import evaluation_service as ev
```

```
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ------------------------------------------------------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
```

```
    print("-" * 60)
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】 （面积占比 %.2f%%）" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
```

```
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码:", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码:", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）:", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
```

```
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3　提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 （0.44, 0.31, 0.25） 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4　不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
```

```
    SECRET_KEY = "test"                                                 11001
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(              11002
        ROOT, "app", "app.db")                                          11003
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              11004
    WTF_CSRF_ENABLED = False                                            11005
    TESTING = True                                                      11006
app = create_app(TestConfig)                                            11007
client = app.test_client()                                              11008
print(LINE)                                                             11009
print("测试 1  未登录应被重定向到登录页")                                  11010
print(LINE)                                                             11011
r = client.get("/project/")                                             11012
print("  /project/ 状态码:", r.status_code, " 重定向到:",                 11013
    r.headers.get("Location"))                                          11014
assert r.status_code == 302 and "login" in                              11015
    r.headers.get("Location", "")                                       11016
print()                                                                 11017
print(LINE)                                                             11018
print("测试 2  登录")                                                    11019
print(LINE)                                                             11020
r = client.post("/login", data={"username": "admin",                    11021
    "password": "admin123"},                                            11022
                follow_redirects=True)                                  11023
body = r.get_data(as_text=True)                                         11024
# 工作台:页首横幅 + 评价流程分区 + 数据概览                              11025
print("  登录后含「评价流程」:", "评价流程" in body)                      11026
print("  含账户 admin:", "admin" in body)                               11027
print("  含页首横幅:", 'class="hero"' in body)                          11028
print("  含数据概览:", 'class="stat-card"' in body)                     11029
assert "评价流程" in body and 'class="hero"' in body                     11030
print()                                                                 11031
print(LINE)                                                             11032
print("测试 3  项目与矿山档案")                                          11033
print(LINE)                                                             11034
r = client.get("/project/")                                             11035
body = r.get_data(as_text=True)                                         11036
print("  项目列表状态码:", r.status_code)                               11037
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)          11038
assert "玉田县典型历史遗留废弃矿山案例" in body                          11039
# 新建项目                                                              11040
client.post("/project/add", data={"name": "测试项目A", "code":          11041
    "T-001"})                                                           11042
r = client.get("/project/")                                             11043
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True11044
    ))                                                                  11045
assert "测试项目A" in r.get_data(as_text=True)                           11046
# 清理测试数据,避免污染演示库                                          11047
with app.app_context():                                                 11048
    from app.models import db, Project                                  11049
    for p in Project.query.filter_by(name="测试项目A").all():           11050
```

```
            db.session.delete(p)
        db.session.commit()
print("    已清理测试项目")
print()
print(LINE)
print("测试 4    微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
        patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
        irst()
        pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("    单元列表含三类单元：", all(t in body for t in ("岩质边坡",
        "碎石边坡", "平台")))
print("    含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5    标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("    含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
        "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("    口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("    含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6    限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    含五项条件：", all(x in body for x in ("政策约束", "区位约束",
        "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
        "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "1",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
```

```
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
```

```
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡")
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
```

```
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
```

```
        rec = save_evaluation(pid, units)
        rid = rec.id
        print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
              % (rid, rec.final_score, rec.grade))
        from app.models import ScoreDetail
        n_detail = ScoreDetail.query.filter_by(result_id=rid).co
        unt()
        print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
      ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
      "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
```

```
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
```

```
      "[不符合]"))                                                    11351
# 2) 勾选后应下发带过期时间的 remember_token                          11352
c = app.test_client()                                                 11353
r = c.post("/login", data={"username": "admin", "password":           11354
      "admin123",                                                     11355
                                  "remember": "y"},                   11356
      follow_redirects=False)                                         11357
cookies = r.headers.getlist("Set-Cookie")                             11358
rt = [h for h in cookies if h.startswith("remember_token=")]          11359
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else          11360
      "[不符合]"))                                                    11361
assert rt, "未下发 remember_token"                                    11362
exp = re.search(r"Expires=([^;]+)", rt[0])                            11363
print("  %s 该 cookie 带过期时间：%s"                                  11364
        % ("[符合]" if exp else "[不符合]", exp.group(1) if exp        11365
      else "无（会话 cookie）"))                                      11366
assert exp, "remember_token 无过期时间，关浏览器即失效"               11367
# 3) 不勾选时不应下发 remember_token                                  11368
c2 = app.test_client()                                                11369
r2 = c2.post("/login", data={"username": "admin",                     11370
      "password": "admin123"},                                        11371
                follow_redirects=False)                               11372
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                 11373
      h.startswith("remember_token=")]                                11374
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2      11375
      else "[不符合]"))                                               11376
assert not rt2                                                        11377
# 4) 携带 remember_token 的新会话应自动登录                           11378
c3 = app.test_client()                                                11379
c3.set_cookie("remember_token",                                       11380
                rt[0].split(";")[0].split("=", 1)[1],                 11381
      domain="localhost")                                             11382
r3 = c3.get("/", follow_redirects=False)                              11383
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"     11384
        % ("[符合]" if r3.status_code == 200 else "[不符合]",         11385
      r3.status_code))                                                11386
assert r3.status_code == 200                                          11387
# ------------------------------------------------------------------- 11388
      ------------------                                              11389
print()                                                               11390
print(LINE)                                                           11391
print("二、删除功能")                                                 11392
print(LINE)                                                           11393
client.post("/login", data={"username": "admin",                      11394
      "password": "admin123"})                                        11395
# 项目：列表页须有删除按钮                                            11396
html = client.get("/project/").get_data(as_text=True)                 11397
has_del = "/delete" in html and "btn--danger" in html                 11398
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))  11399
assert has_del, "项目列表缺少删除按钮"                                11400
```

```
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
    project_id = patch.project_id
html = client.get("/project/%d/patches" %
    project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
```

```
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
```

```
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
```

```
        "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
    ------------------
print()
```

```
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
```

```
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
```

```
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
#  报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
```

```
                                ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
```

```
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
```

```
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
```

```
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
```

```
      (r.headers.get("Location") or ""),
     "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
      rr = c.get("/report/%d%s" % (pid2, suffix),
      follow_redirects=False)
      ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
      %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
      add_data(pid3)
with app.app_context():
      pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
      "evaluate",
     "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
     "直接访问 /report/<id> 被退回，且不产生任何评价结果",
     "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
      pr = wf.report_progress(pid3)
      rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
      rr.status_code)
ok(("%.4f" % rec.final_score) in body,
     "报告数值取自评价记录，而非重新计算",
     "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete()，
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
```

```
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
```

```
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
                follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
            follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
                data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
# ==== 测试/test_weights.py ====
# -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    )))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
```

```
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
```

```
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
```

```
        label)                                                          12201
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",          12202
    "土壤指标 %.2f" % w["l1"]["土壤指标"])                               12203
pid = build_case()                                                      12204
base_score = run_eval(pid)                                              12205
print("   默认权重下的评价值 F = %.4f" % base_score)                       12206
print()                                                                 12207
print(LINE)                                                             12208
print("二、保存 AHP 算出的权重后，评价应改用新权重")                        12209
print(LINE)                                                             12210
with app.app_context():                                                 12211
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）      12212
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                 12213
    0.20),                                                              12214
                                ("terrain", 0.10)])                     12215
    n_items = WeightItem.query.count()                                  12216
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)     12217
with app.app_context():                                                 12218
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)               12219
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",   12220
    label2)                                                             12221
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",               12222
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])                              12223
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")    12224
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                     12225
    "未覆盖的二级权重仍保留论文取值",                                    12226
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                      12227
new_score = run_eval(pid)                                               12228
print("   新权重下的评价值 F = %.4f" % new_score)                          12229
ok(abs(new_score - base_score) > 1e-4,                                  12230
    "权重变化确实改变了评价结果",                                        12231
    "%.4f → %.4f" % (base_score, new_score))                            12232
print()                                                                 12233
print(LINE)                                                             12234
print("三、结果应记录所用权重来源")                                        12235
print(LINE)                                                             12236
import json                                                             12237
with app.app_context():                                                 12238
    rec = EvaluationResult.query.filter_by(patch_id=pid) \              12239
          .order_by(EvaluationResult.id.desc()).first()                 12240
    stored = json.loads(rec.unit_scores)                                12241
ok(any(u.get("weight_source") for u in stored),                         12242
    "评价记录中标注了权重来源",                                          12243
    stored[0].get("weight_source") or "")                               12244
print()                                                                 12245
print(LINE)                                                             12246
print("四、恢复默认后应回到论文取值")                                      12247
print(LINE)                                                             12248
with app.app_context():                                                 12249
    ws.reset_weights(UNIT_TYPE)                                         12250
```

```
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
```

```
# ==== app/views/__init__.py ====                                    12301
# -*- coding: utf-8 -*-                                              12302
```